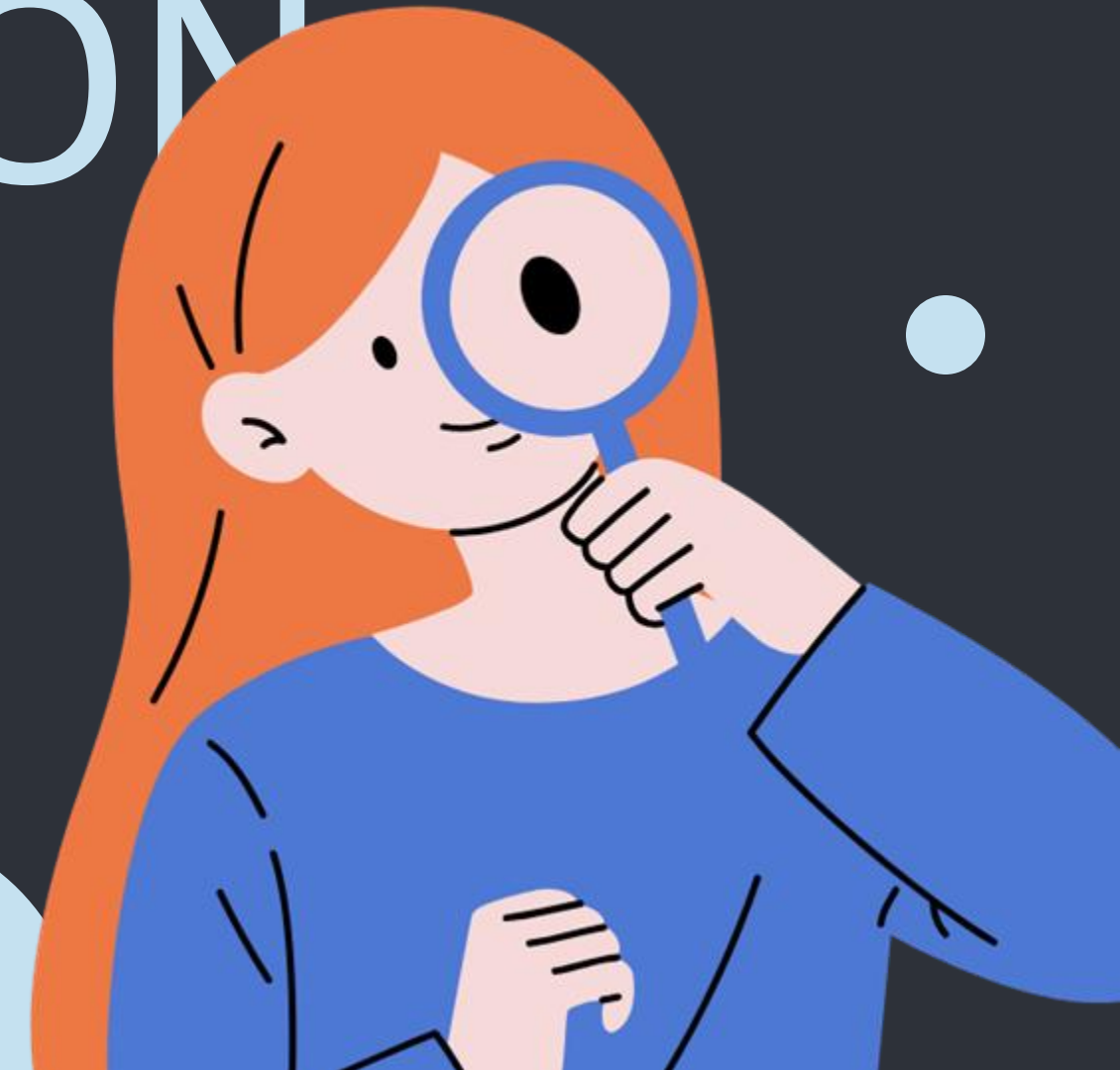
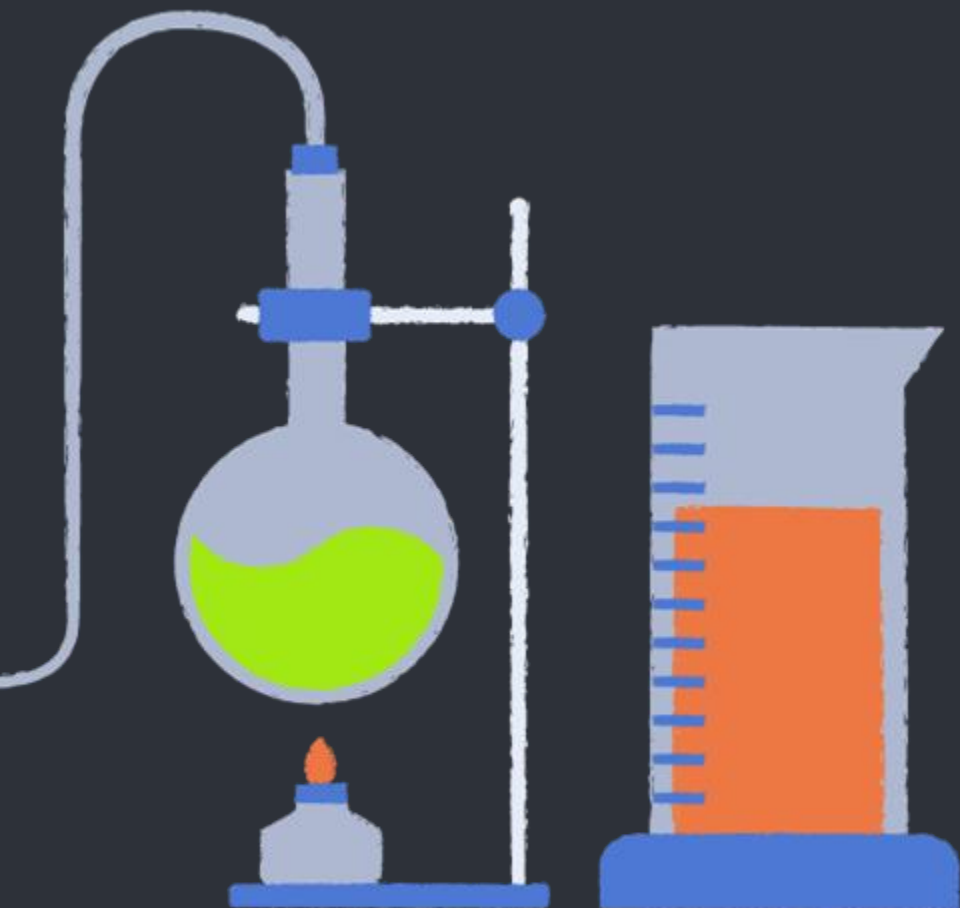


Class 10th One-Shot series

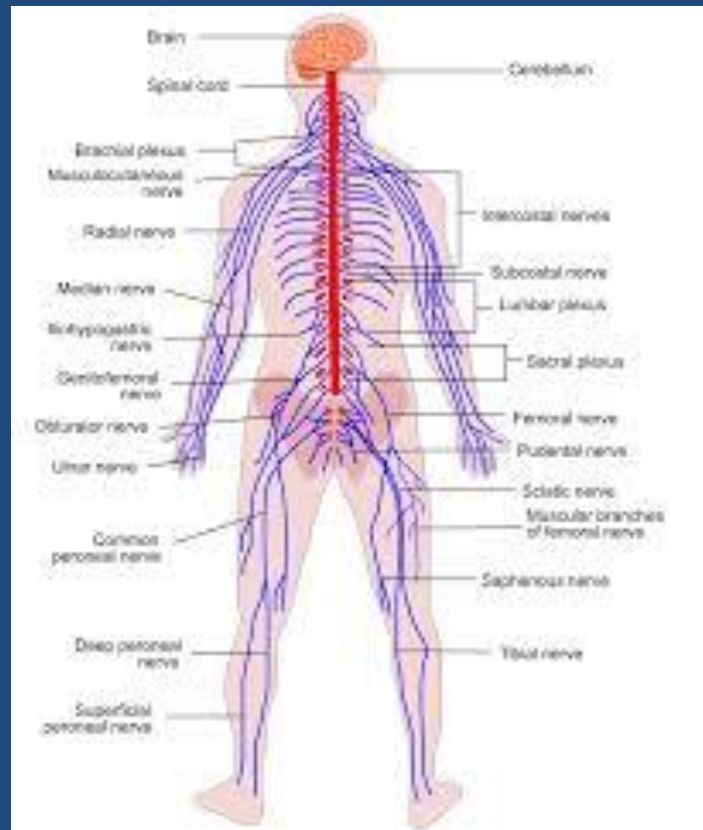
CONTROL AND

COORDINATION BIOLOGY

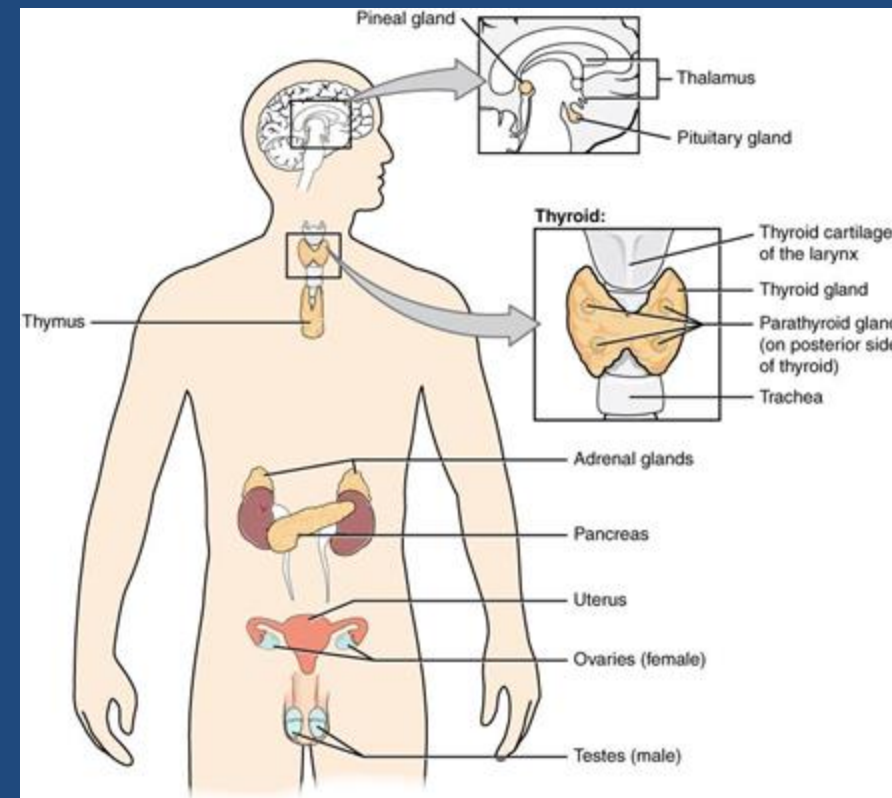




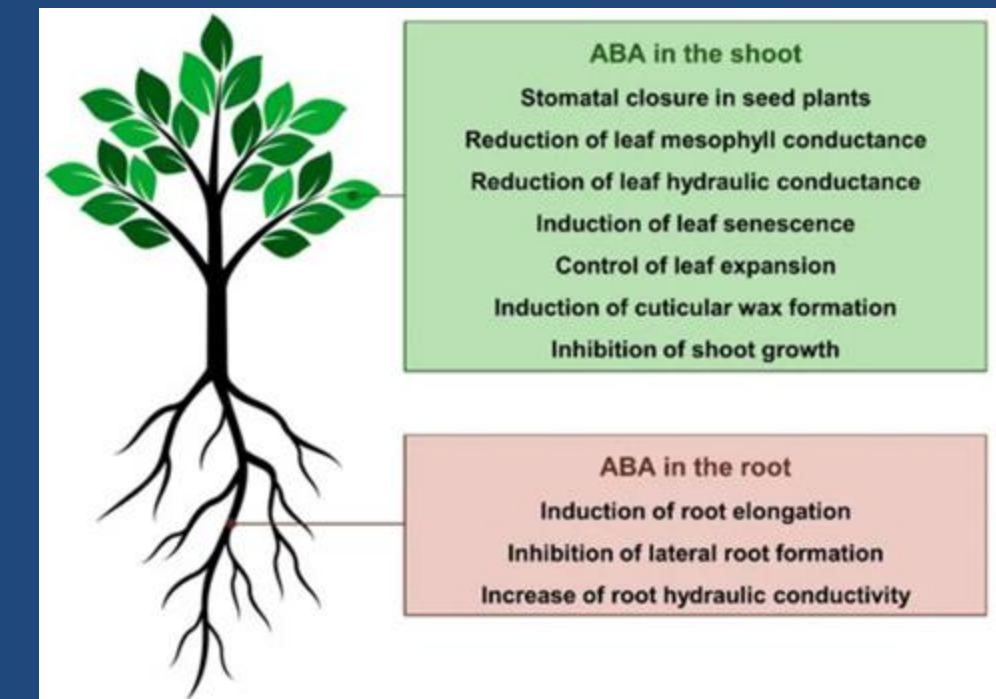
TOPICS COVERED:



Human Nervous System

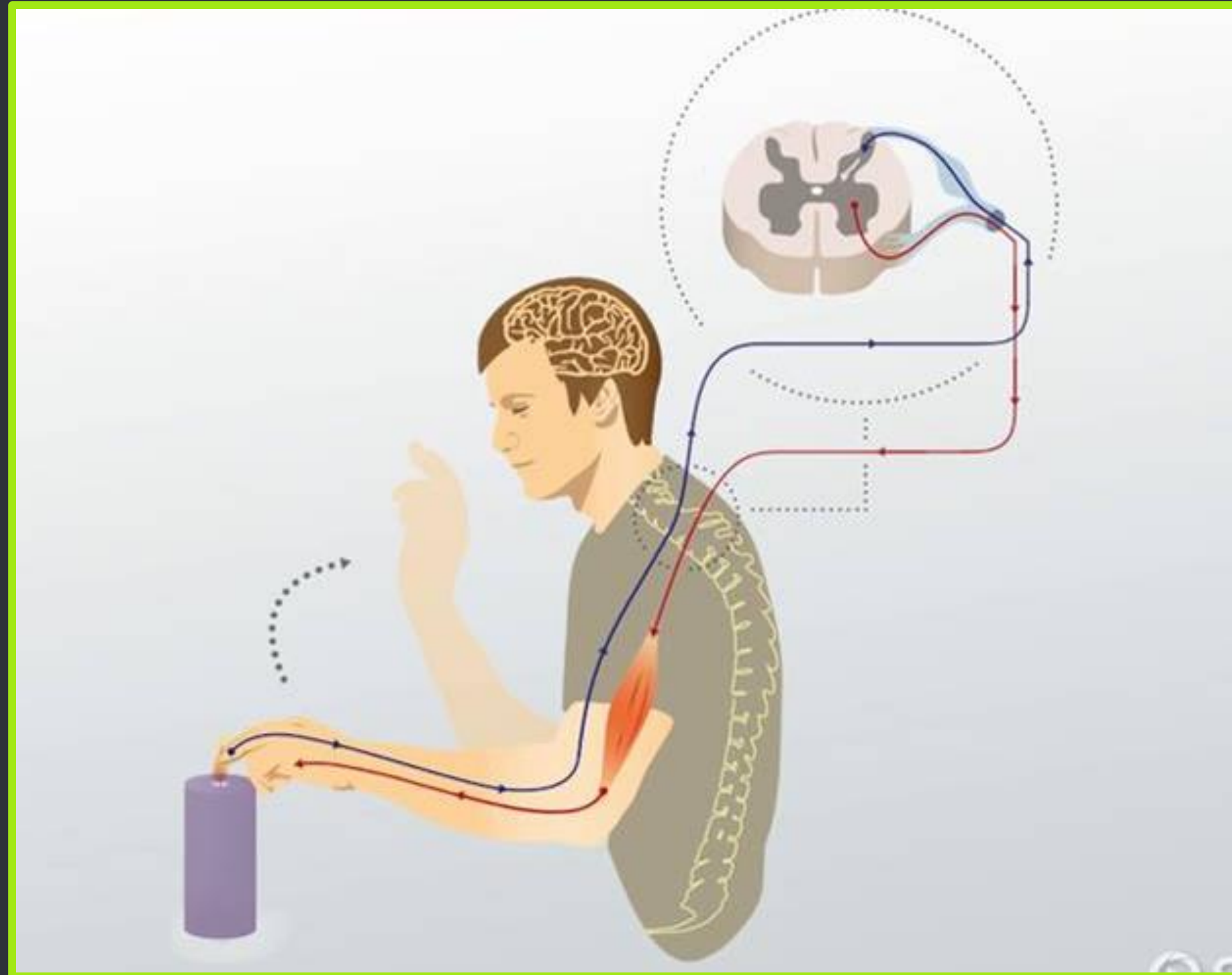


Endocrine System



Plants : Movements and Hormones

CONTROL AND COORDINATION



CONTROL AND COORDINATION





CONTROL AND COORDINATION



NERVOUS SYSTEM
(Neurons)

ENDOCRINE SYSTEM
(Hormones)



Human Nervous System

Central Nervous
System (CNS)

Peripheral Nervous
System (PNS)





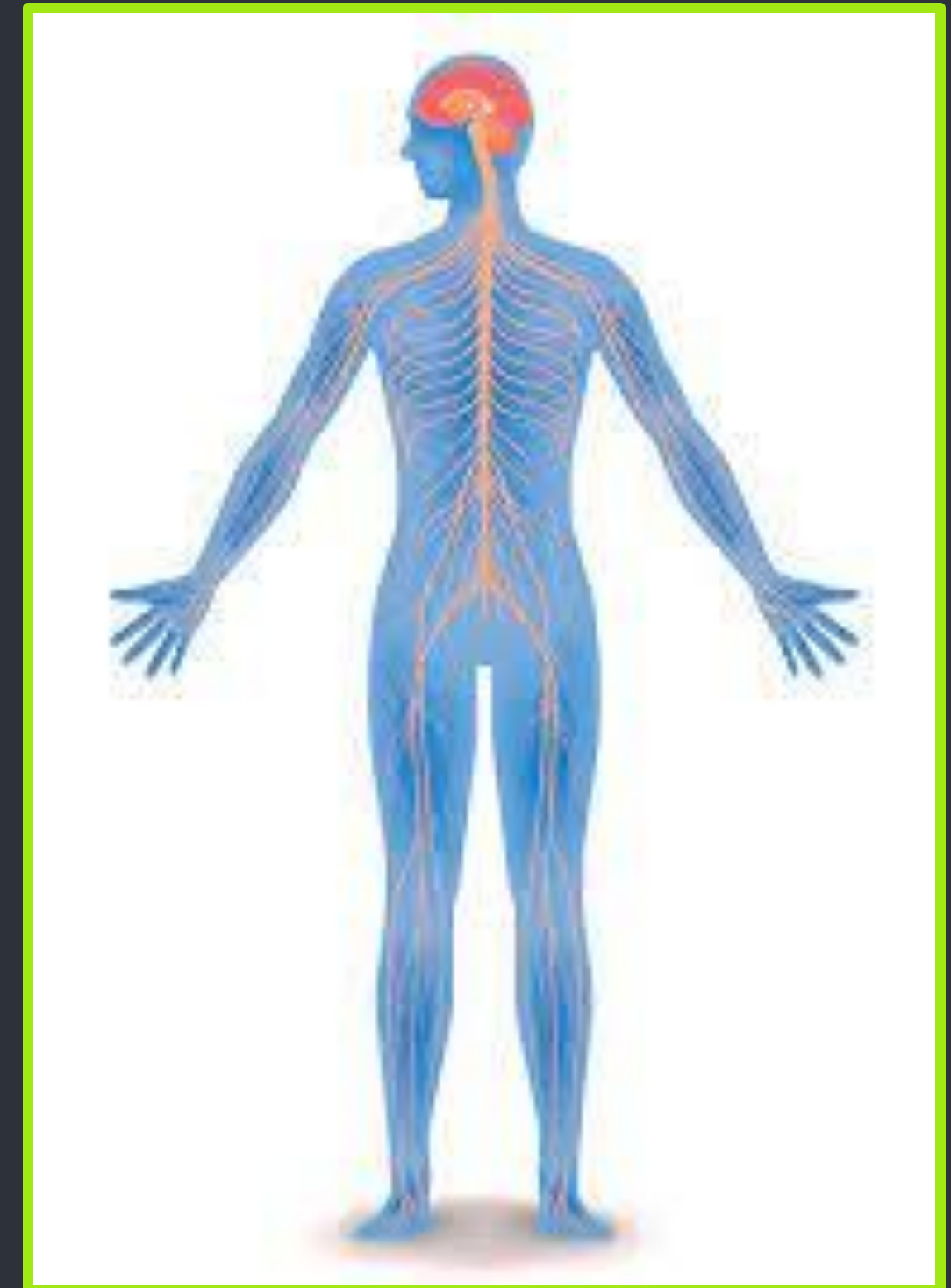
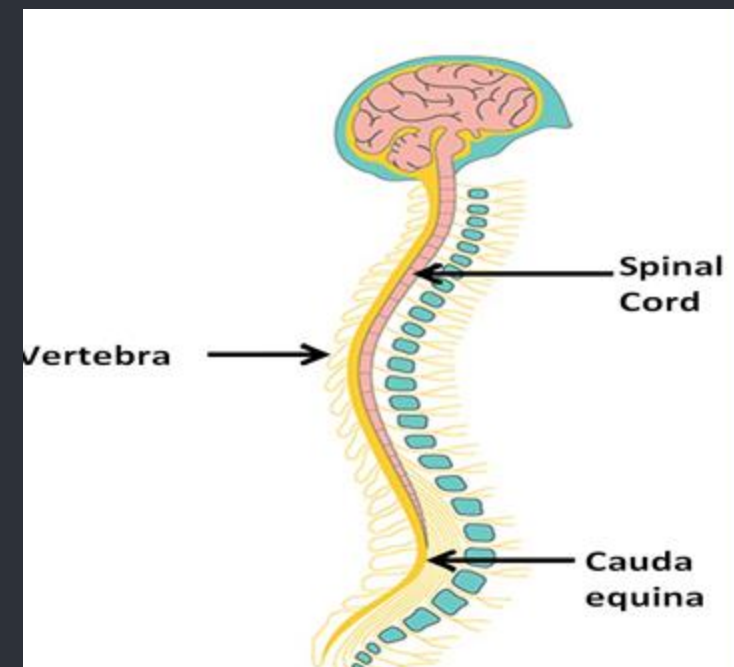
Human Nervous System

Central Nervous System (CNS)

Peripheral Nervous System (PNS)

Brain

Spinal Cord



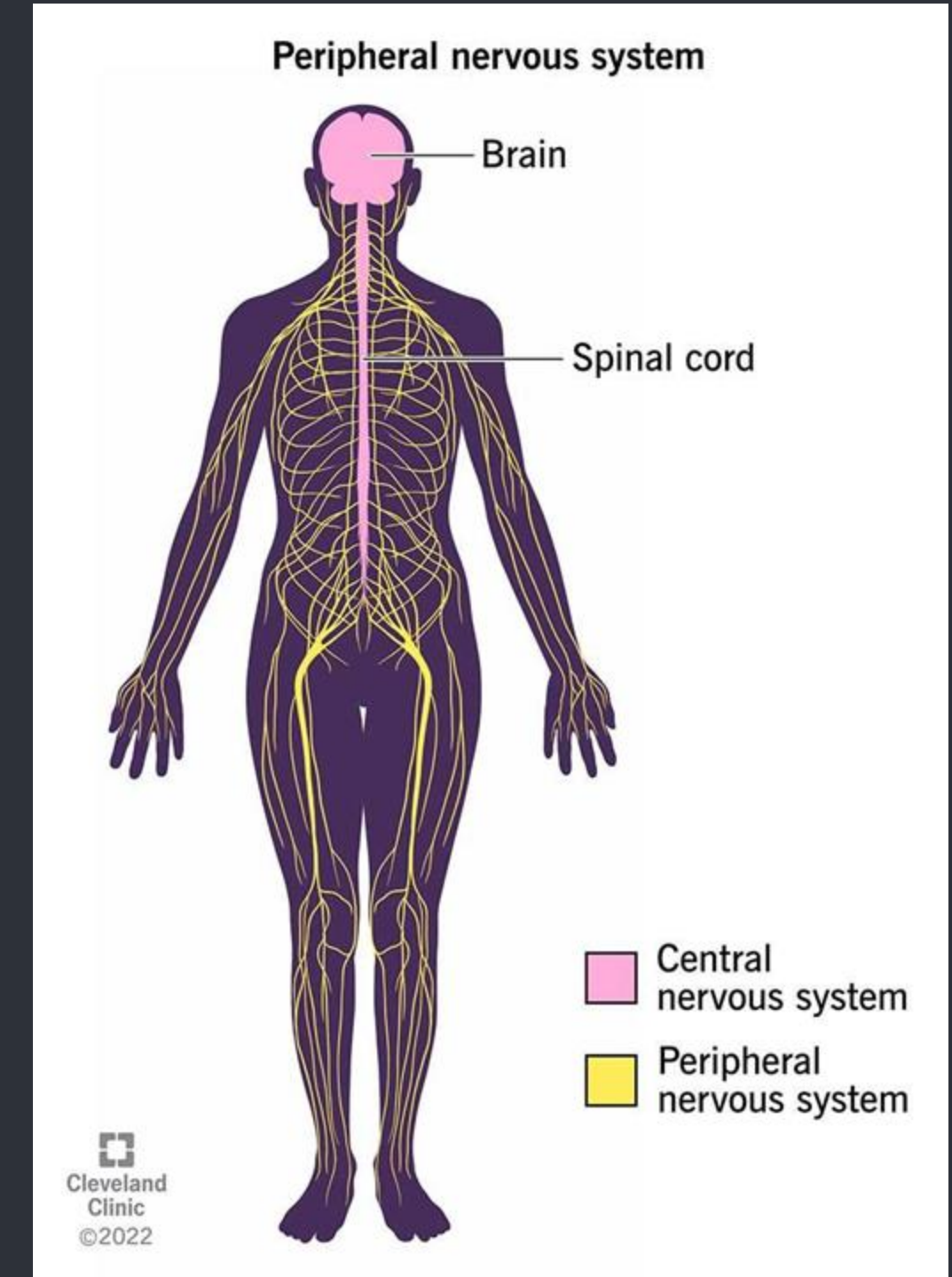
Peripheral Nervous System



Reflex Action:

It is quick, sudden and immediate response of the body to a stimulus.

Example - Knee jerk, withdrawal of hand on touching an object.





Peripheral Nervous System

Three types of Nerves:

- **Spinal nerves** arise from the spinal cord along most to the length of the spinal cord and spread throughout the body (except the head).
- **Cranial nerves** arise from the brain and spread throughout the head. They carry both sensory and motor neurons.
- **The visceral nerves** arise from the spinal cord. They are connected to the internal organs of the body. They carry both sensory and motor neurons.



STIMULI

The factors which bring response or change in the environment.

Example : Light, Heat, Cold, Sound, Smell, Touch etc.





Receptor and Effector

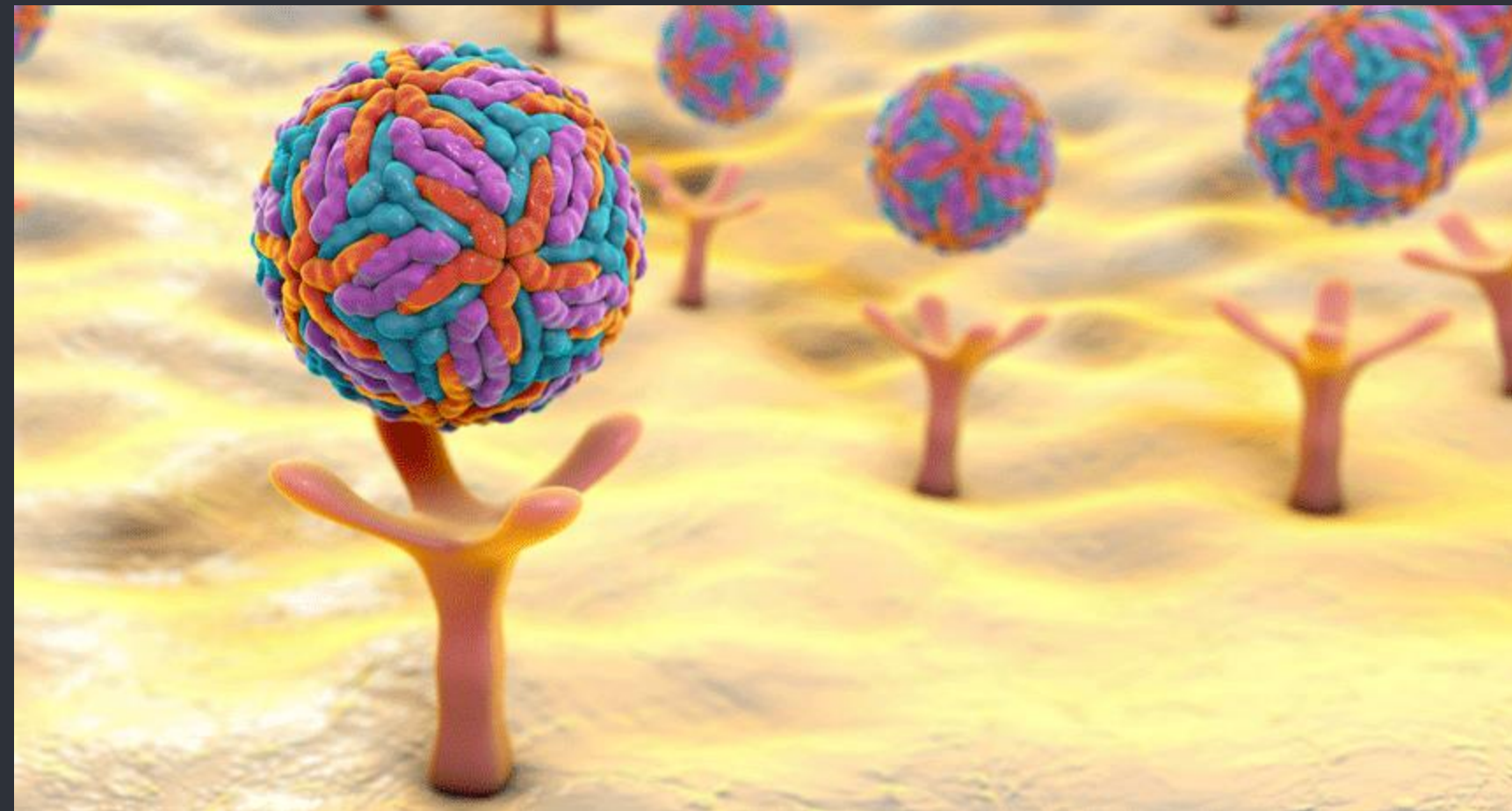
- A receptor is a cell (or a group of cells) in a sense organ which is sensitive to a particular type of stimulus. Example: Nose and ears.
- An effector is a part of the body which can respond to a stimulus according to the instructions sent from the nervous system (spinal cord and brain). Example: Glands and muscles.



Receptors



These are special tips of some nerve cells that detect information from the environment. These are located in our sense organs.





RECEPTORS

1. Ear

- It acts as phono receptors
- It helps in hearing and maintaining the balance of the body.





RECEPTORS

2. Eyes

- It acts as photoreceptors
- It helps in seeing.





RECEPTORS

3. Nose

- It acts as olfactory receptors. It helps in detection of the smell.





RECEPTORS

4. Tongue

- It acts as Gustatory receptors. It helps in detection of taste.





RECEPTORS

5. Skin

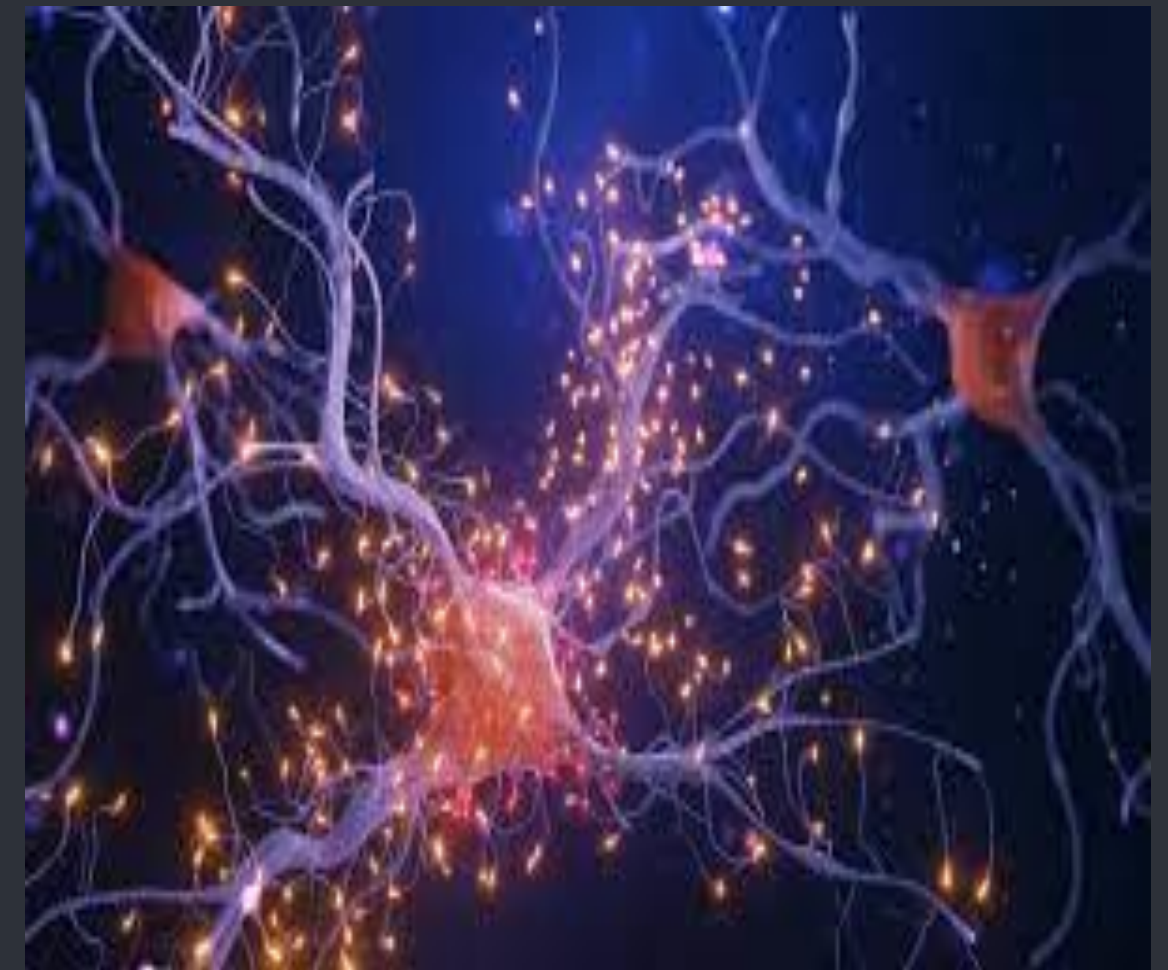
It acts as thermoreceptors. It helps in feeling heat or cold and touch.





Neuron

- **Structural and functional unit of Nervous System.**
- **Neuron is a highly specialized cell.**
- **Responsible for the transmission of signals to and from the different parts of the body.**
- **Structural and functional unit of nervous system.**
- **Longest cell in human body.**





Neuron

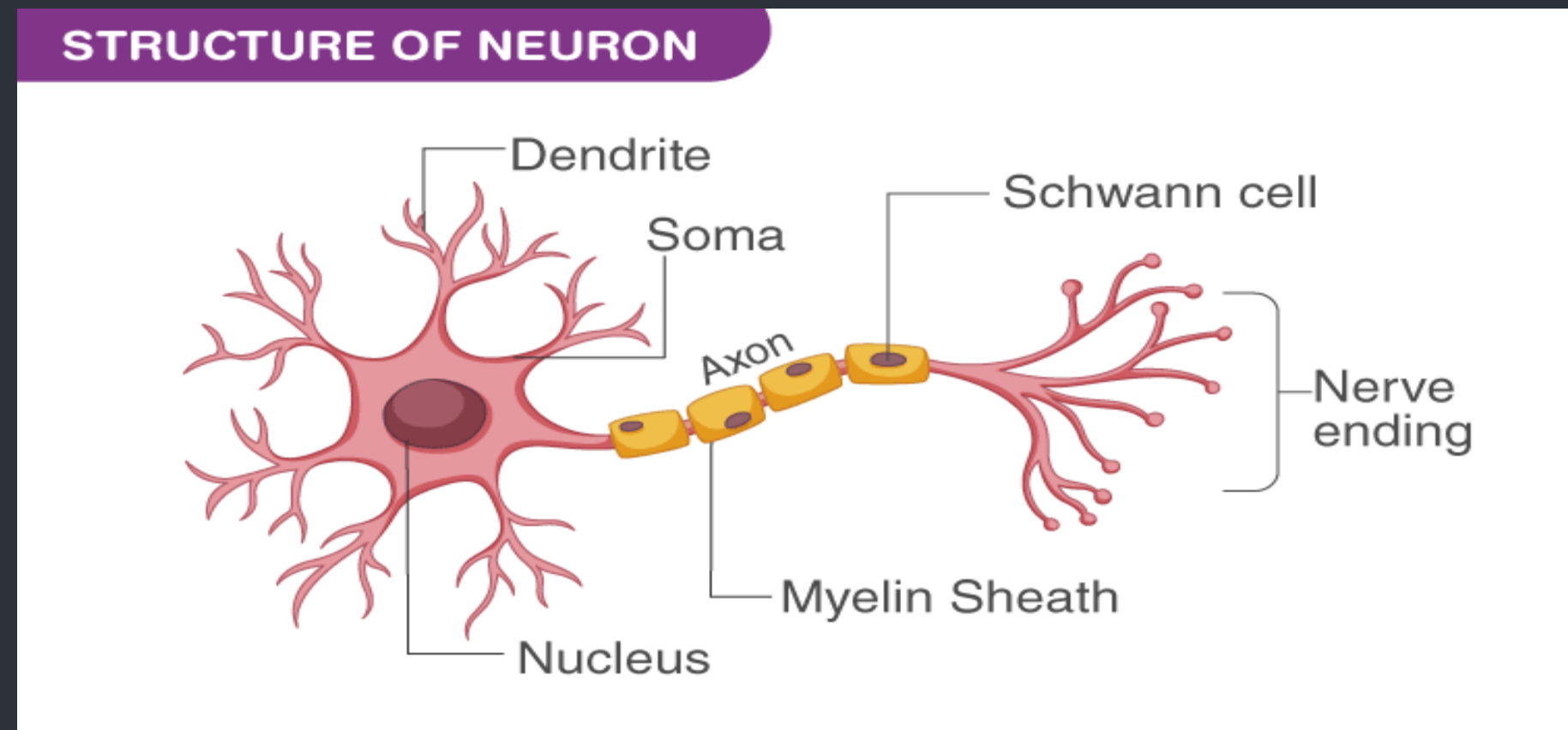
Parts of Neuron:

1. Dendrite :

It acquires information and receive impulse from other neurons.

2. Cell Body

The information acquired by it travels as an electric impulse.



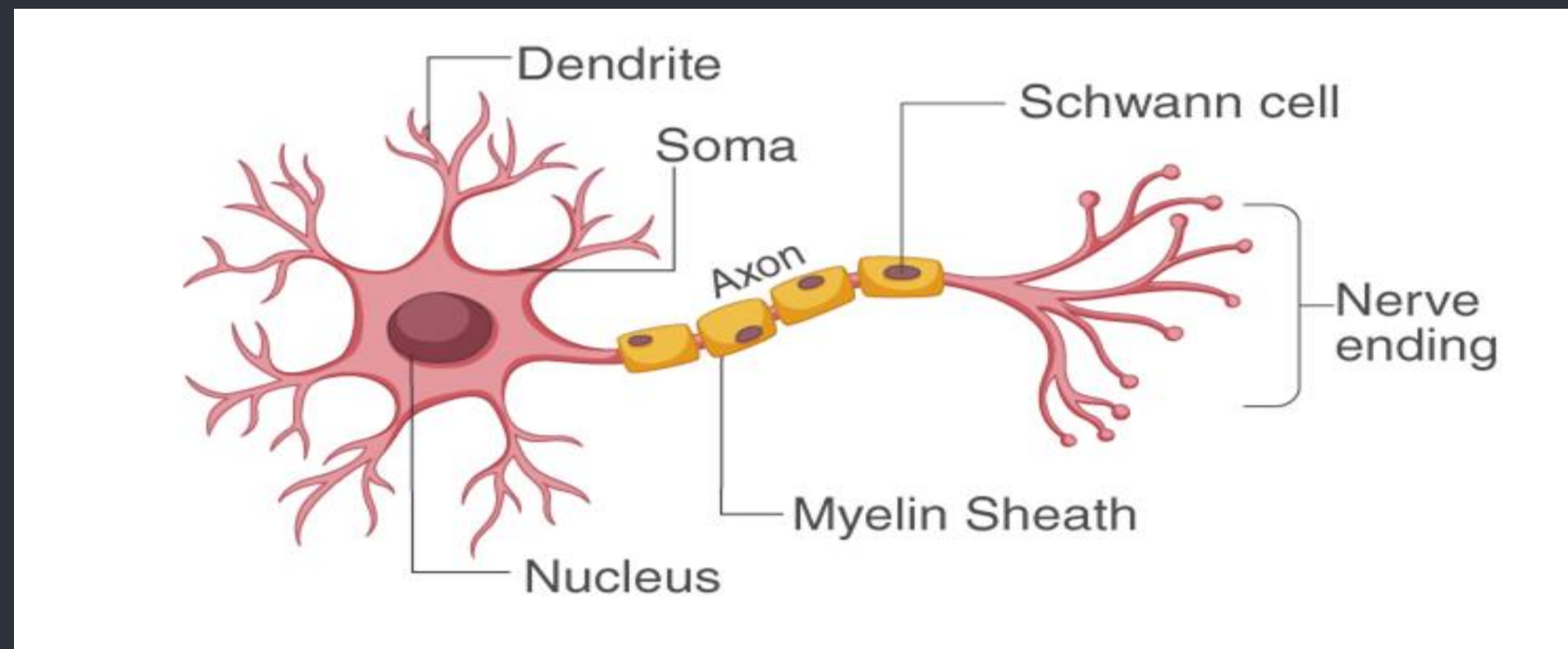


Neuron

Parts of Neuron:

3. Axon:

- Longest fiber on the cell body is called axon.
- It transmits electrical impulse from cell body to dendrite of next neuron.
- It may be myelinated or non - myelinated Impulse transmission is faster in myelinated neurons.

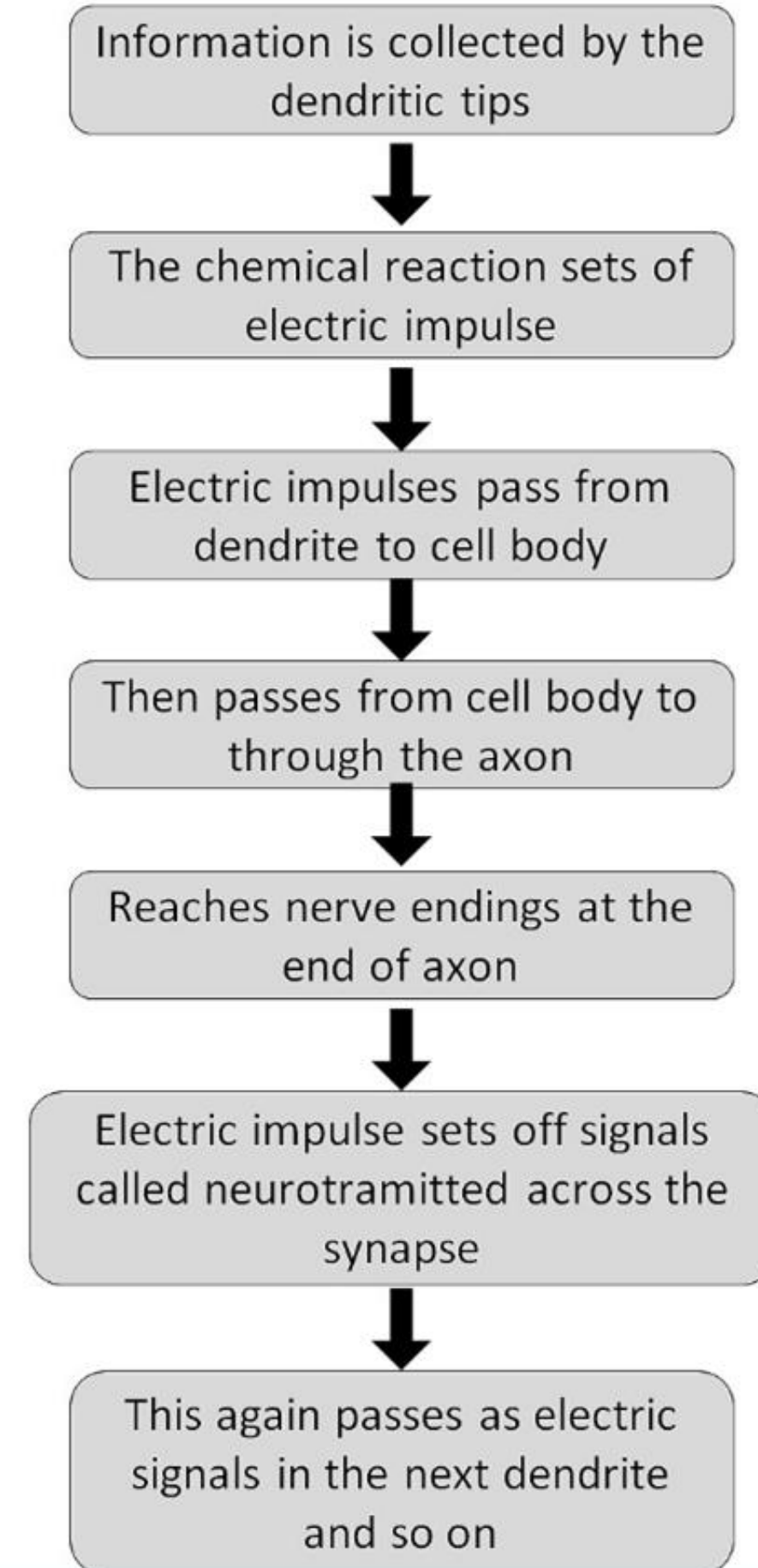




Functions of Neuron

- The information or signals from receptors is acquired at the end of the dendritic tip of a nerve cell as chemical reaction that creates an electric impulse.
- This impulse travels from the dendrite to the cell body and then at the end of the axon.
- There is a gap between two neurons called (Synapse). These chemicals cross the gap and transfers the same signal to next neuron.
- Chemicals are released at the end of the the axon by the effect of electrical impulse.

How exactly do these signals pass through and between neurons?





Nerve Impulse

- Nerve impulses are wave of electrical and chemical signals carried along nerves or neurons.
- Nerve impulses are initiated at receptor cells as a result of stimuli from the environment.



Types of Neuron

**Sensory neurons
(afferent neuron)**

They help in transmitting nerve impulse from receptor to central nervous system.

**Motor neurons
(efferent neurons)**

They help in transmitting nerve impulse from central nervous system to a muscle or gland..

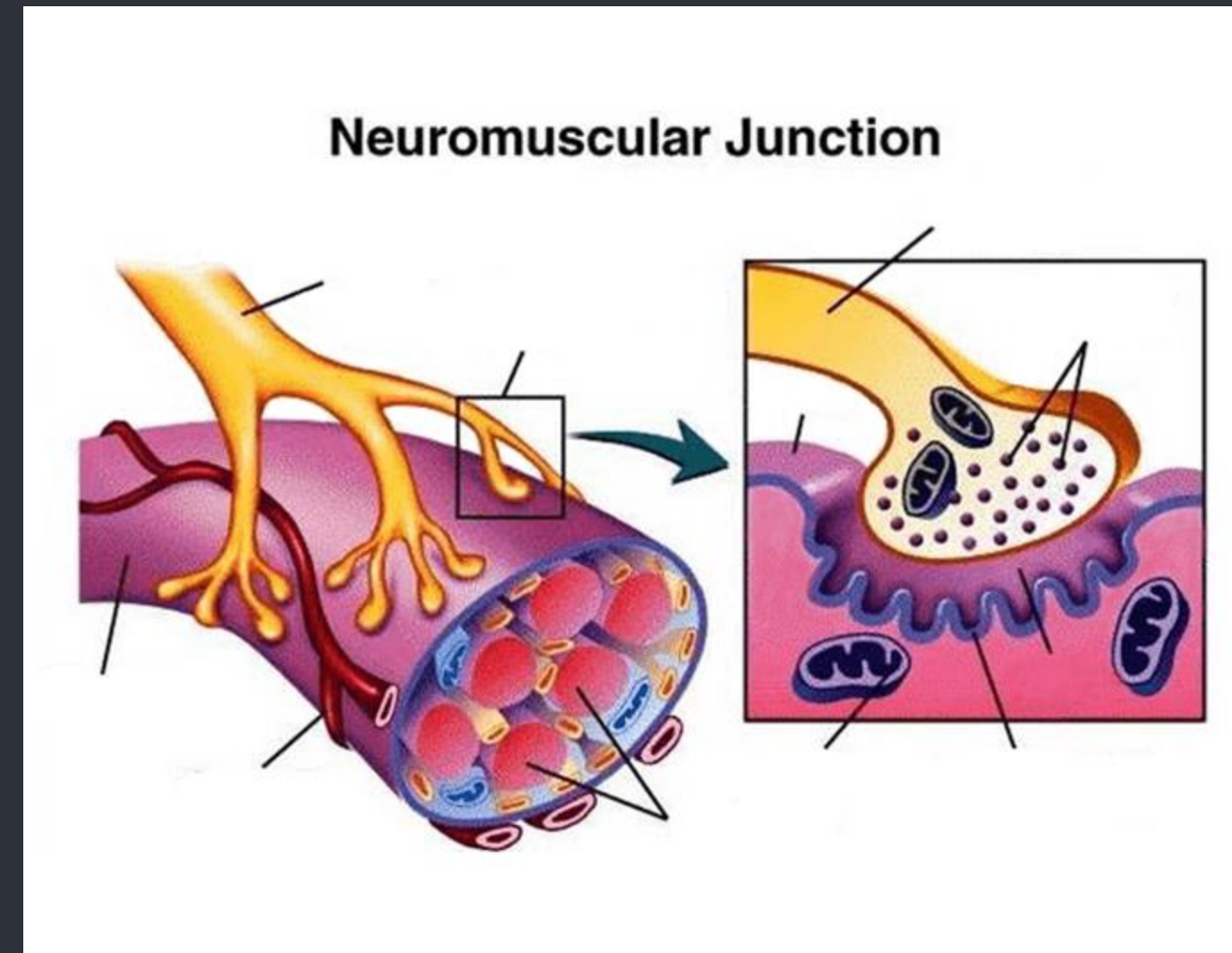
**Interneurons
(relay or associated neuron)**

They connect sensory and motor neuron.



Neuromuscular Junction

- The neuromuscular junction (NMJ) is a synaptic connection between the terminal end of a motor nerve and a muscle.





Voluntary actions

- These actions can be controlled by our own will.
 - Thinking involved.
 - Brain involved.

eg. Walking
Writing
Dancing
Jumping

Involuntary actions

- These actions cannot be controlled by our own will.
 - Thinking not involved.
 - Brain is involved.

eg. Pumping of Blood
Peristaltic movements
Contraction relaxation of blood vessels

Reflex actions

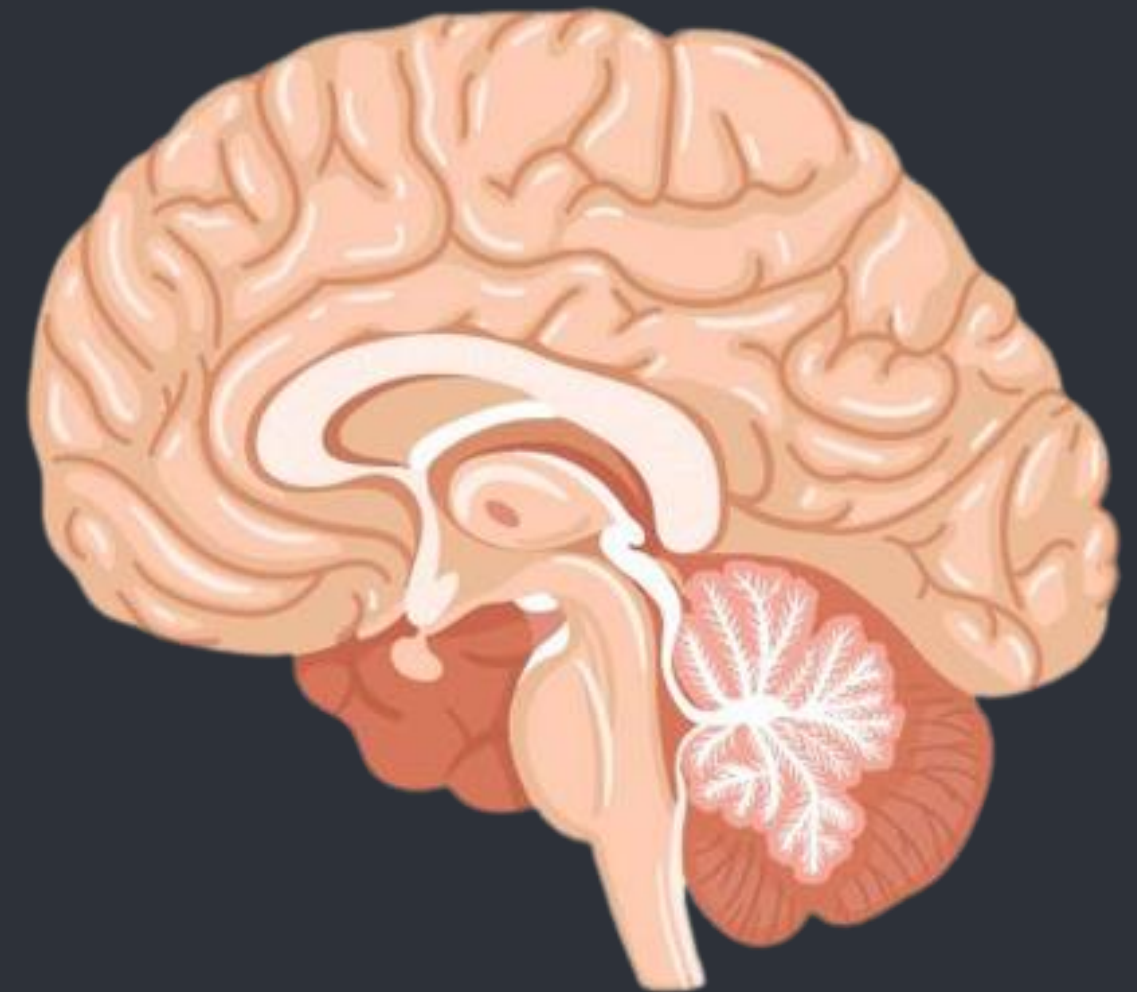
- These actions can not be controlled by our own will.
- Thinking is not involved.
- Spinal Cord is involved.

eg. Withdrawal of hand when touch an hot object.



Brain

- Highest coordinating centre in the body.
- Located inside the skull.
- Protected by a box called Cranium.
- The brain is surrounded by 3 membranes called meninges.
- The space between the membranes is filled with Cerebra Spinal fluid.
- The fluid protects the brain from mechanical shocks.



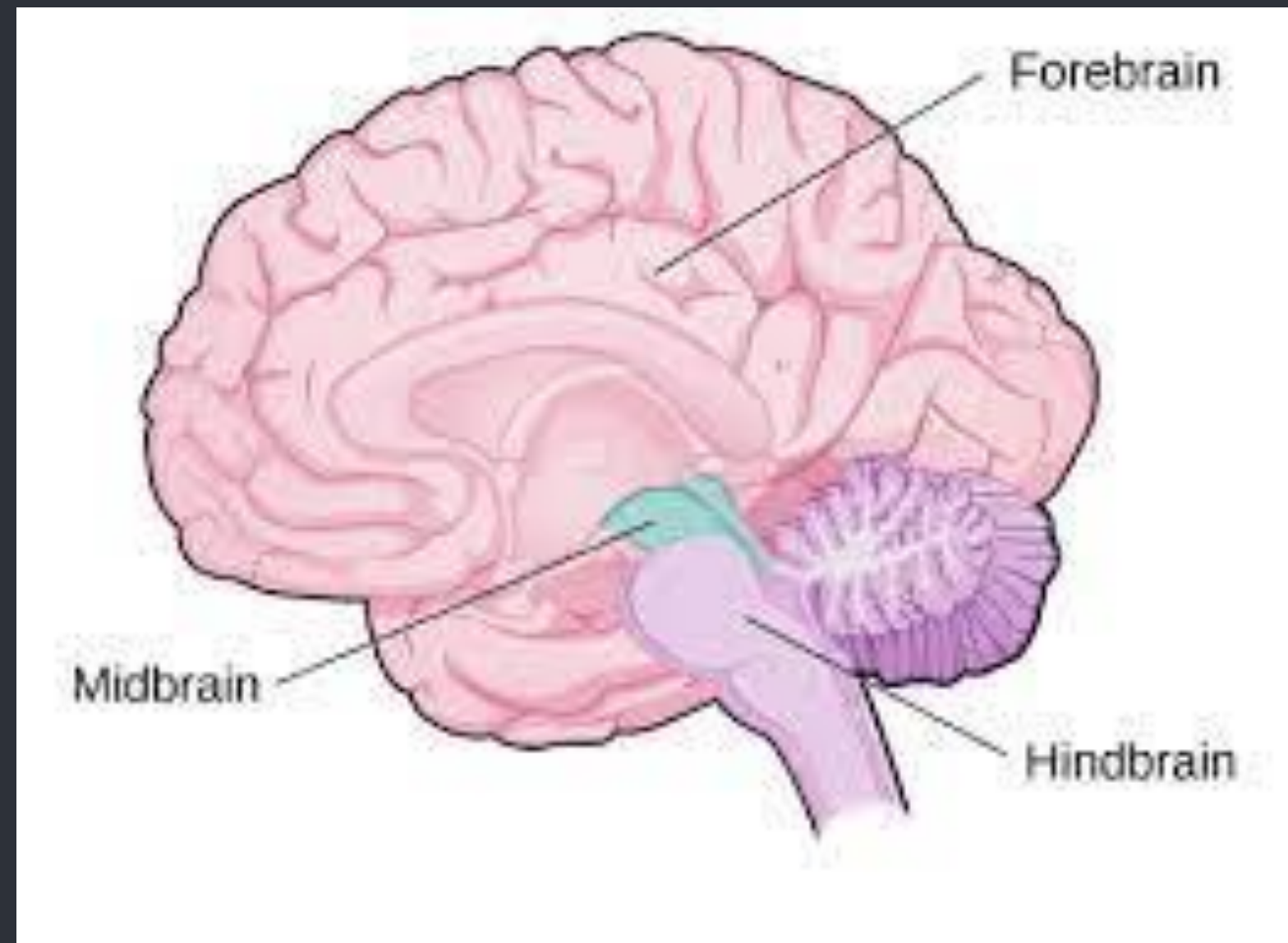


Three regions of Brain

1. Fore Brain

2. Mid Brain

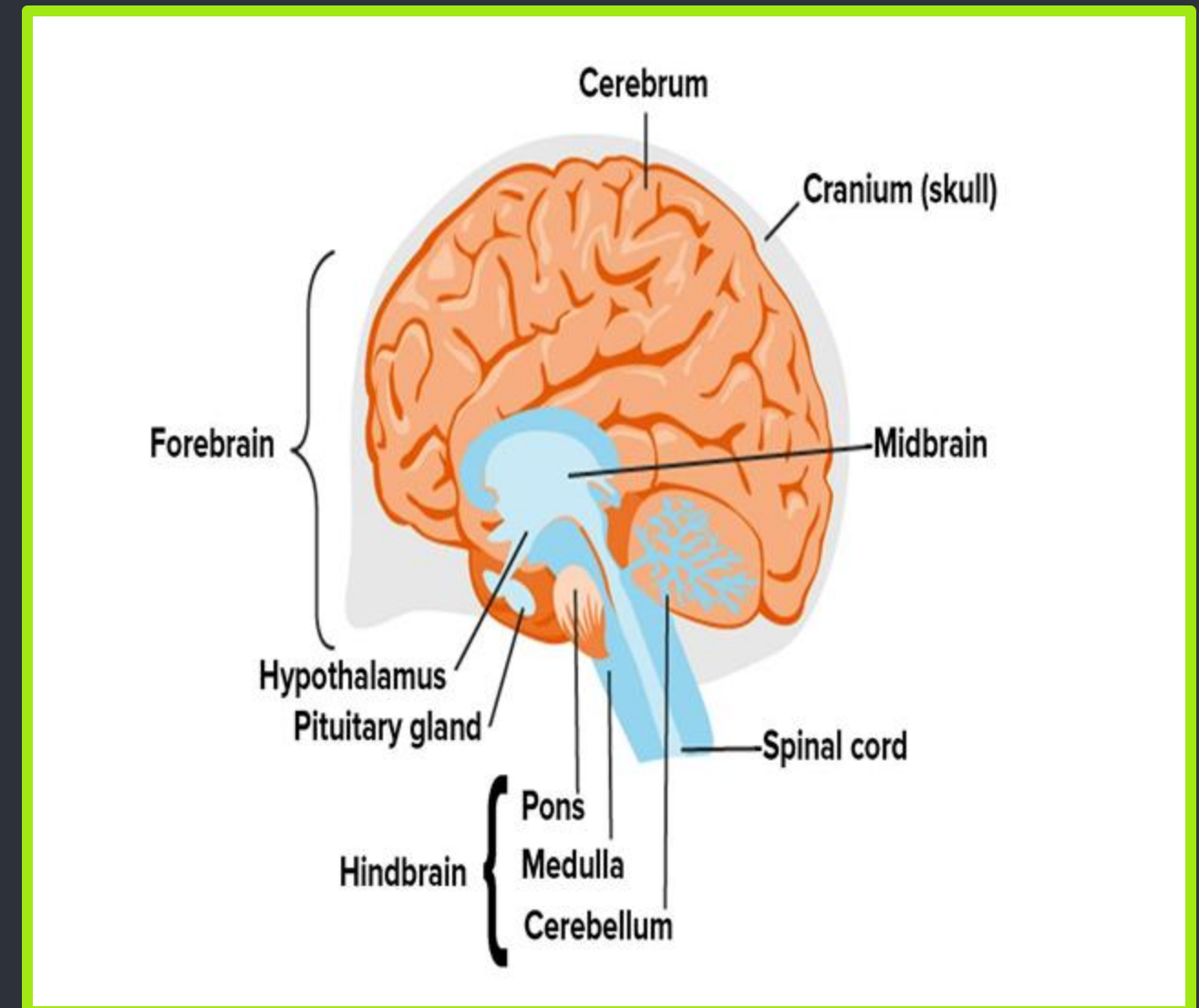
3. Hind Brain





Fore Brain

- It mainly consists of Cerebrum.
- Cerebrum is the main thinking part of the brain.
- Site of learning, reasoning, intelligence, personality and memory.
- It also controls thoughts, sensations, actions and movements.
- Information from sense organs like (eyes, ears, nose, tongue or skin) is received in cerebrum.
- It also has motor areas from which instructions are sent to muscles to do voluntary actions.





Fore Brain

Forebrain consists of :

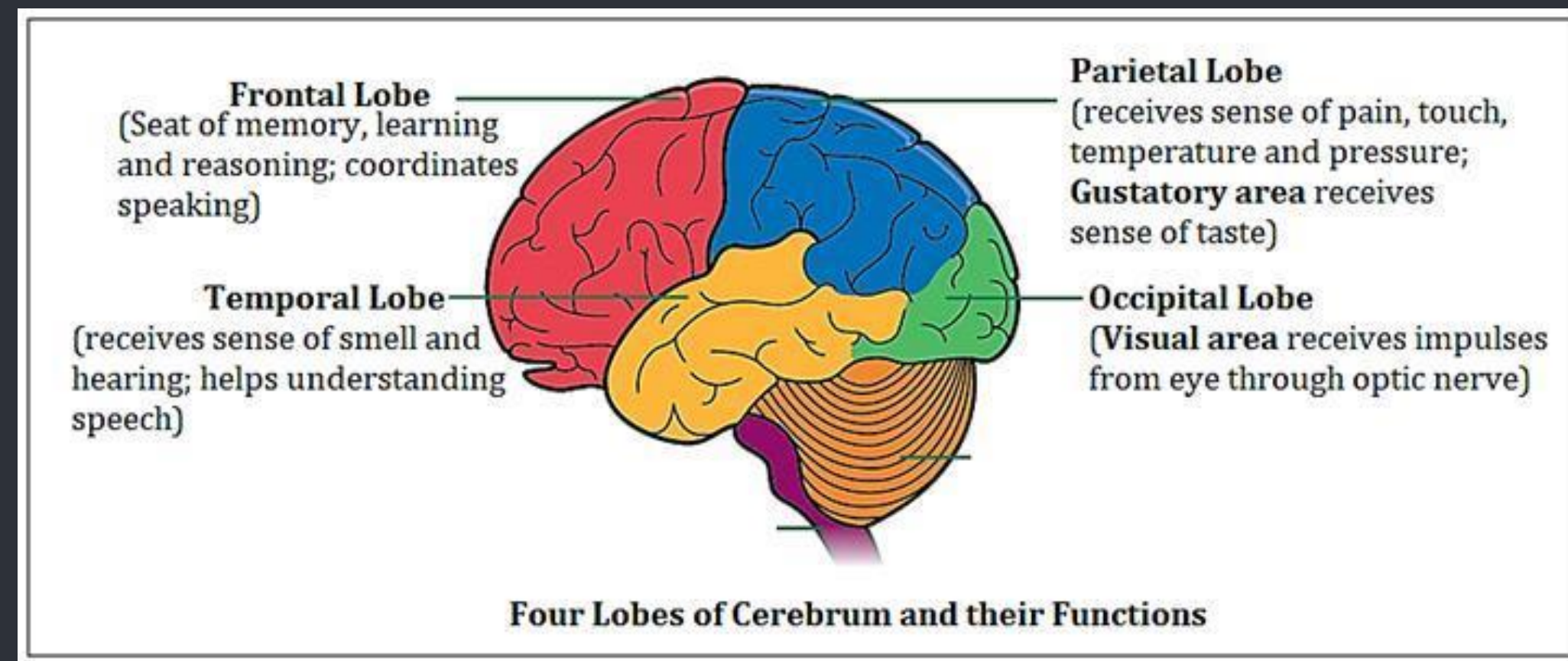
1. Cerebrum (largest part)
 2. Thalamus
 3. Hypothalamus
- (Diencephalon)



Fore Brain

Cerebrum

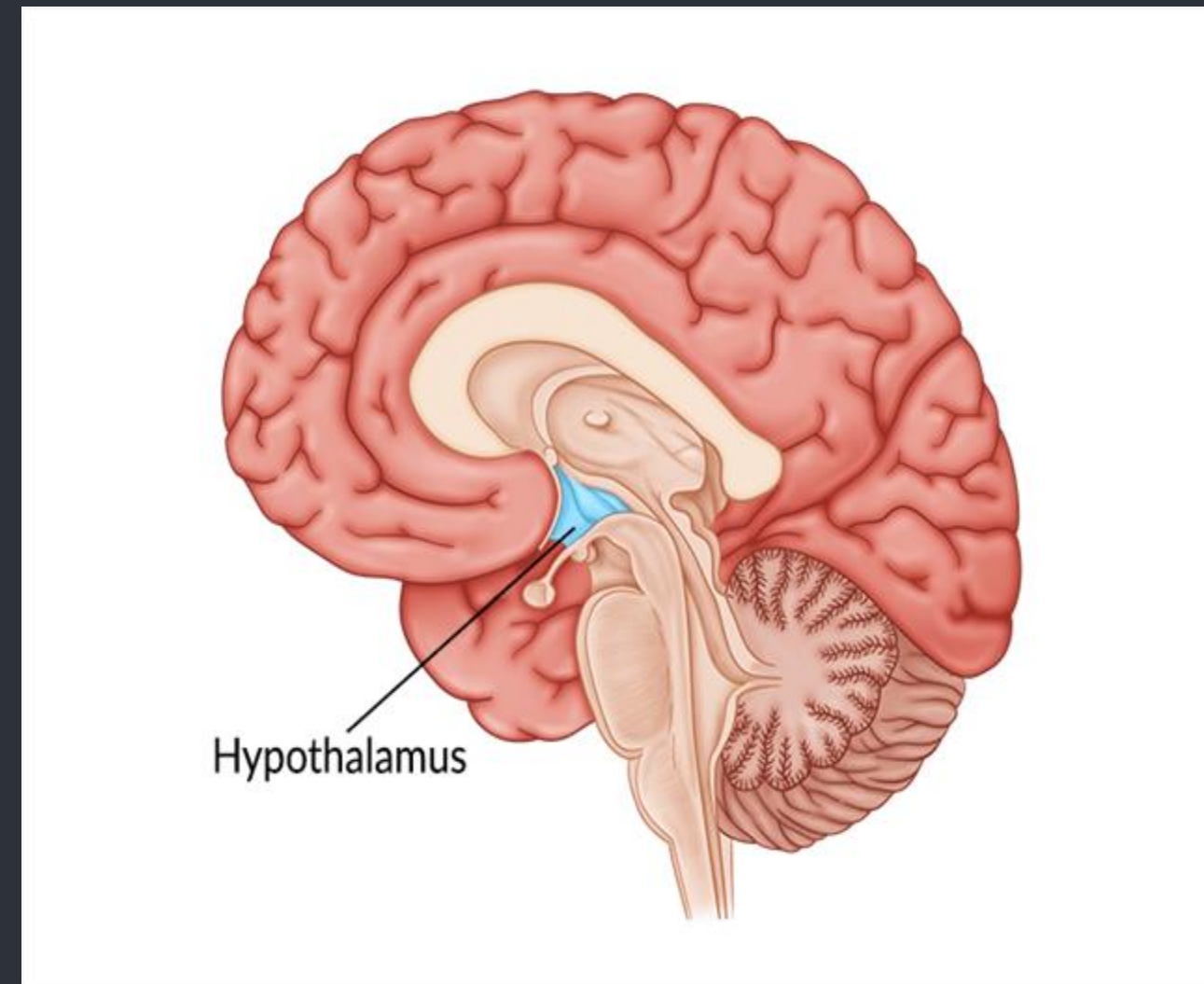
- Seat of mental abilities, controls thinking, memory, reasoning, perception, emotions, and speech.
- Interprets sensations and responds to pain, cold, heat, and pressure.





Mid Brain

It is composed of the hypothalamus.

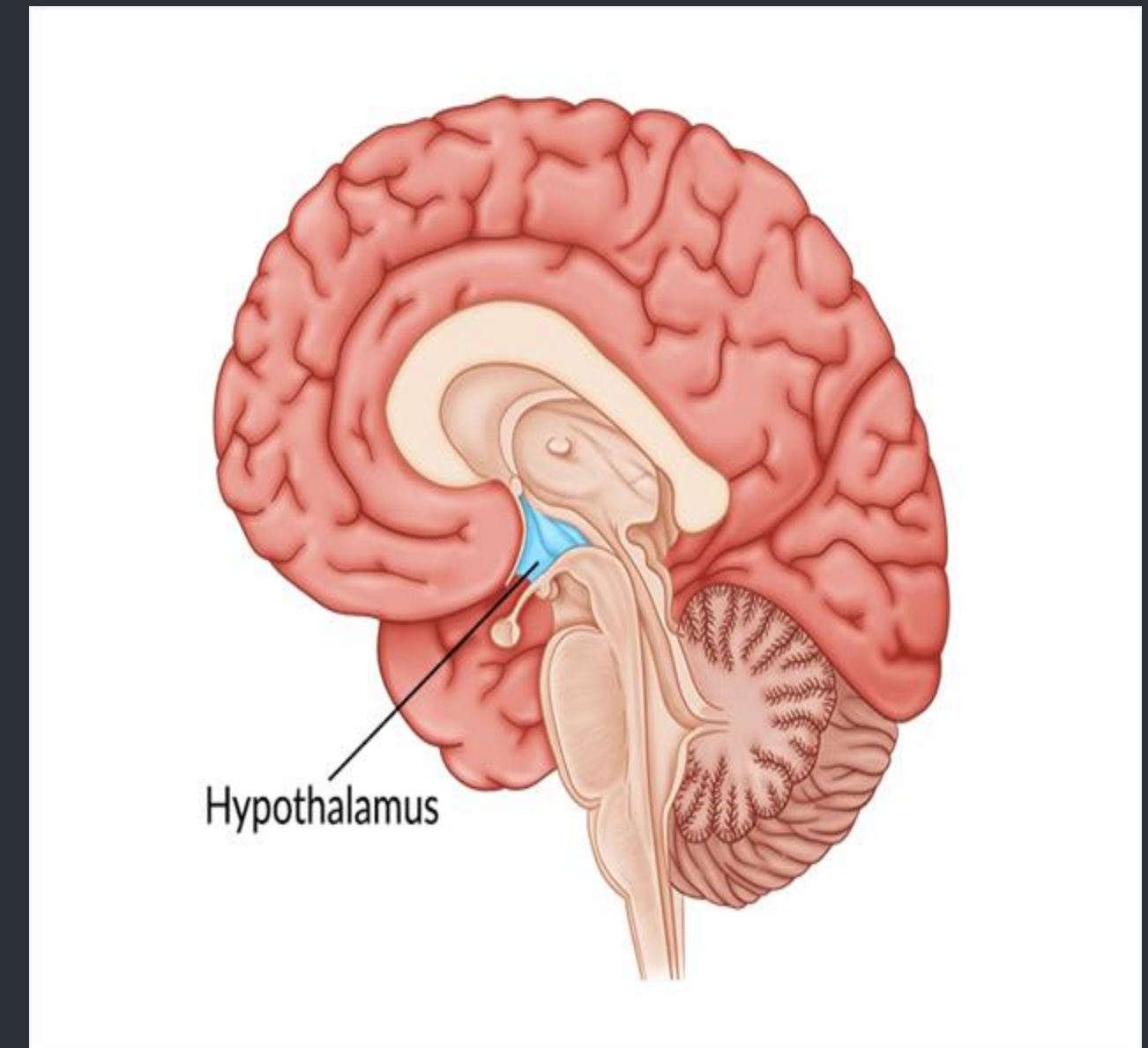




Hypothalamus

- It lies at the base of the cerebrum.
- It controls sleep and wake cycle of the body.
- It also controls the urges for eating and drinking.

h

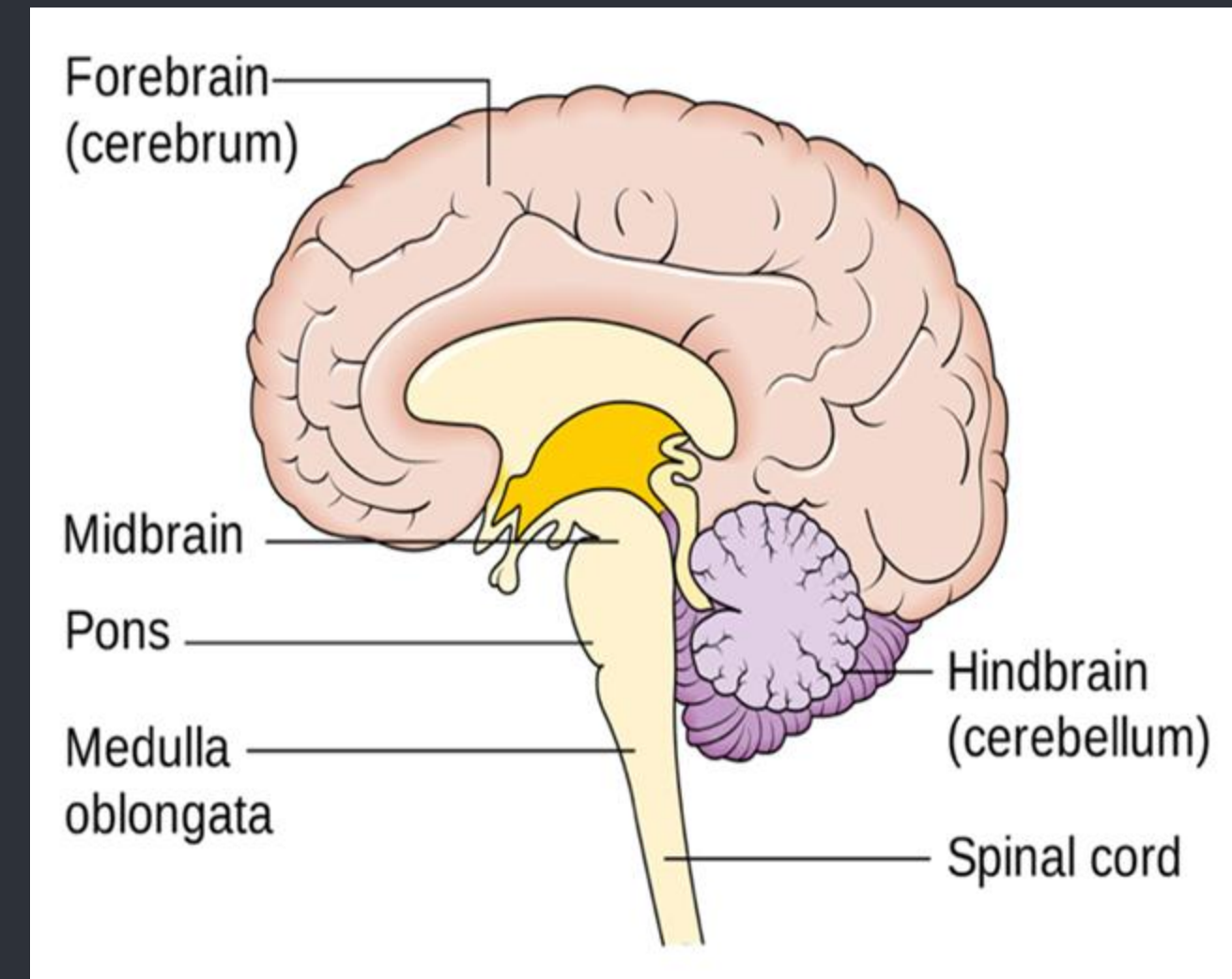




Hind Brain

It is composed of three parts:

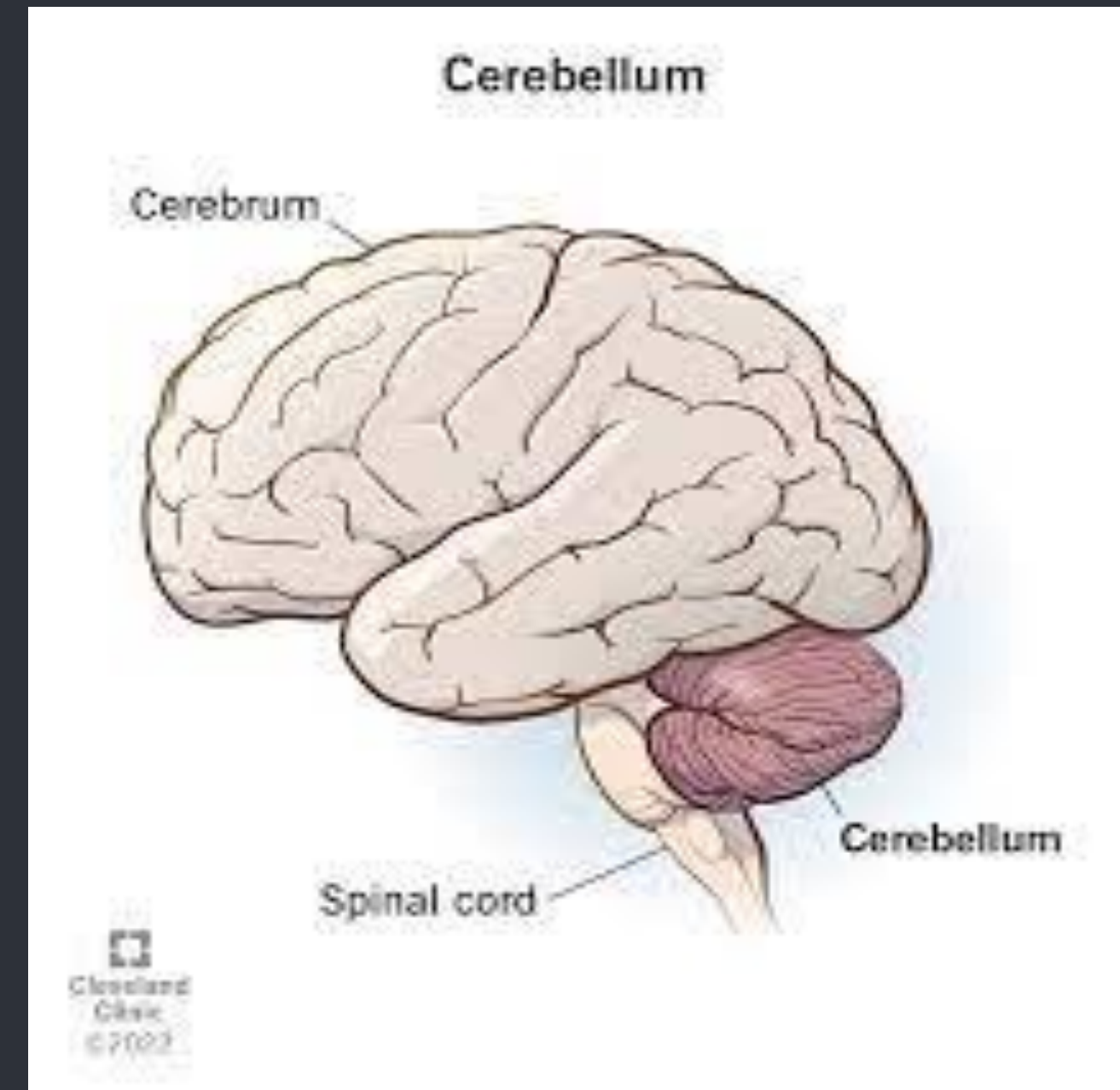
- Cerebellum
- Medulla
- Pons





Cerebellum

- It lies below cerebrum.
- It coordinates the motor functions.
- It controls posture and balance.
- It controls voluntary activities.





Medulla

- It forms the brain stem.
- It lies at the base of the brain and continues into the spinal cord.
- It controls involuntary functions like hearing, heart beating and Respiration, salivation, vomiting.





Pons

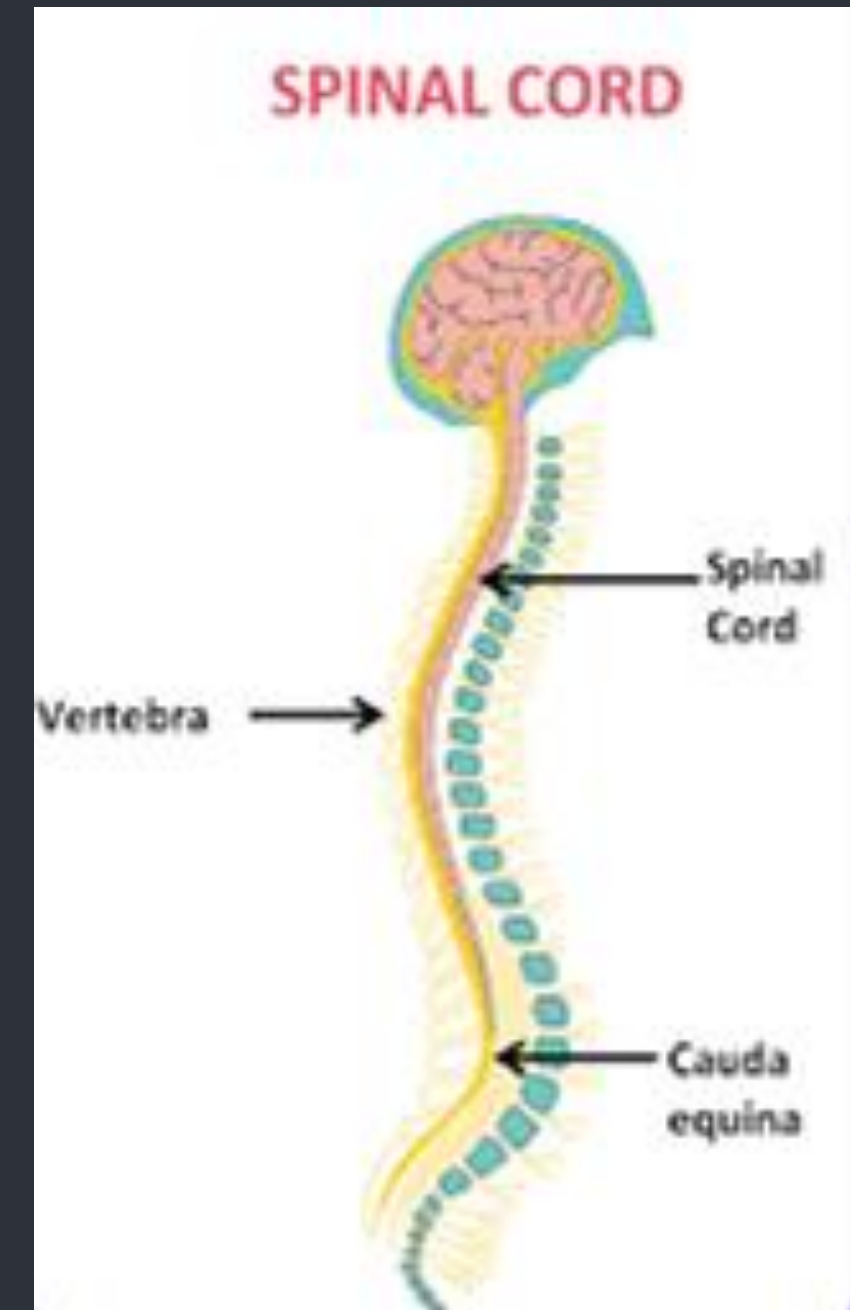
- It also controls involuntary actions.
- It regulates respiration.





Spinal Cord

- A cylindrical structure
- Begins in continuation with medulla
- It is enclosed in a bony case called vertebral column.
- It is surrounded by membranes called meninges
- 31 pairs of nerves arise from the Spinal Cord.
- It is concerned with spinal reflexes.
- It helps in conduction of nerve impulses to and from the brain.





Important Questions



Q. How are nervous organs protected?

1. Protection of Brain:

- The brain is placed in a bony box called cranium and is a part of the skull.
- Brain is wrapped in three separate membranes called meninges.
- The space present between these layers is filled with cerebro-spinal fluid(CSF).
- This fluid-filled balloon acts as a shock absorber/ spring/ cushion and protects the brain from injuries and shocks.



2. Protection of Spinal Cord :

- Like brain, spinal cord is also wrapped in spinal meninges and consist of Cerebrospinal fluid.
- The spinal cord is protected by the vertebral column or backbone.
- The vertebral column is formed by 33 individual bones called vertebrae.



1. How does process of reflex action take place in human body?

The process of reflex action is like

- The stimuli is sensed by the sensory receptors and sensory neurons generate impulses.
- These impulses reach the spinal cord
- The relay neuron in the spinal cord passes signals between neurons.
- The motor neurons pass the signals for response to the organ that needs to respond to stimulus.

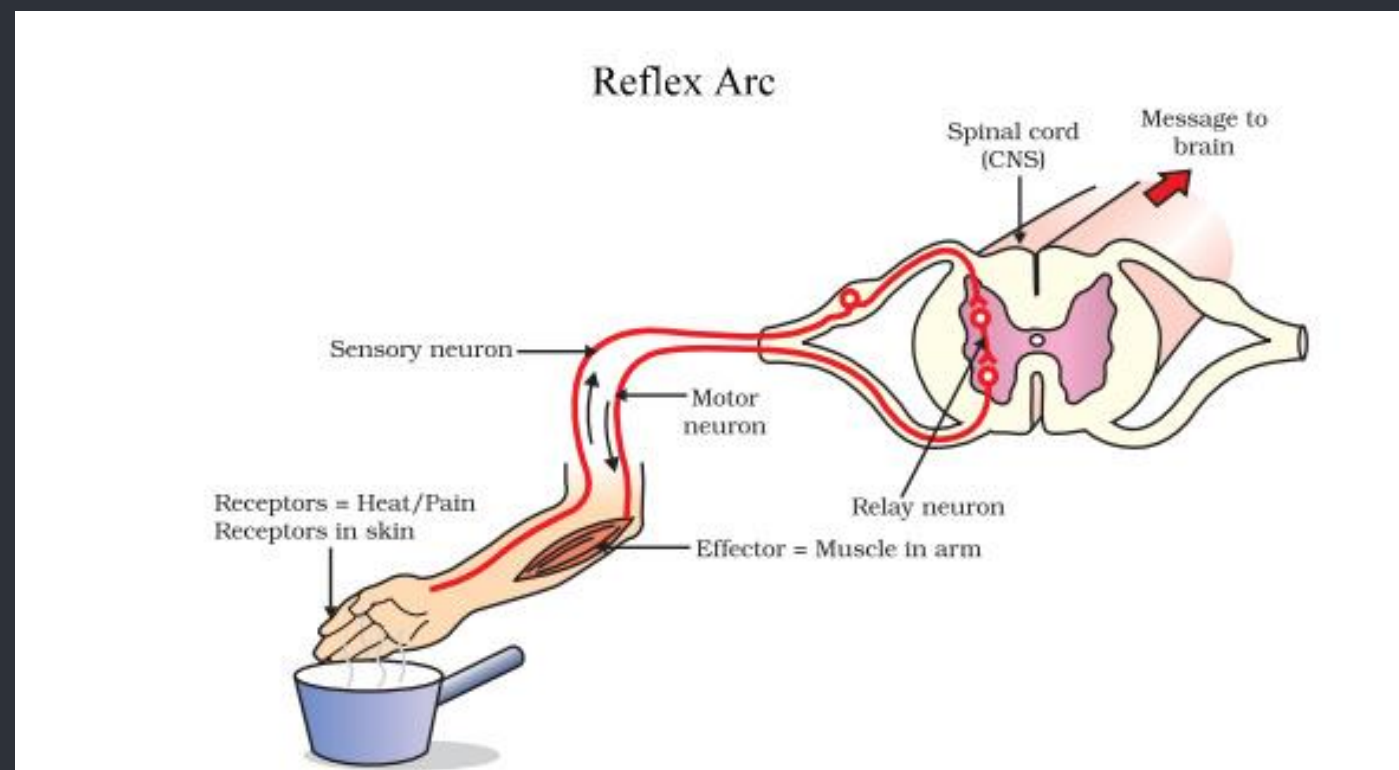
The information input also goes on to reach the brain. But it is not involved in reflex action.



Reflex Arc

The pathway through which nerve impulses pass during reflex action is called Reflex Arc.

Heat (Stimulus) → Receptors (Skin) → Spinal organs → Effector Organ (Muscles) → Response (Hand withdrawal)



Response : It is final reaction after the reflex action.



2. What is the role of the brain in reflex action?

There is no direct involvement of brain in reflex actions.

When Reflex actions are generated in spinal cord the information also reaches brain. This helps the brain to record this event and remember it for future use.

Brain helps the person to get awareness of the stimulus and prevent himself from that situation again in the future.

THREE TYPES OF RESPONSES



1. Voluntary

Controlled by forebrain. Example – talking, writing etc



THREE TYPES OF RESPONSES



2. Involuntary

Controlled by mid & hind brain. Example – heart beat, vomiting, respiration



THREE TYPES OF RESPONSES

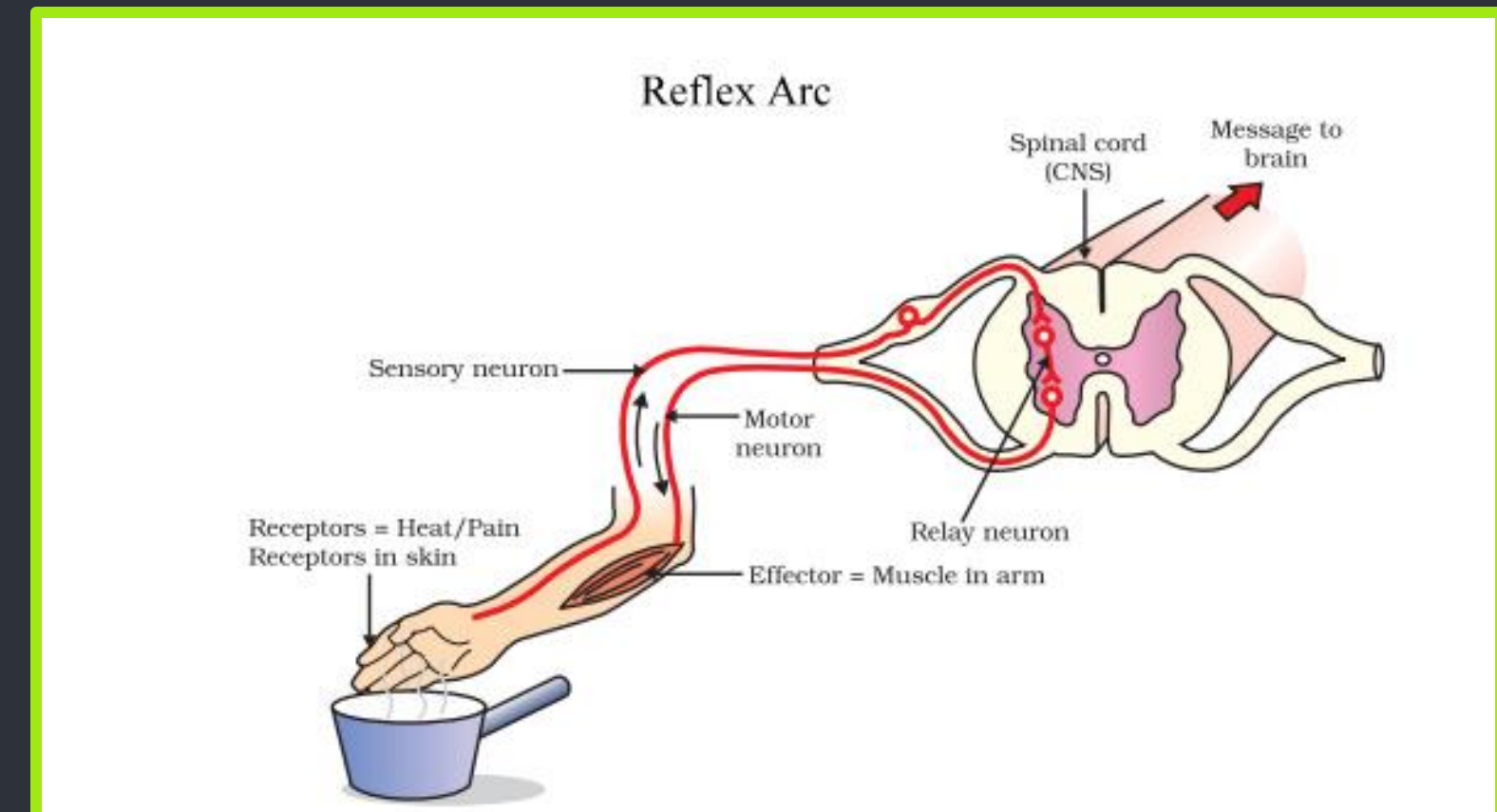


3. Reflex action

Controlled by spinal cord. Example – withdrawal of hand on touching hot object.

NEED FOR REFLEX ACTIONS

- In situations like touching a hot object, pinching etc. a sudden response is required.
- Reflex action is generated from SPINAL CORD instead of brain.
- So that time taken for action is reduced.



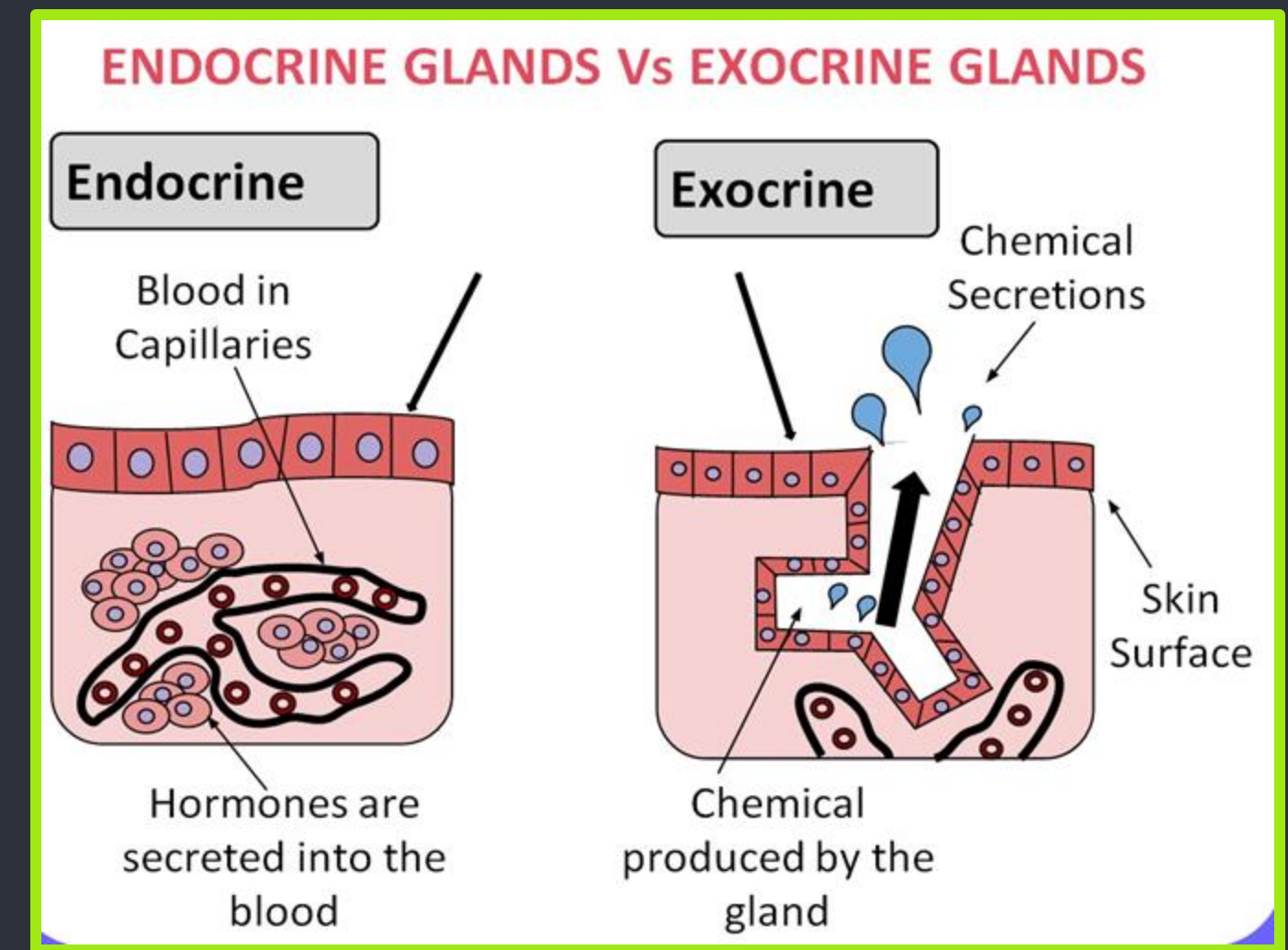
GLANDS



A gland is a structure which secretes a specific substance (or substances) in the body. A gland is made up of a group of cells or tissue.

There are two types of glands in the body:

- 1) exocrine glands
- 2) endocrine glands

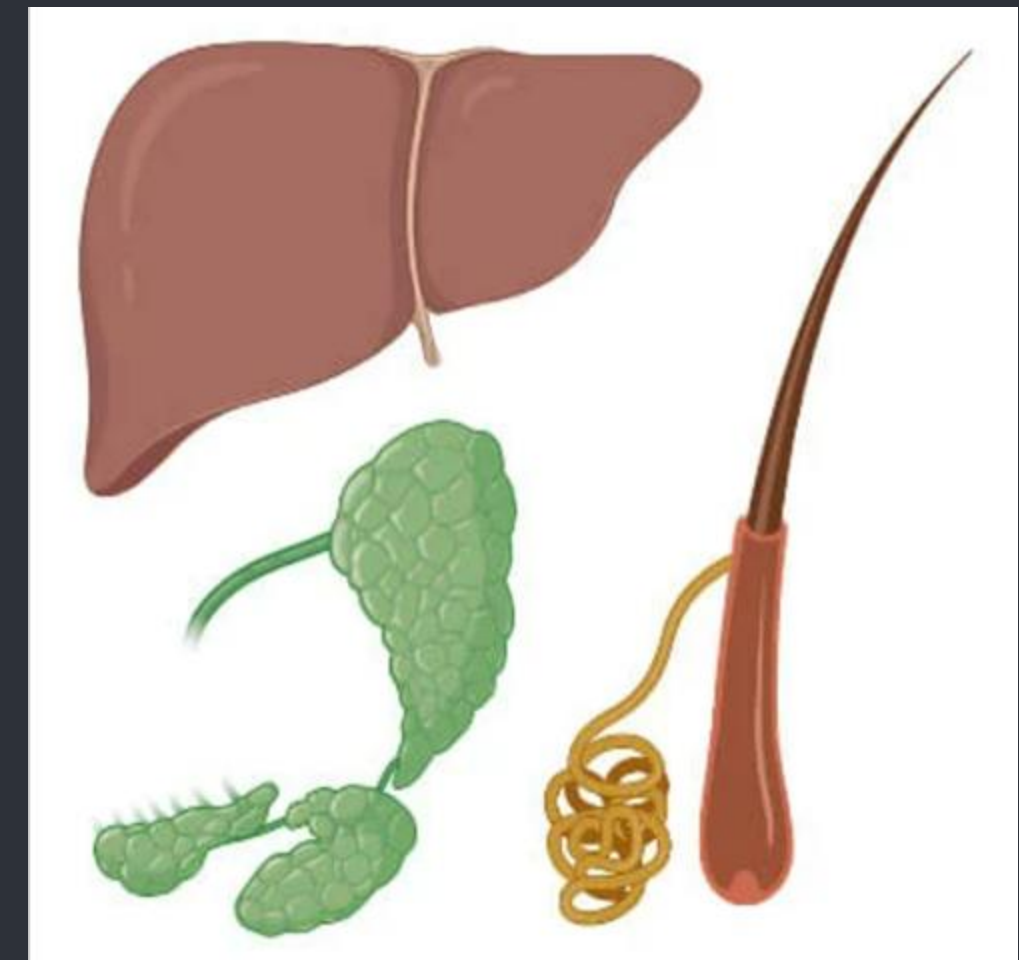




GLANDS

1. Exocrine Glands

- Secretes its product into a duct (or tube).
- For example, the salivary gland secretes the saliva into a duct called salivary duct.



GLANDS



2. Endocrine Glands

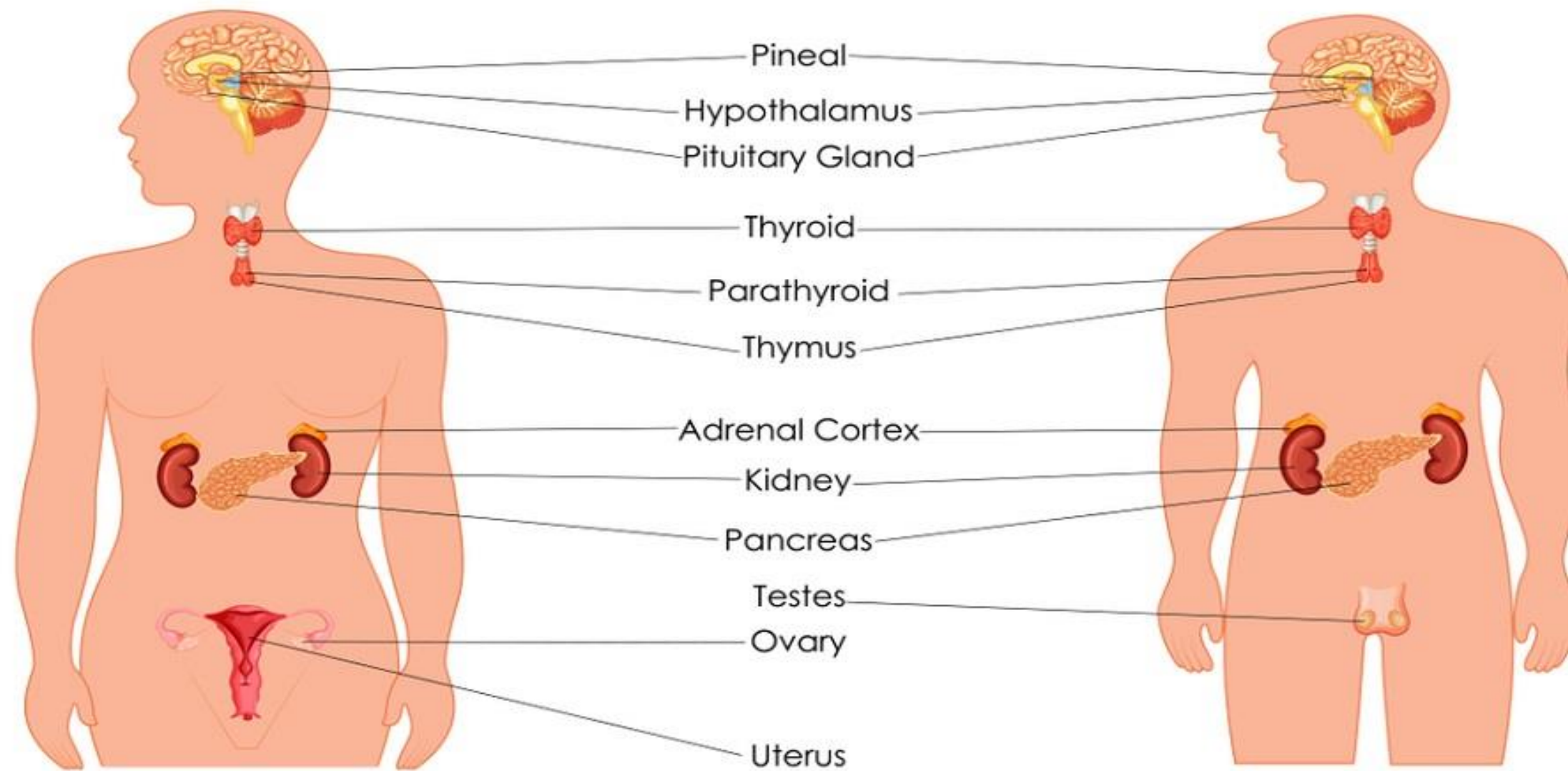
- A group of endocrine glands which produces various hormones are called endocrine glands.
- The endocrine glands present in the human body are :
Pineal gland; Hypothalamus gland; Pituitary gland;
Thyroid gland; Parathyroid glands; Thymus; Pancreas;
Adrenal glands; Testes (only in males) and Ovaries (only in females).



GLANDS



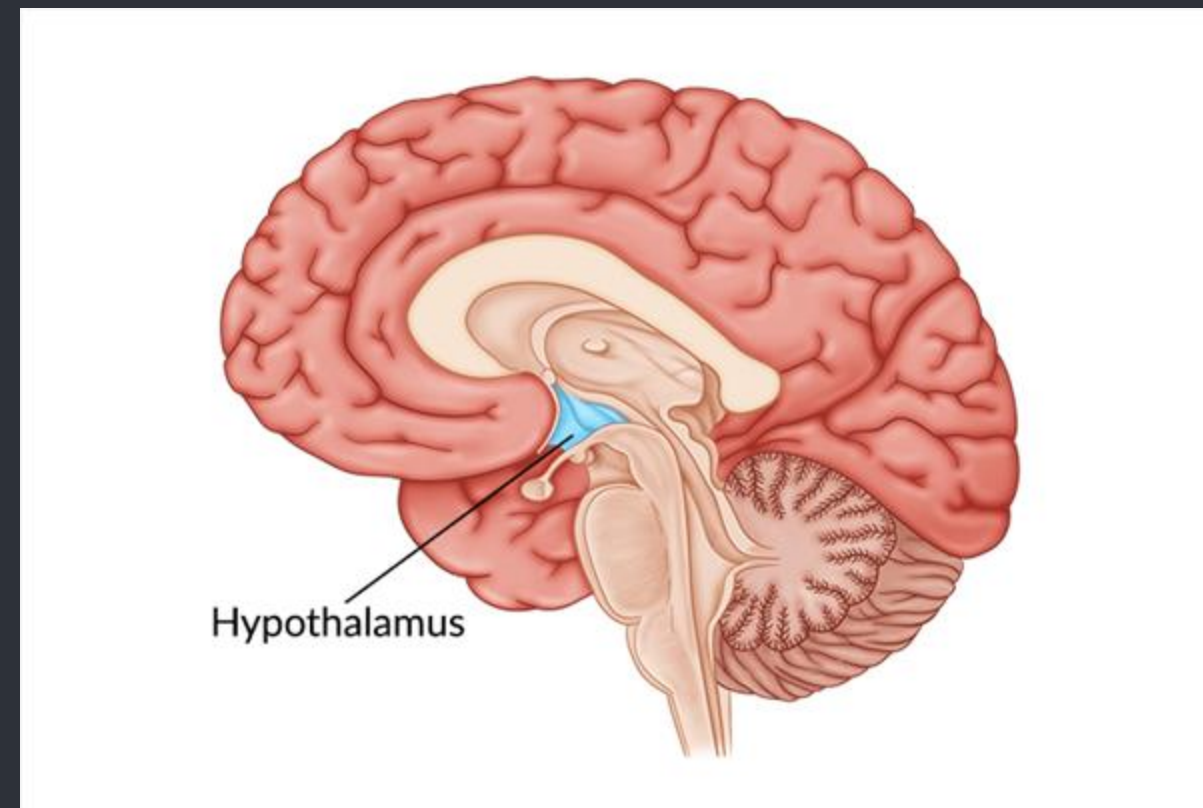
Endocrine System Male-Female





HYPOTHALAMUS

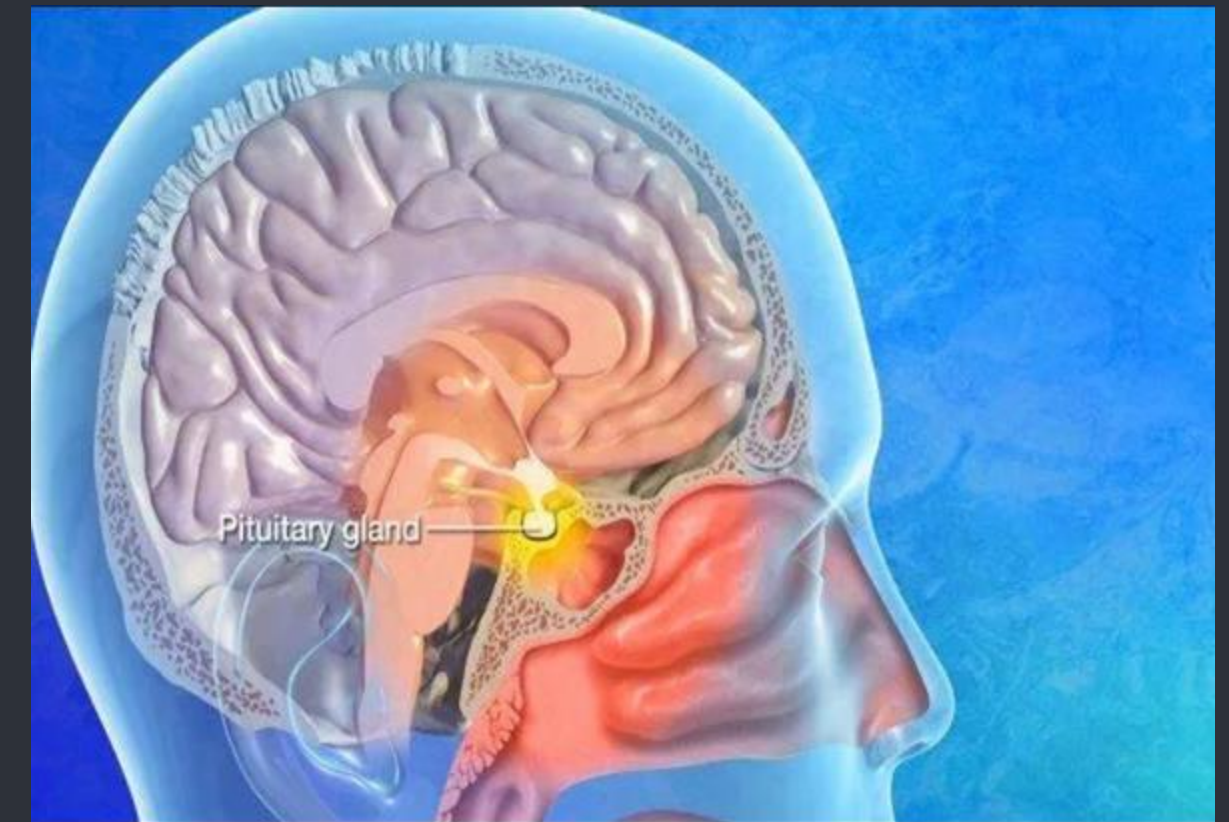
Hypothalamus gland is present in the brain. Hypothalamus produces 'releasing hormones' and 'inhibitory hormones'. The function of hypothalamus is to regulate the secretions of hormones from pituitary gland. That is how, hypothalamus controls the pituitary hormones.





PITUITARY GLAND

- Pituitary gland is present just below the brain.
- The pituitary gland secretes a number of hormones.
- One of the hormones secreted by pituitary gland is growth hormone (or human growth hormone).
- The growth hormone controls the growth of the human body.
- For example, growth hormone controls the development of bones and muscles.





PITUITARY GLAND

A person having a deficiency of growth hormone in childhood remains very short and becomes a dwarf.

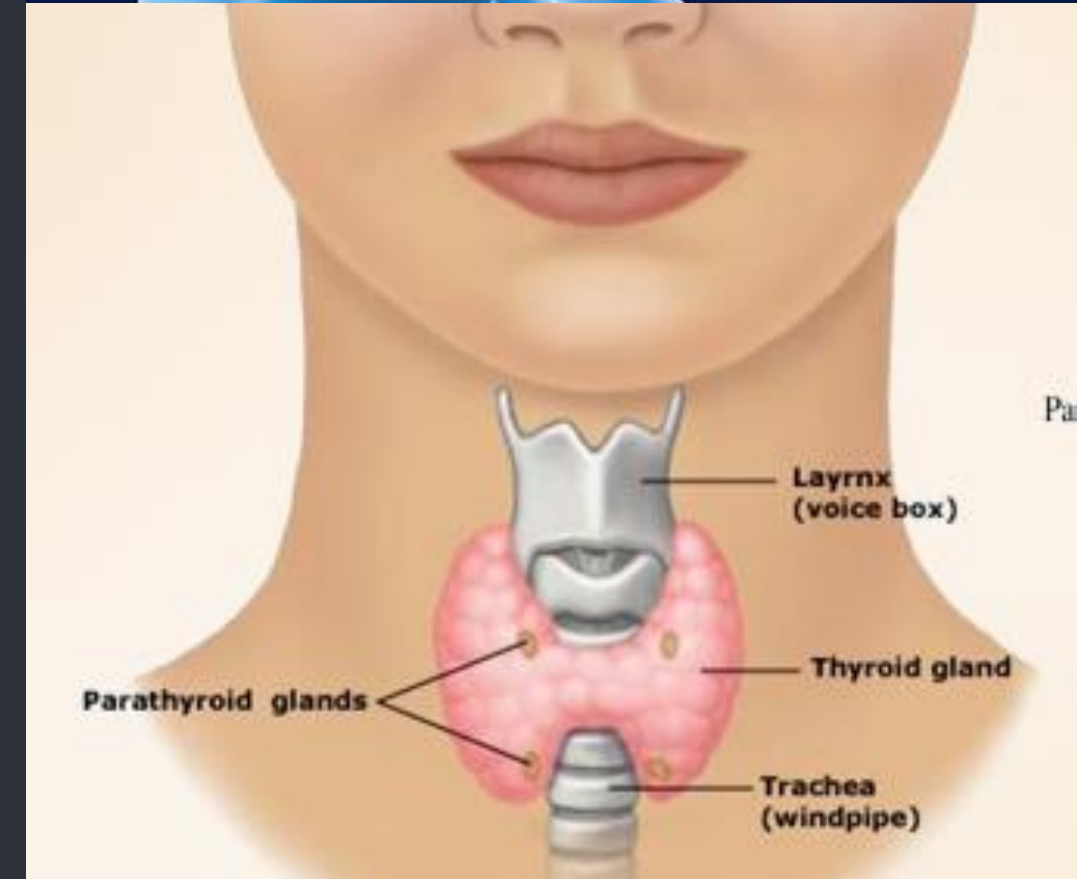
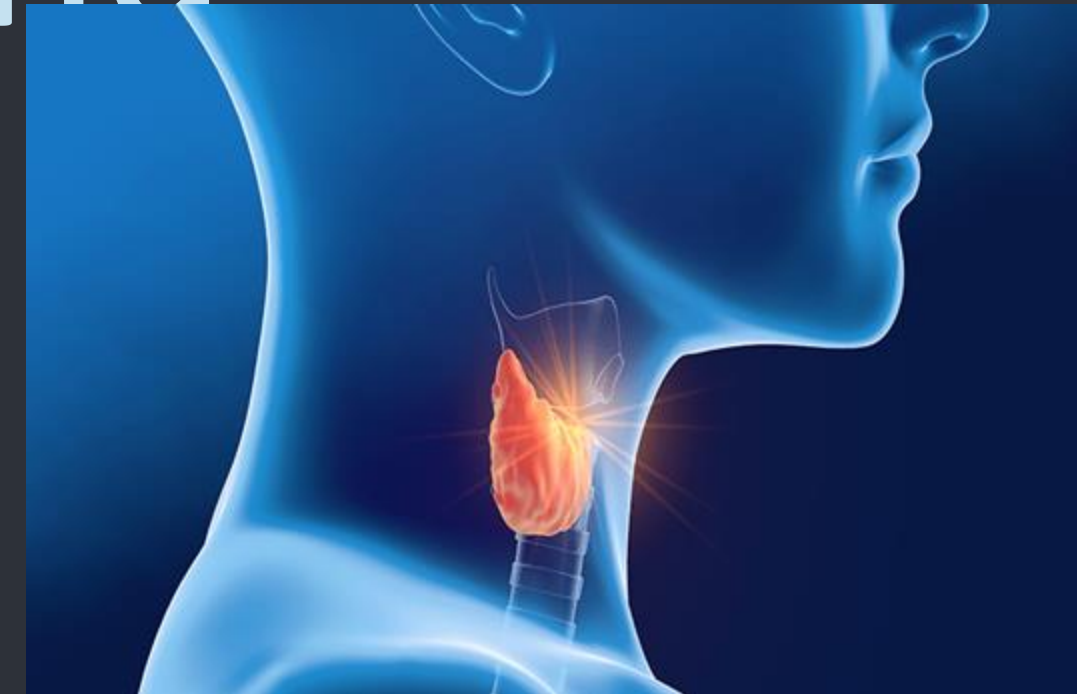
On the other hand, a person having too much growth hormone becomes very tall (or a giant).



Thyroid Gland



- Thyroid gland is attached to the wind pipe in our body.
- Thyroid gland makes a hormone called thyroxine (which contains iodine).
- The function of thyroxine hormone is to control the rate of metabolism of carbohydrates, fats and proteins in the body





GOITRE

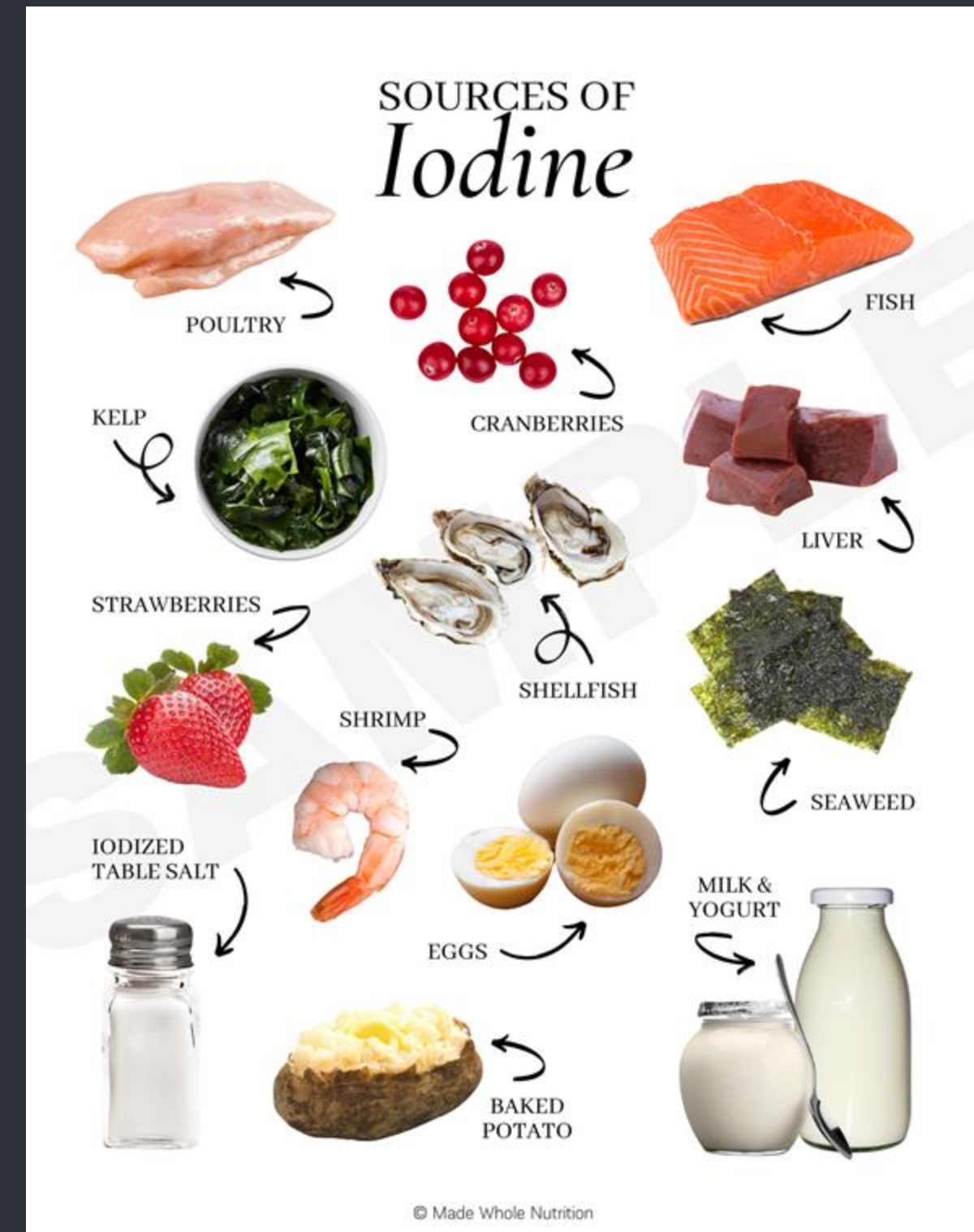
- Iodine is necessary for the making of thyroxine hormone in the body.
- The deficiency of iodine in the diet of a person produces less thyroxine hormone and causes a disease known as goitre.



SOURCES OF IODINE



Seaweed, Fish, shellfish, Table salts labeled “iodized”, Dairy (milk, cheese, yogurt), Eggs, Beef liver, Chicken.

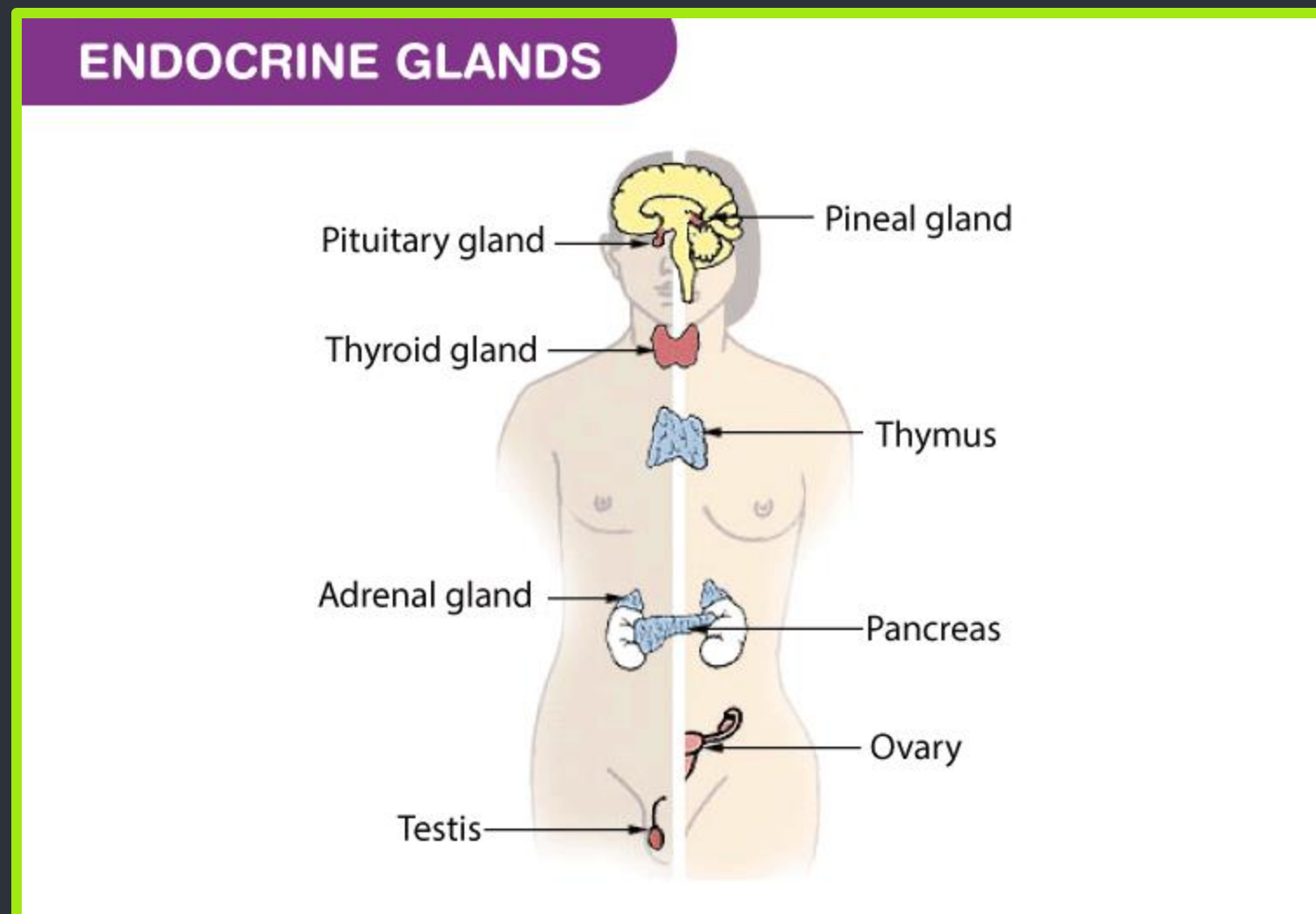




What are Hormones?

Hormones are chemical substances that act like messenger molecules in the body.

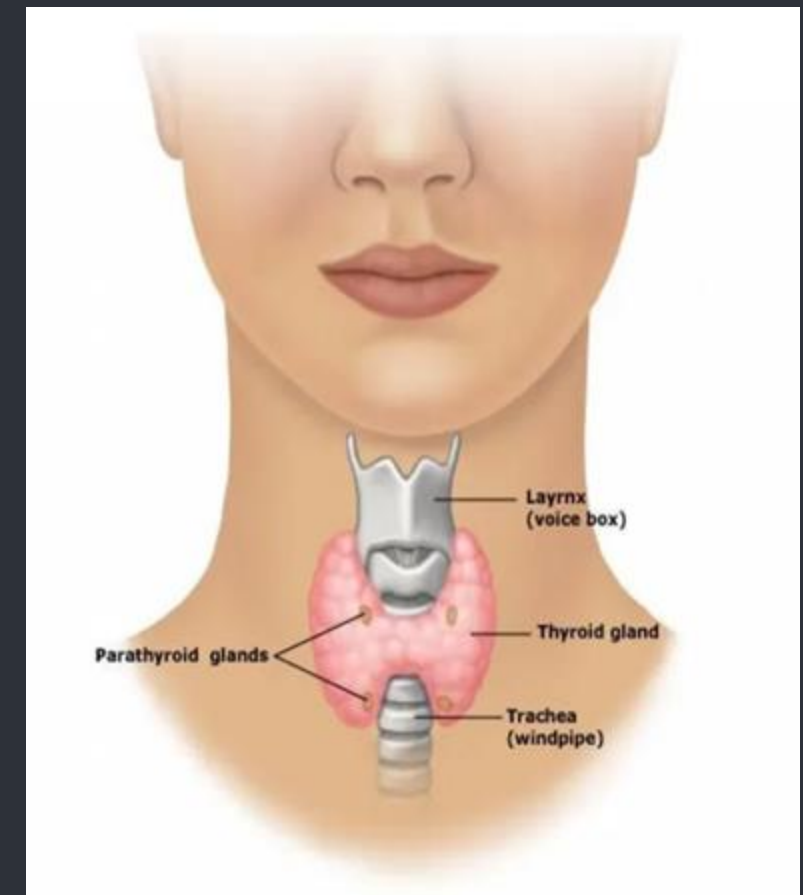
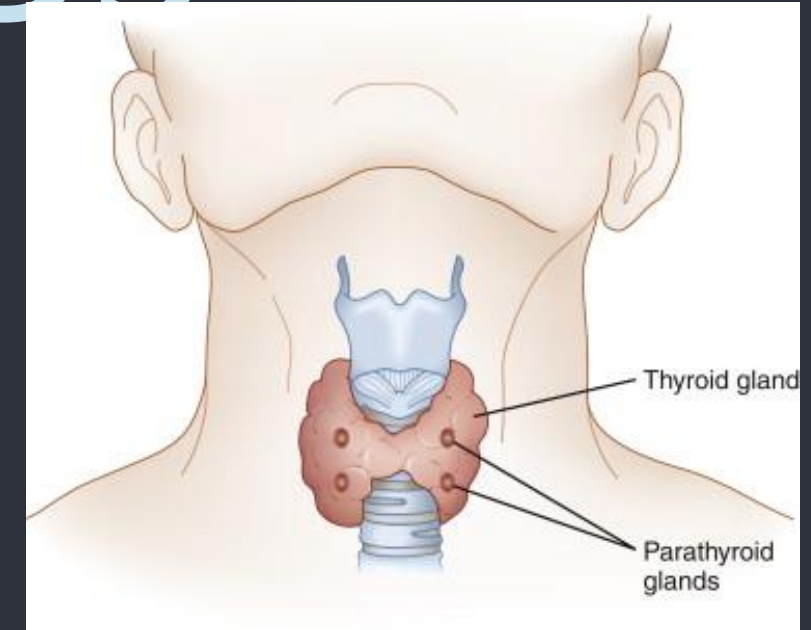
Hormones are secreted in small quantities by **ENDOCRINE GLANDS**.





PARATHYROID GLANDS

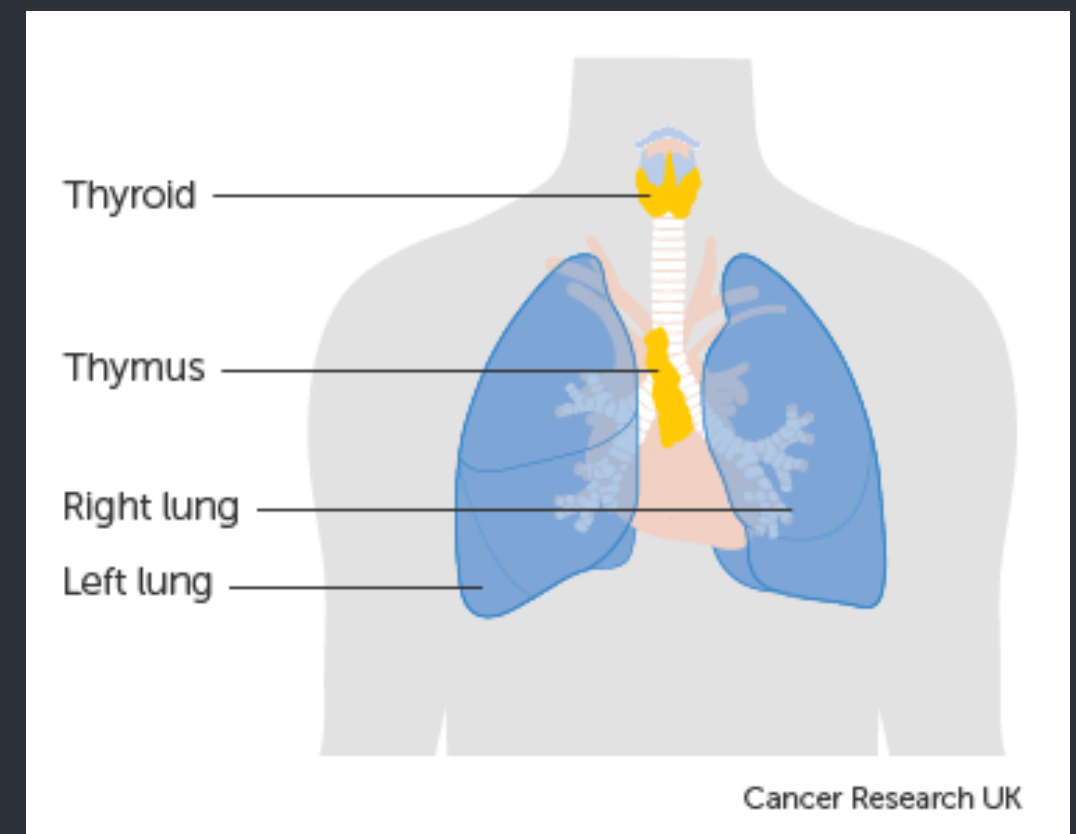
- There are four small **parathyroid glands** which are embedded in the thyroid gland.
- Parathyroid glands secrete a hormone called **parathormone**.
- The function of parathormone hormone is to regulate **calcium and phosphate** levels in the blood.



Thymus Gland



- Thymus gland lies in the lower part of the neck and upper part of chest. Thymus gland secretes thymus hormone which plays a role in the development of the immune system of the body.
- Thymus gland is large in young children but shrinks after puberty (or sexual maturity).

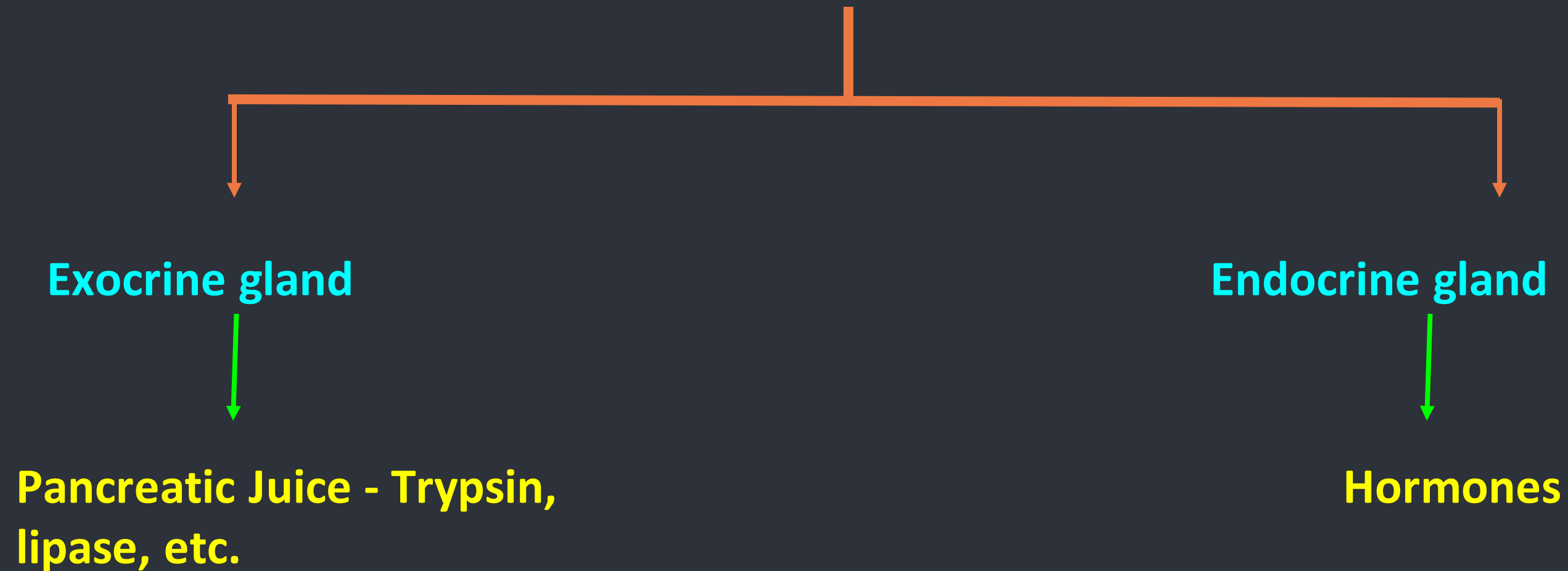




Pancreas

(Dual gland/ mixed gland/ heterocrine gland)

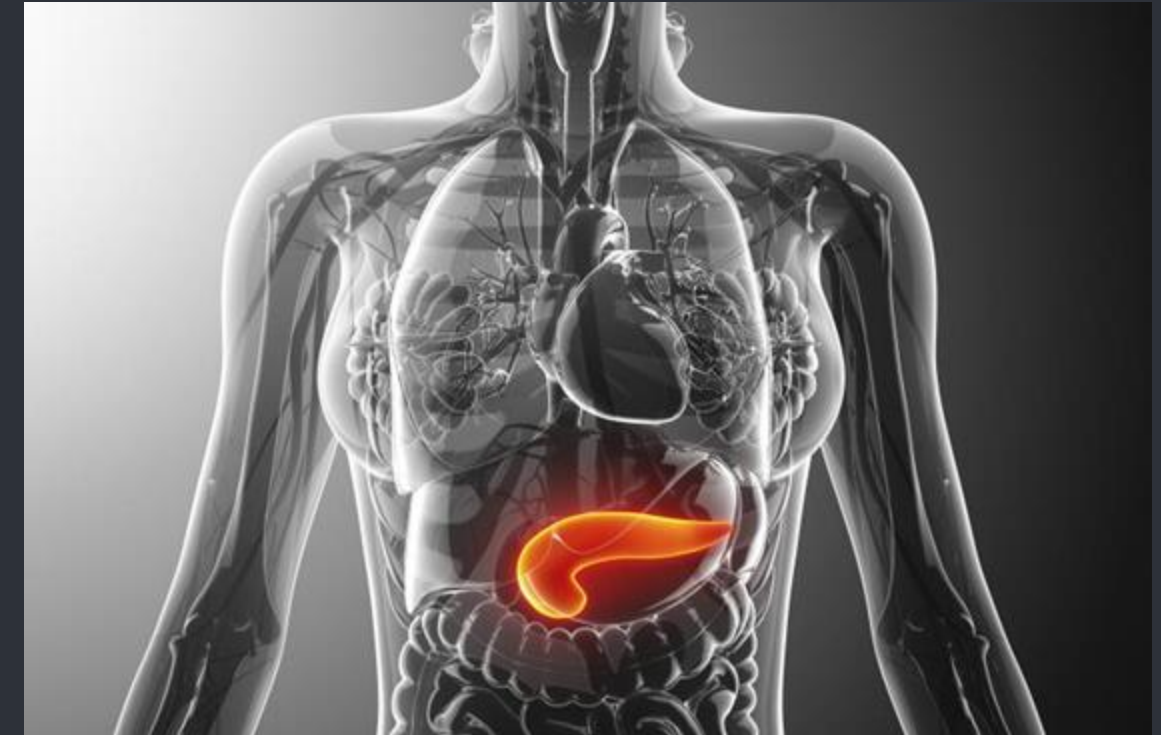
Function as both:





Pancreas

- The pancreas is just below the stomach in the body.
- Pancreas secretes the hormone called insulin.
- The function of insulin hormone is to lower the blood sugar level (or blood glucose level).





Pancreas





DIABETES

- Deficiency of insulin hormone in the body causes a disease known as diabetes.
- Diabetes is characterized by large quantities of sugar in the blood (and even urine).
- The insulin hormone controls the metabolism of sugar.

DIAGNOSING DIABETES			
	A1C (percent)	Fasting Plasma Glucose (mg/dL)	Oral Glucose Tolerance Test (mg/dL)
Diabetes*	6.5 or above	126 or above	200 or above
Prediabetes	5.7 to 6.4	100 to 125	140 to 199
Normal	About 5	99 or below	139 or below



ADRENAL GLANDS

- There are two adrenal glands which are located on the top of two kidneys.
- The adrenal glands secrete **adrenaline hormone**.
- The function of adrenaline hormone is to regulate heart rate, breathing rate, blood pressure and carbohydrate metabolism.





ADRENAL GLANDS

- Adrenaline hormone is secreted in small amounts all the time but in large amounts when a person is frightened or excited.
- When adrenaline is secreted in large amounts it prepares our body for action.
- It speeds up heartbeat and breathing, raises blood pressure and allows more glucose (carbohydrate) to go into the blood to give us a lot of energy quickly to fight or flight (run away).
- Adrenal glands are often called 'glands of emergency'.



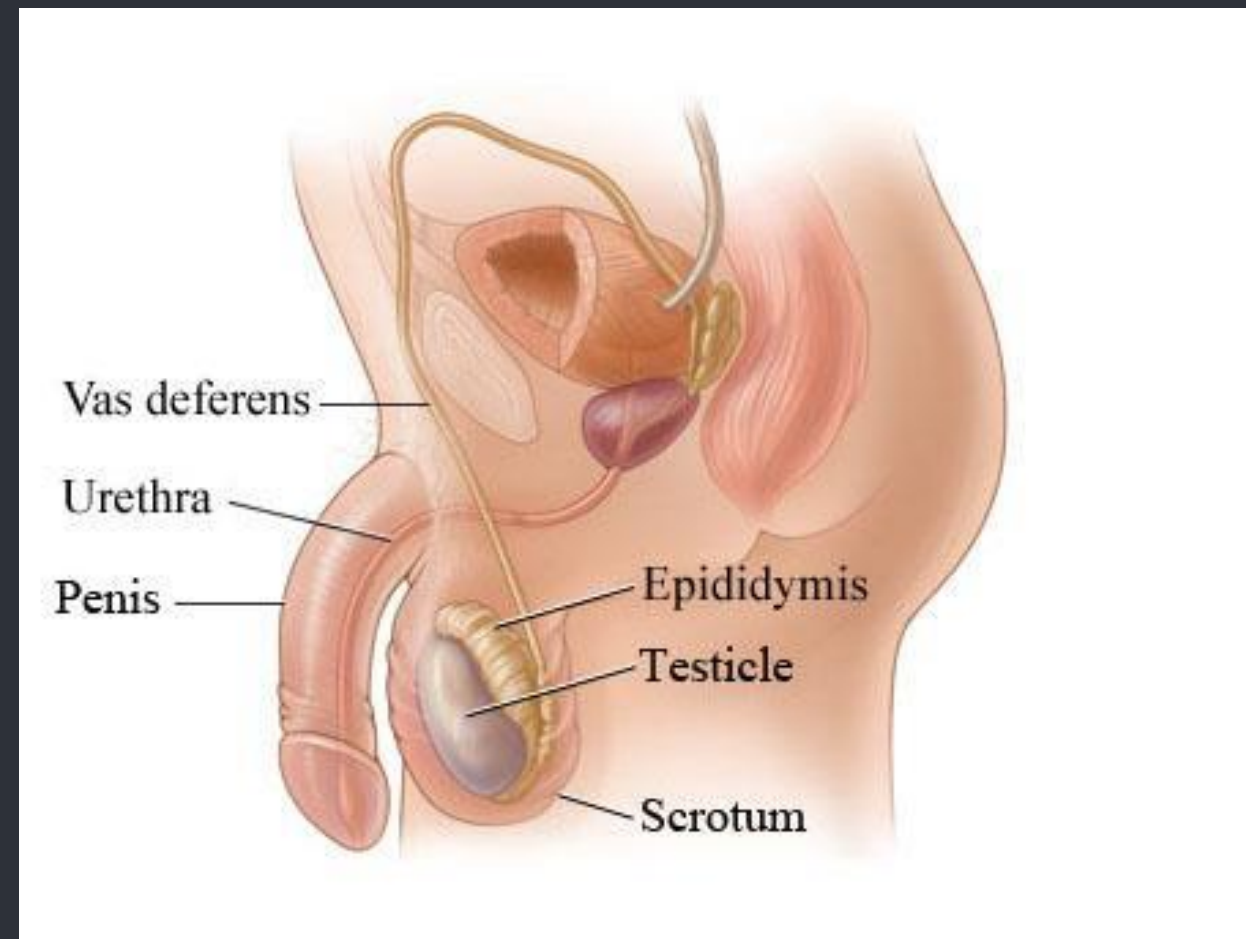
TESTES





Testes

- Testes are the glands which are present only in males (men).
- Testes make male sex hormones called testosterone.
- The testes also make the male gametes called sperms.





TESTES

- The function of testosterone hormone is to control the development of male sex organs and male features such as deeper voice, moustache, beard, and more body hair (than females).
- All these changes caused by testosterone are associated with male puberty which the boys attain at an age of 13 to 14 years.





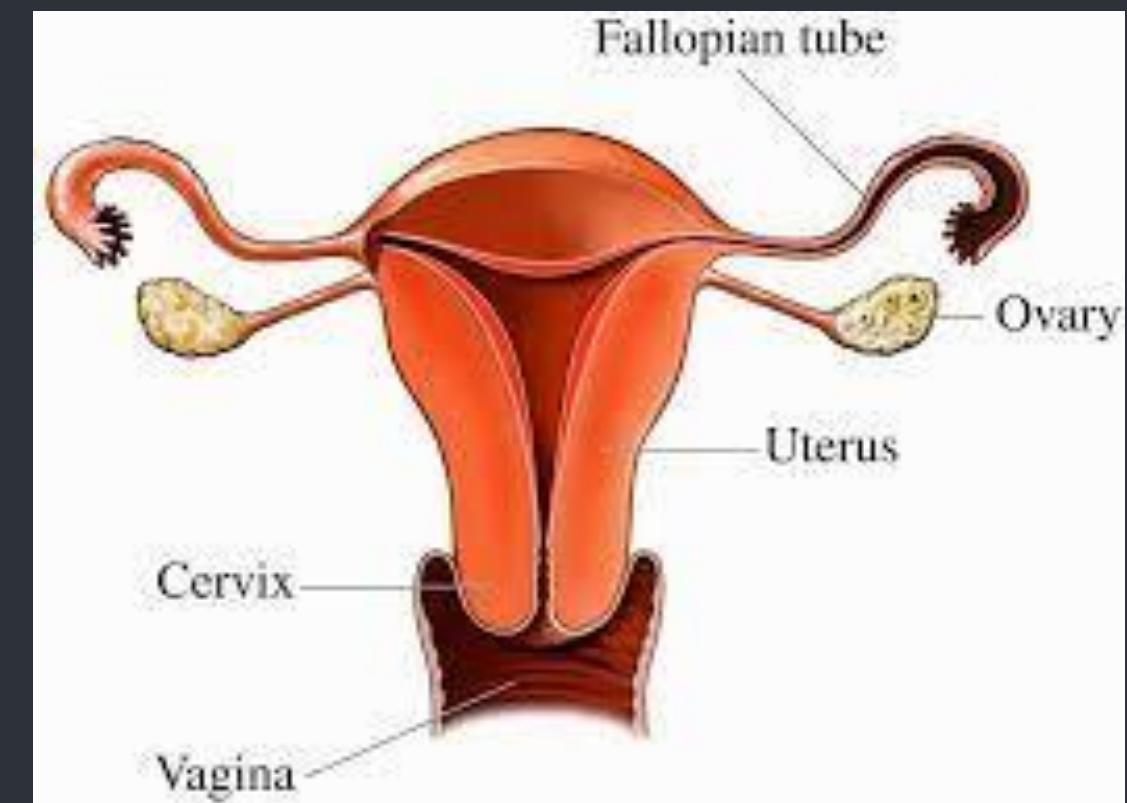
Ovaries





Ovaries

- Ovaries are the glands which are present only in females (women).
- Ovaries make two female sex hormones called oestrogen and progesterone.
- The function of oestrogen hormone is to control the development of female sex organs, and female features such as feminine voice, soft skin and mammary glands (breasts).





OVARIES

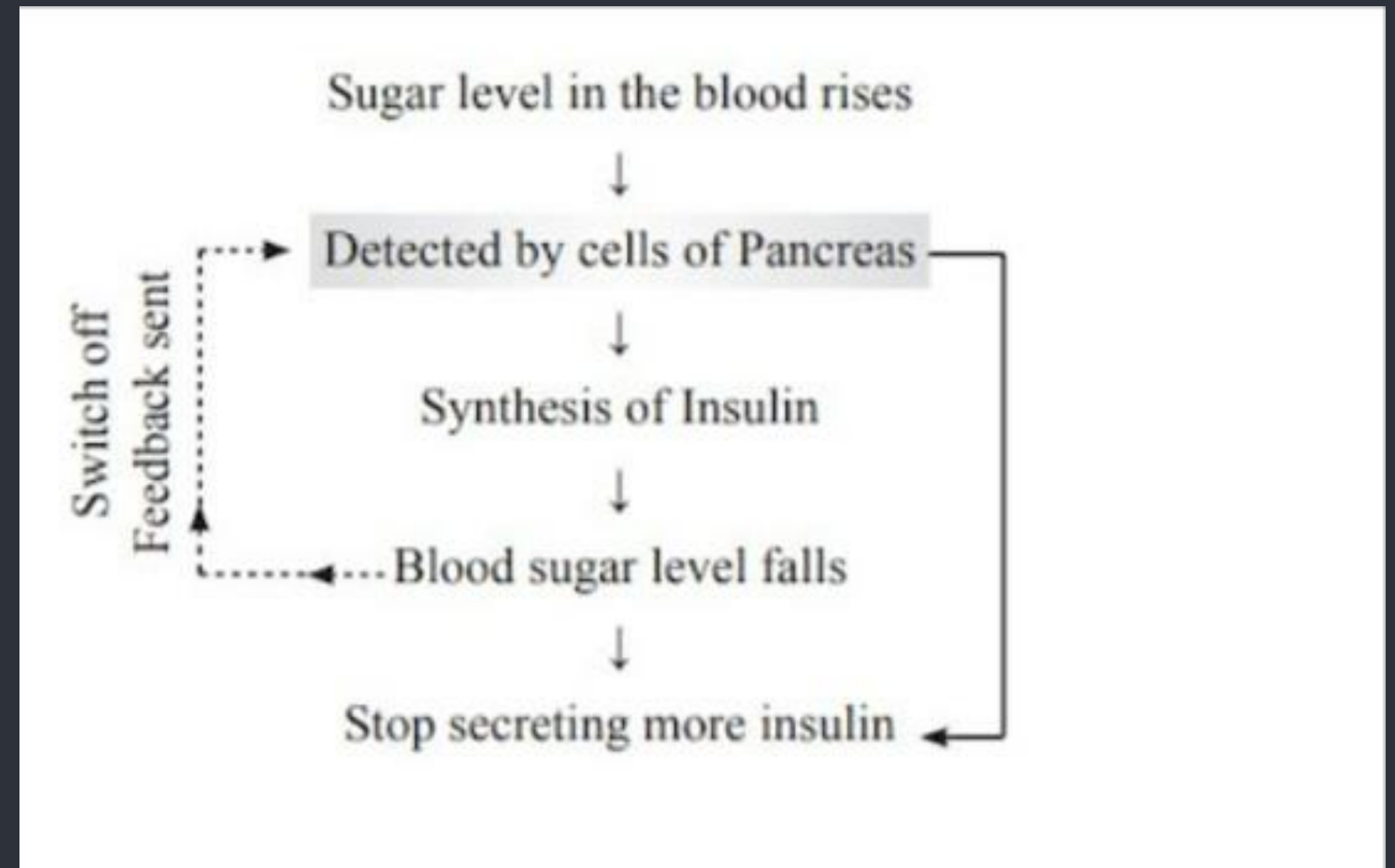
- All these changes caused by oestrogen are associated with female puberty which the girls attain at an age of 10 to 12 years.
- The functions of progesterone hormone is to control the uterus changes in menstrual cycle.
- It also helps in the maintenance of pregnancy. The ovaries also make the female gametes called ova (or eggs).





FEEDBACK MECHANISM

- The excess or deficiency of hormones has a harmful effect on our body.
- For example, the deficiency of insulin hormone results in a disease called diabetes whereas excess of insulin in the body can lead to coma.





FEEDBACK MECHANISM

- The timing and amount of hormones released by various glands are controlled by the 'feedback mechanism' which is in-built in our body. For example, if the sugar level in the blood rises too much, they are detected by the cells of pancreas which respond by producing and secreting more insulin into blood. And as the blood sugar falls to a certain level, the secretion of insulin is reduced automatically.



CONTROL AND COORDINATION IN PLANTS

- The plants do not have a nervous system and sense organs like eyes, ears, or nose, etc., like the animals, but they can still sense things.
- The plants coordinate their behaviour against environmental changes by using hormones.



CONTROL AND COORDINATION IN



PLANTS

Movements

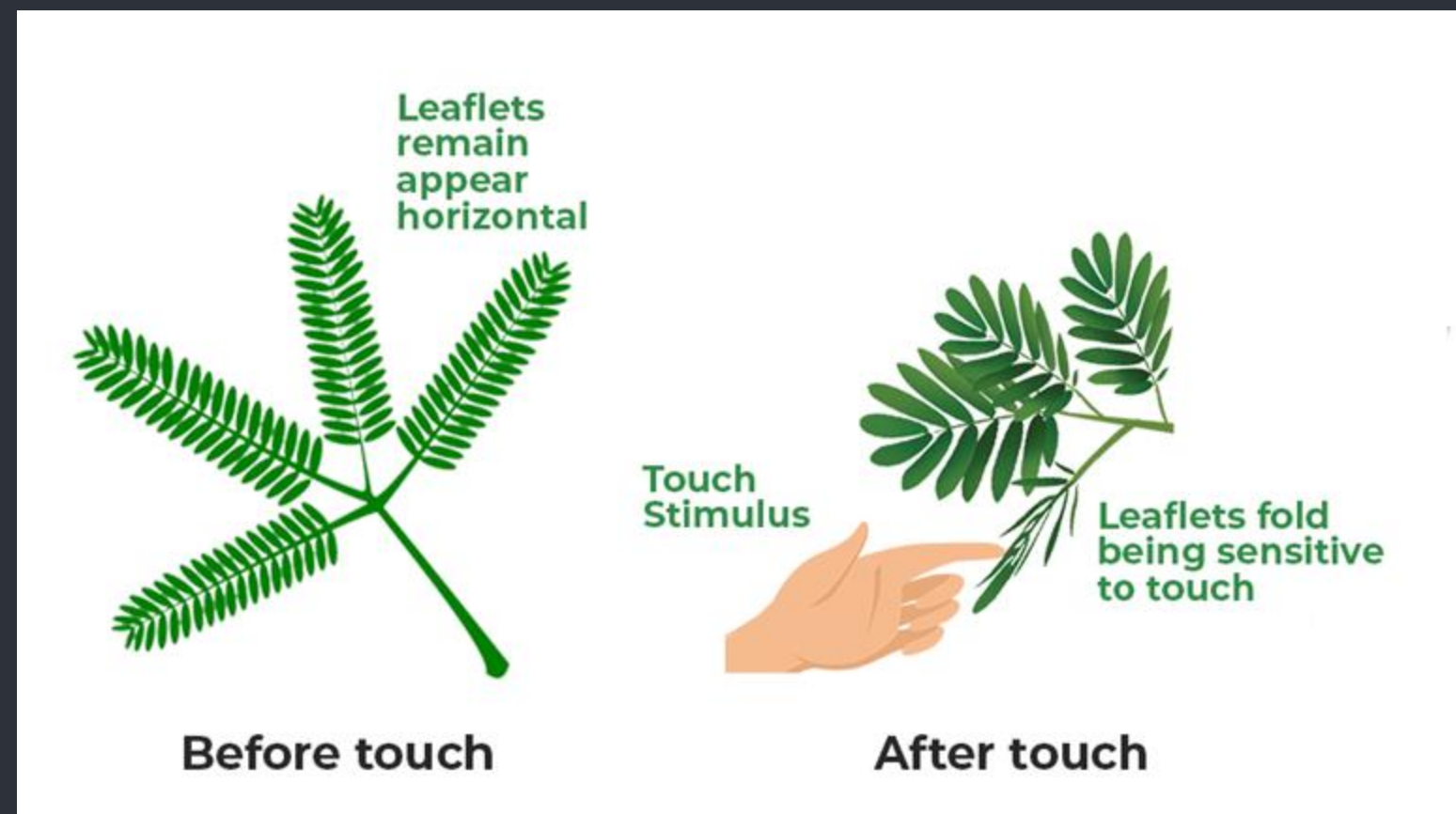
Hormones



CONTROL AND COORDINATION IN

PLANTS

- The plants do not have a nervous system and sense organs like eyes, ears, or nose, etc., like the animals, but they can still sense things.
- The plants coordinate their behaviour against environmental changes by using hormones.





RESPONSE IN PLANTS

**Leaves of touch - me - not plant
drooping on touching**

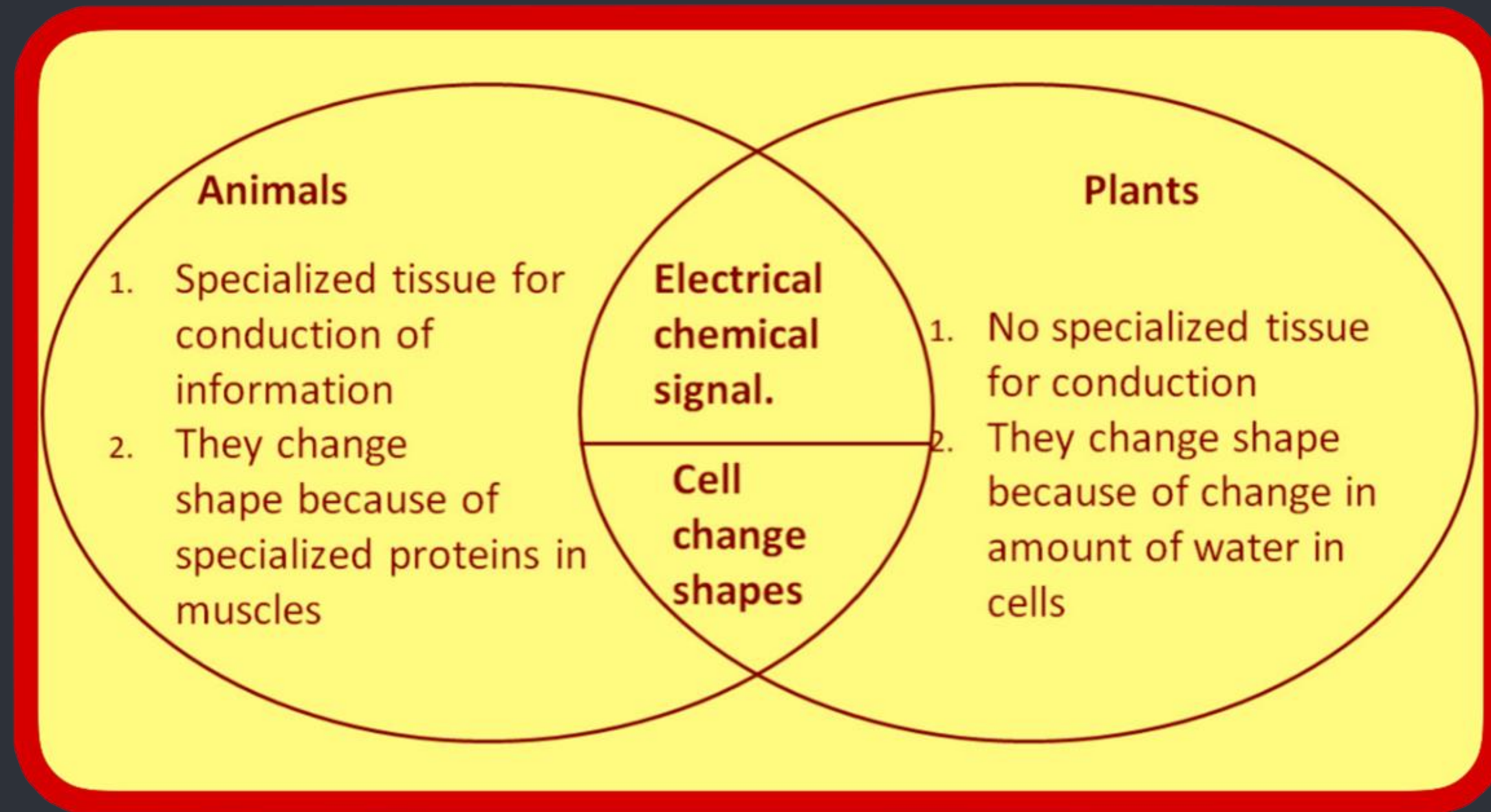
Not related to Growth

**Root and Shoot growing
downward and upward
respectively**

Related to Growth

COMPARING RESPONSE TO STIMULUS IN

ANIMALS AND PLANTS





PLANT HORMONES (OR PHYTOHORMONES)

- The control and coordination in plants is done by plant hormones (or phytohormones).
- The growth of a plant can be divided into three stages:
 1. Cell division
 2. Cell enlargement
 3. Cell differentiation (or cell specialization)



PLANT HORMONES (OR

PHYTOHORMONES)

- There are four major types of plant hormones (or phytohormones) which are involved in the control and coordination in plants.
- 1) Auxins 2) Gibberellins 3) Cytokinin's 4) Absciscic acid (ABA)

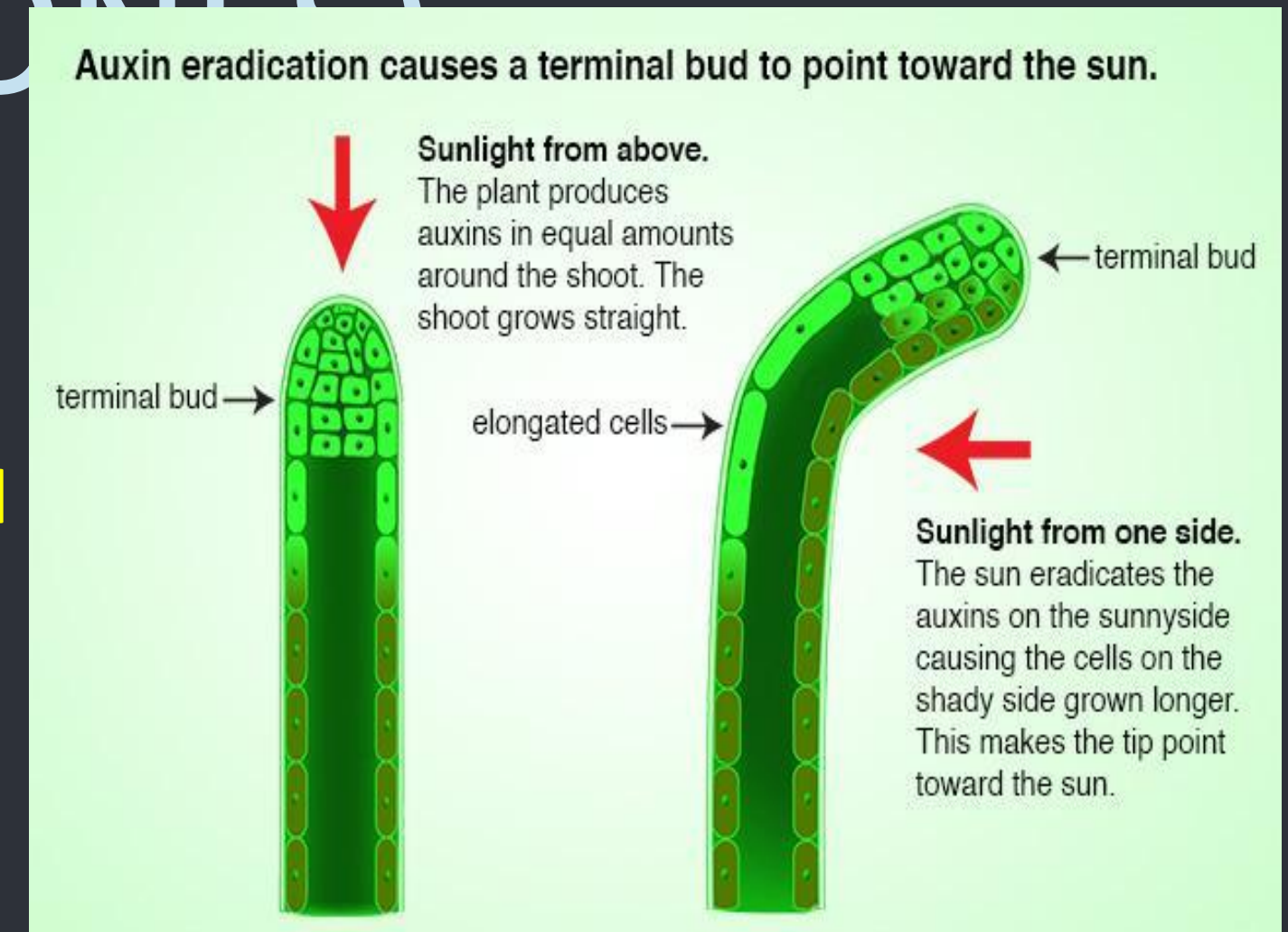
						
	Germination	Growth to Maturity	Flowering	Fruit Development	Abscission	Seed Dormancy
Gibberellin	✓	✓	✓	✓	✗	✗
Auxin	✗	✓	✓	✓	✗	✗
Cytokinins	✗	✓	✓	✓	✗	✗
Ethylene	✗	✗	✓	✓	✓	✗
Abscisic Acid	✗	✗	✗	✗	✓	✓



PLANT HORMONES (OR PHYTOHORMONES)

1. Auxins

- Auxins are the plant hormones which promote cell enlargement and cell differentiation in plants. Auxins also promote fruit growth.
- Auxin hormone controls a plant's response to light and gravity.
- Auxin is made by cells at the tip of stems and roots.





PLANT HORMONES (OR PHYTOHORMONES)

1. Auxins

- Auxin moves away from light, and towards gravity.
- Auxin speeds up growth in stem but it slows down growth in roots.
- Synthetic auxins are applied in agriculture and horticulture.

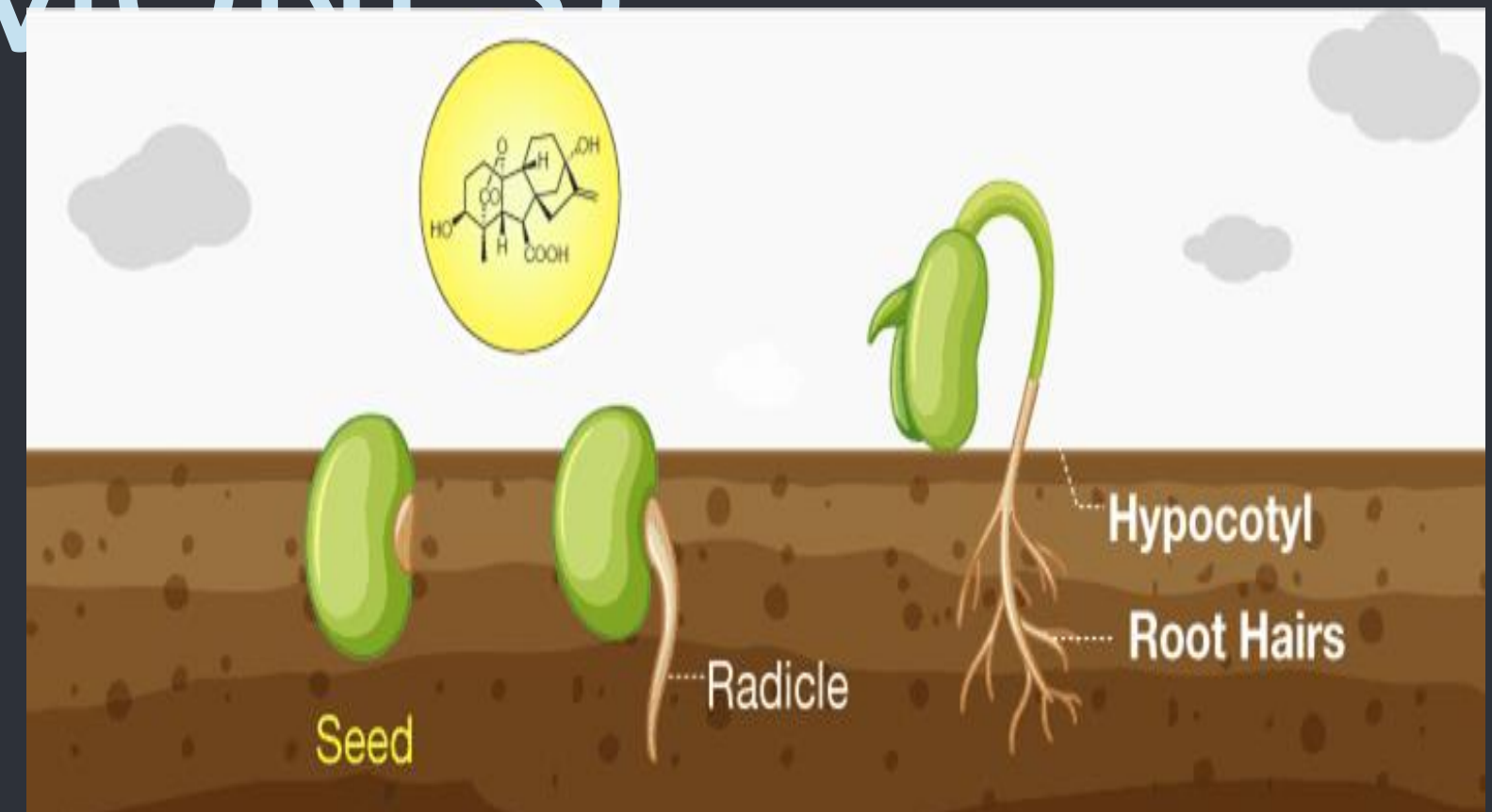


PLANT HORMONES (OR

2. Gibberellins

- Gibberellins are plant hormones which promote cell enlargement and cell differentiation in the presence of auxins.
- Gibberellins help in breaking the dormancy in seeds and buds. They also promote growth in fruits.
- Gibberellin hormone is involved mainly in shoot extensions. Gibberellin stimulates elongation of shoots of various plants.

PHYTOHORMONES)





PLANT HORMONES (OR PHYTOHORMONES)

3. Cytokinin's

- Cytokinin's are the plant hormones which promote cell division in plants.
- Cytokinin's also help in breaking the dormancy of seeds and buds. They delay the ageing in leaves.
- Cytokinin's promote the opening of stomata. They also promote fruit growth.





PLANT HORMONES (OR

4. Abscissic Acid (ABA) PHYTOHORMONES)

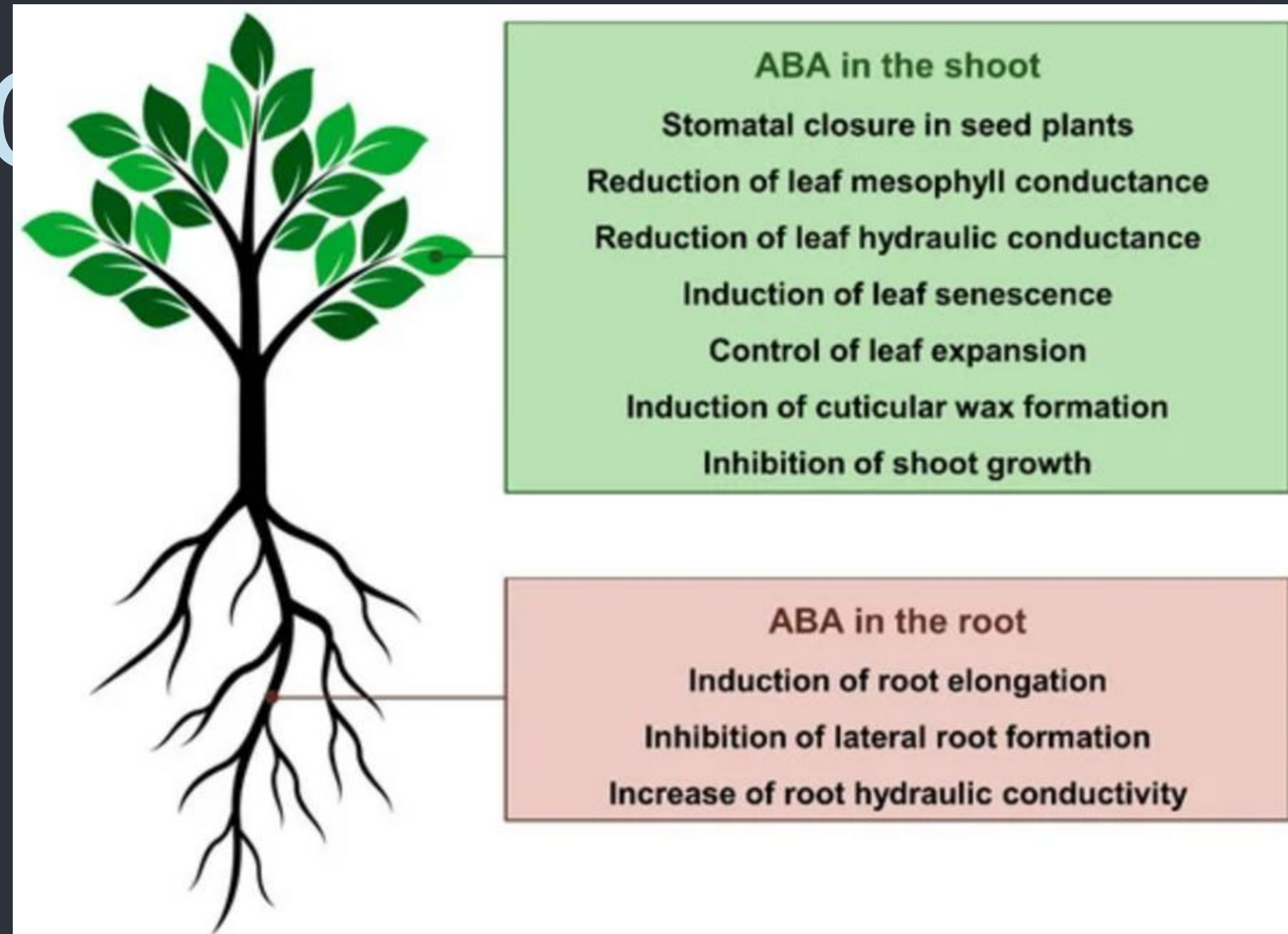
- Abscissic acid is a plant hormone which functions mainly as a growth inhibitor.
- It promotes the dormancy in seeds and buds (this is the opposite of breaking of dormancy). It also promotes the closing of stomata.
- It promotes the wilting and falling of leaves (which is called abscission). It also causes the detachment of flowers and fruits from the plants.



PLANT HORMONES (OR

4. Abscissic Acid (ABA)

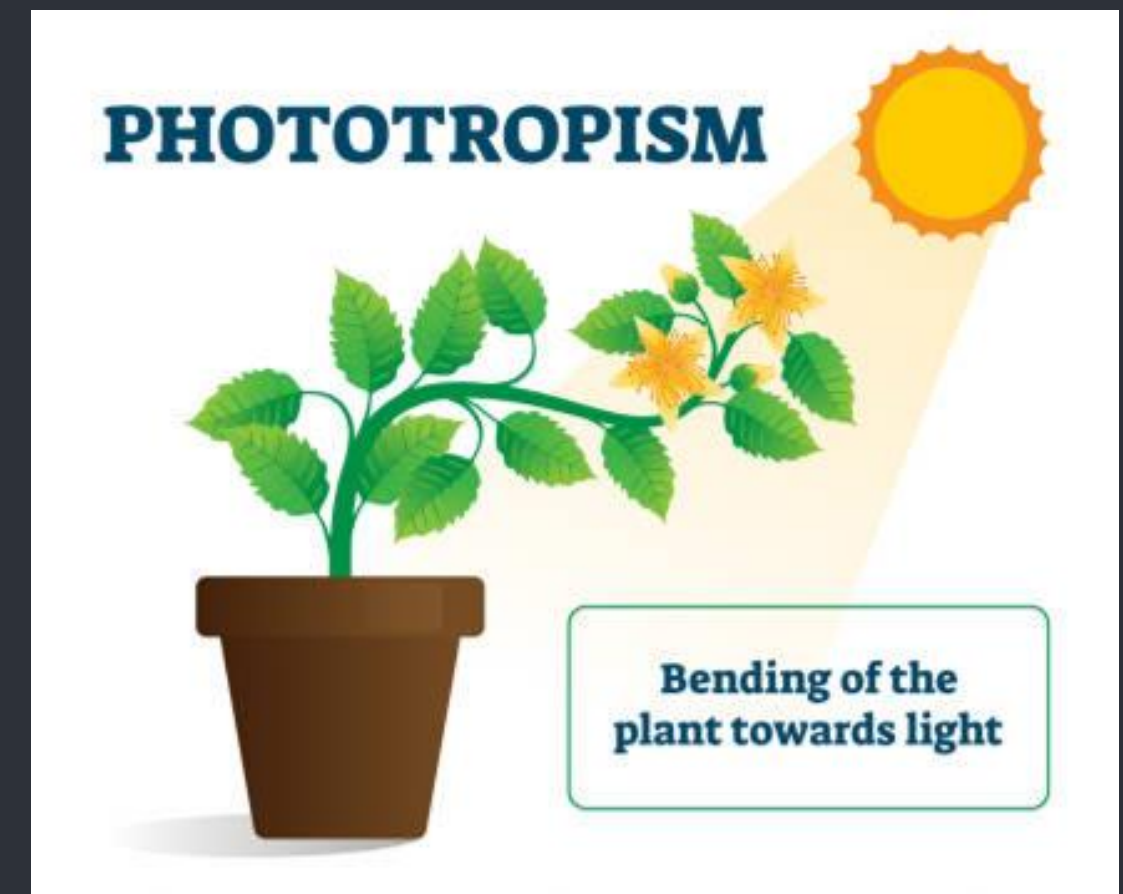
PHYTO



TROPISMS (OR TROPIC MOVEMENTS)



- A growth movement of a plant part in response to an external stimulus in which the direction of stimulus determines the direction of response is called tropism.
- Thus, tropism is a directional movement of the part of a plant caused by its growth.
- If the growth (or movement) of a plant part is towards the stimulus, it is called positive tropism.
- If the growth (or movement) of a plant part is away from the stimulus, then it is called negative tropism.

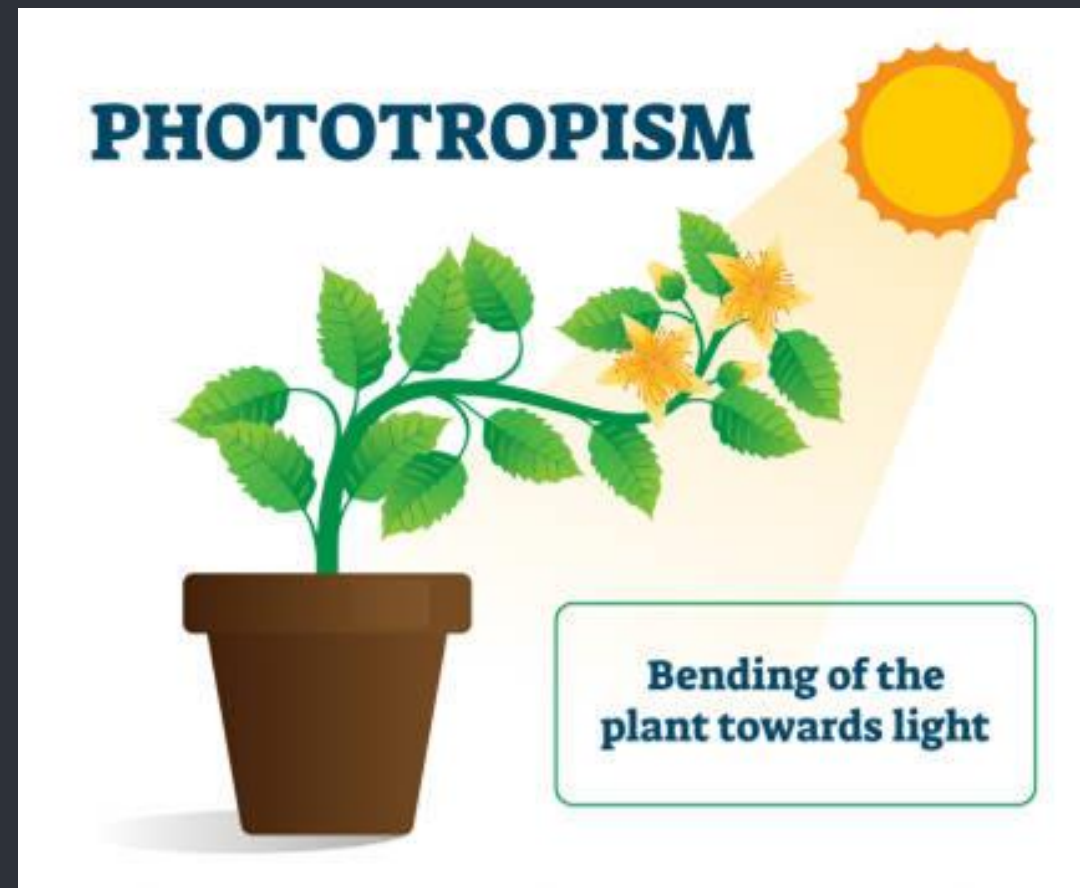




TYPES OF TROPISMS

1. Phototropism:

- The movement of a plant part in response to light is called phototropism
- If the plant part moves towards light, it is called positive phototropism.

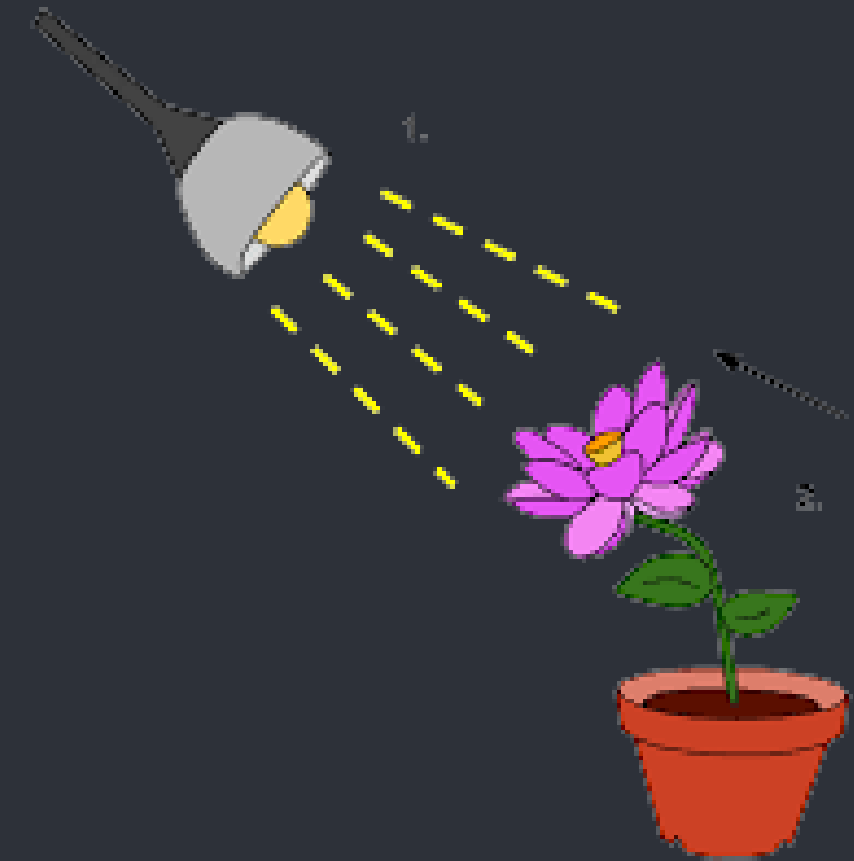




TYPES OF TROPISMS

1. Phototropism:

- On the other hand, if the plant part moves away from light, then it is called negative phototropism.
- The stem (or shoot) of a growing plant bends towards light, so the stem (or shoot) of a plant shows positive phototropism.
- On the other hand, the roots of a plant move away from light, so the roots of a plant show negative phototropism.

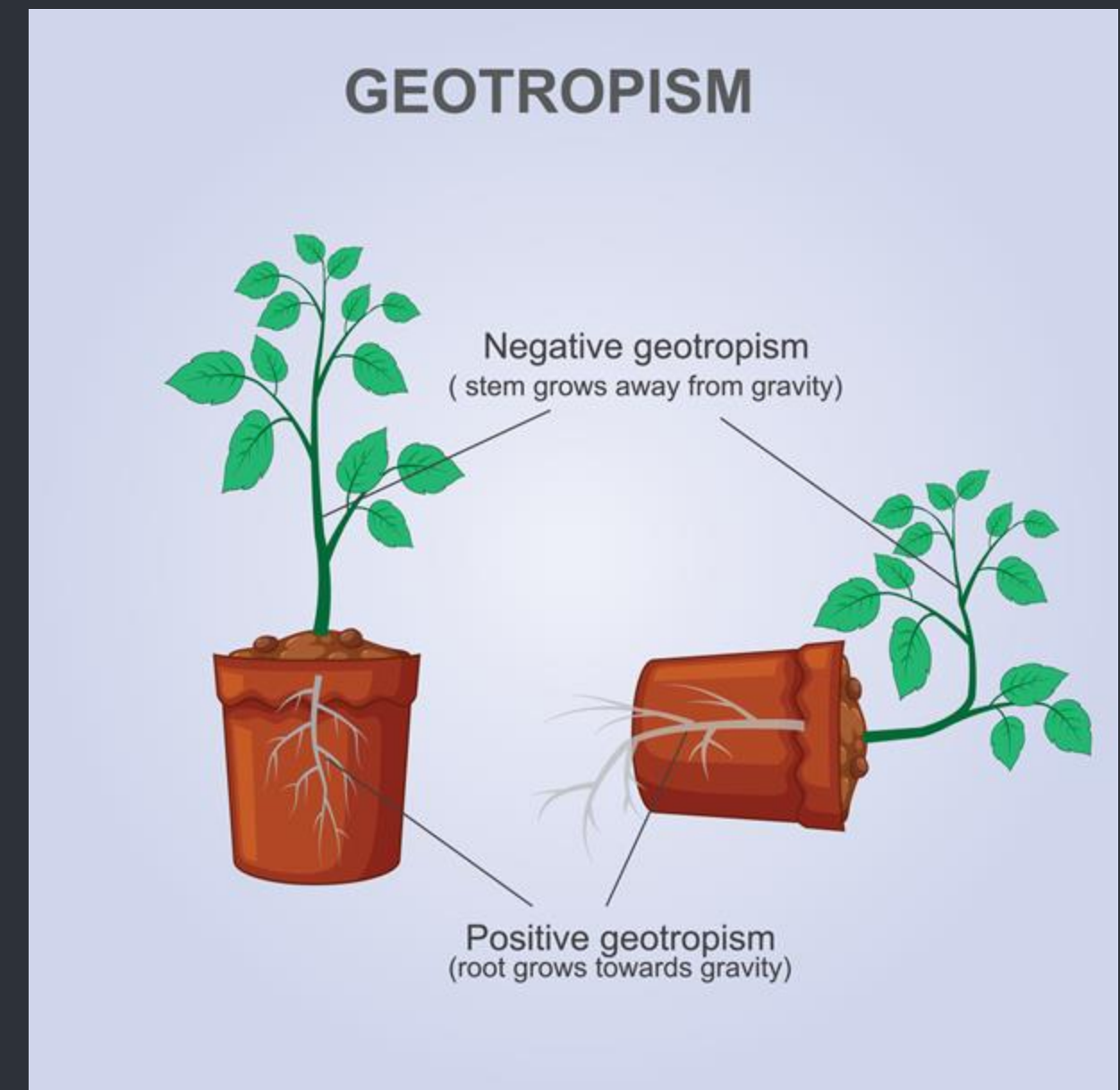




TYPES OF TROPISMS

2. Geotropism

- The movement of a plant part in response to gravity is called geotropism.
- If the plant part moves in the direction of gravity, it is called positive geotropism.
- On the other hand, if the plant part moves against the direction of gravity, it is negative geotropism.

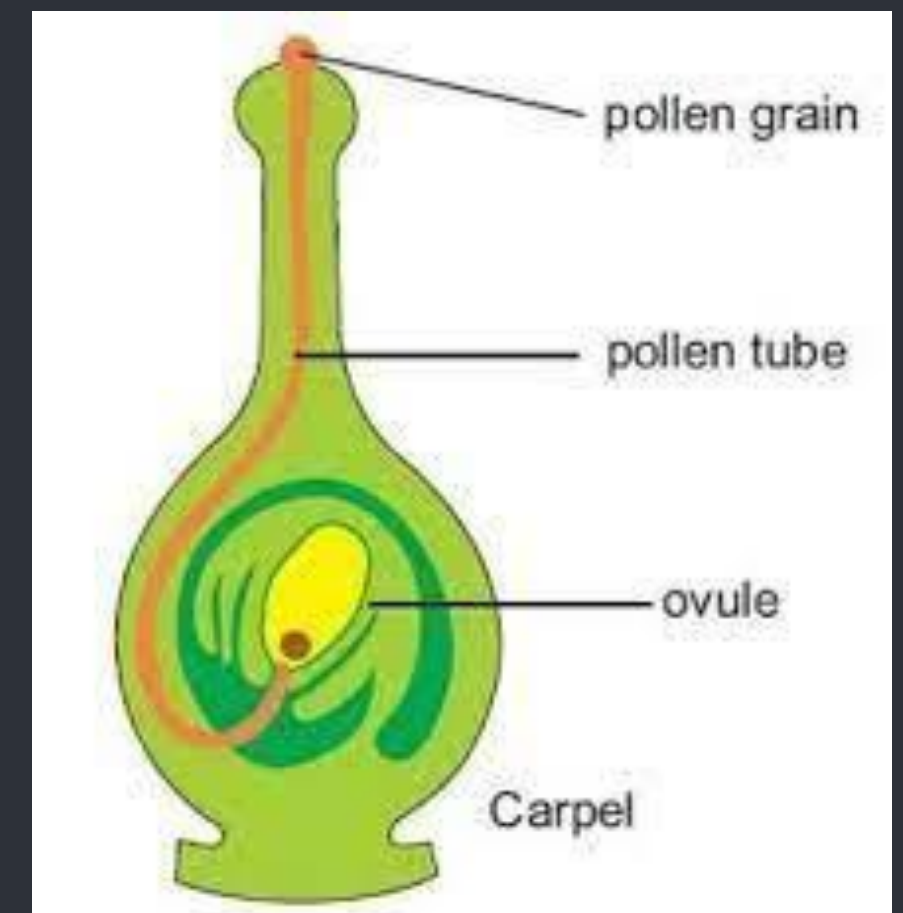




TYPES OF TROPISMS

3. Chemotropism

- The movement of a plant part in response to a chemical stimulus is called chemotropism.
- If the plant part shows movement (or growth) towards the chemical, it is called positive chemotropism.
- If the plant part shows movement (or growth) away from the chemical, then it is called negative chemotropism.

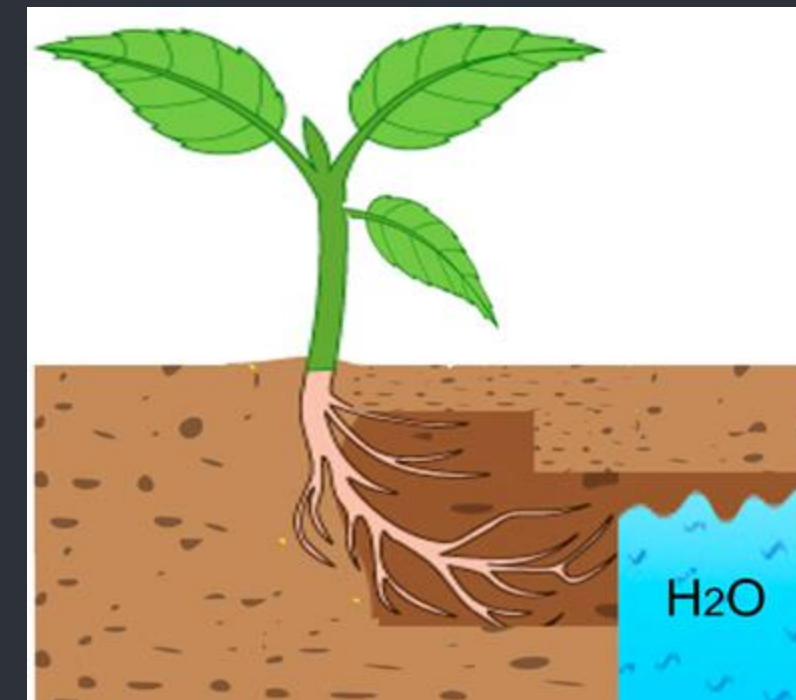
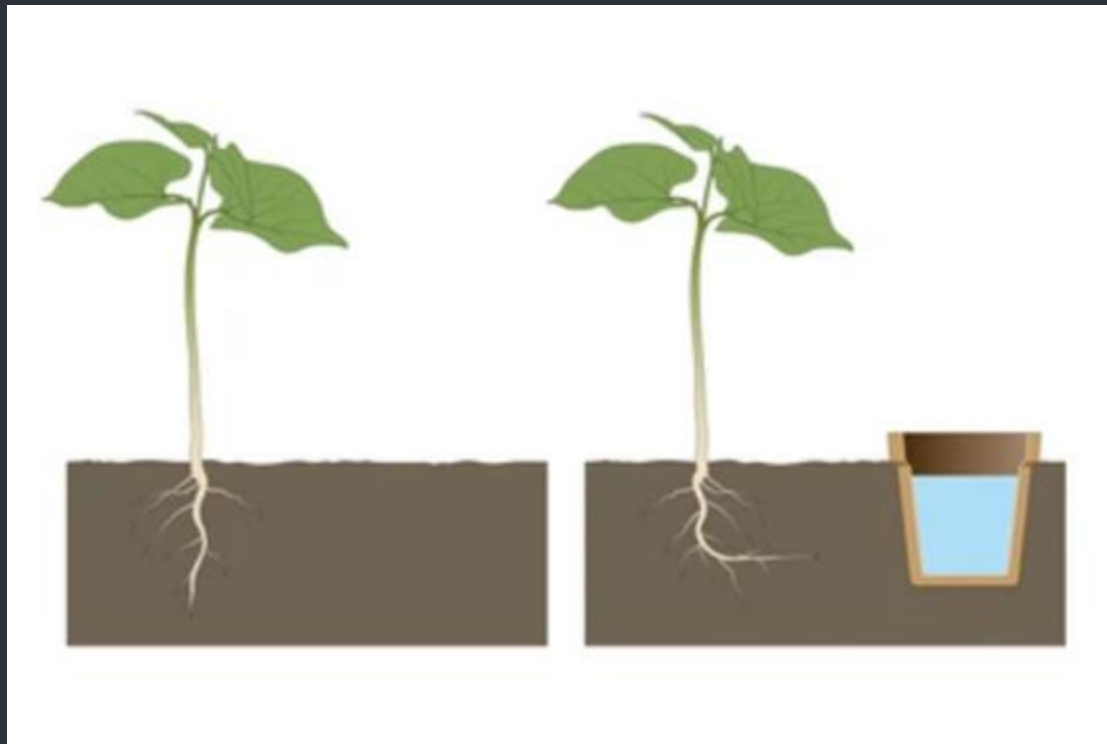




TYPES OF TROPISMS

4. Hydrotropism

- The movement of a plant part in response to water is called **hydrotropism**.
- If the plant part moves towards water, it is called **positive hydrotropism**.
- If the plant part moves away from water, then it is called **negative hydrotropism**.





TYPES OF TROPISMS

5. Thigmotropism

- The directional growth movement of a plant part in response to the touch of an object is called thigmotropism.
- The climbing parts of the plants such as tendrils grow towards any support which they happen to touch and wind around that support. So, tendrils of plants are positively thigmotropism.





RESPONSE OF PLANTS TO LIGHT:

PHOTOTROPISM

- Plants need sunlight, so the stems (or shoots) respond to sunlight by growing towards it.
- This observation shows that the stem of plant responds to light and bends towards it.
- The plant stem responds to light and bends towards it due to the action of 'auxin hormone'.

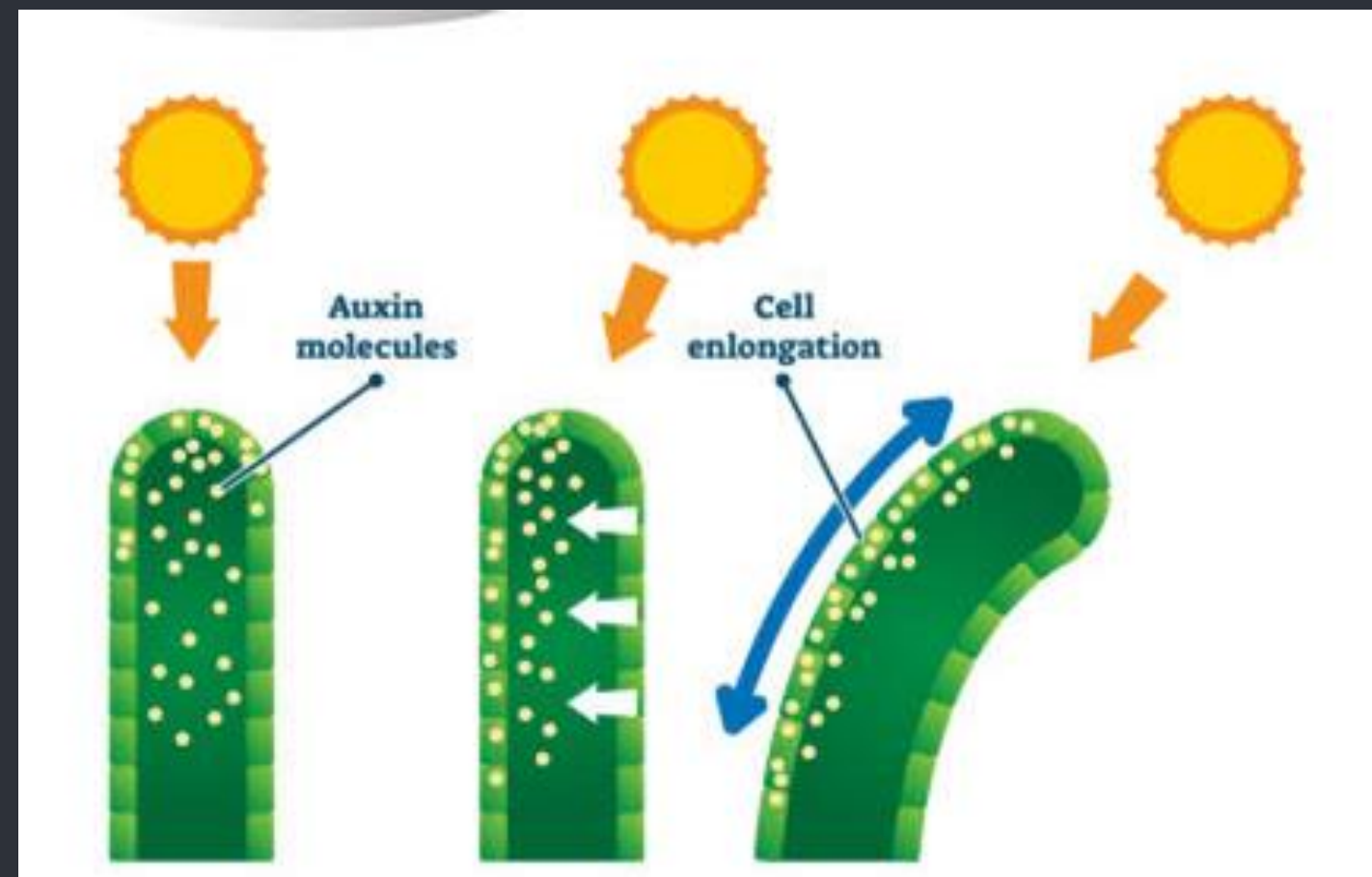




RESPONSE OF PLANTS TO LIGHT:

PHOTOTROPISM

- When sunlight comes from above, then the auxin hormone present in the tip of the stem spreads uniformly down the stem. Due to the equal presence of auxin, both the sides of the stem grow equally rapidly. And the stem grows straight up.

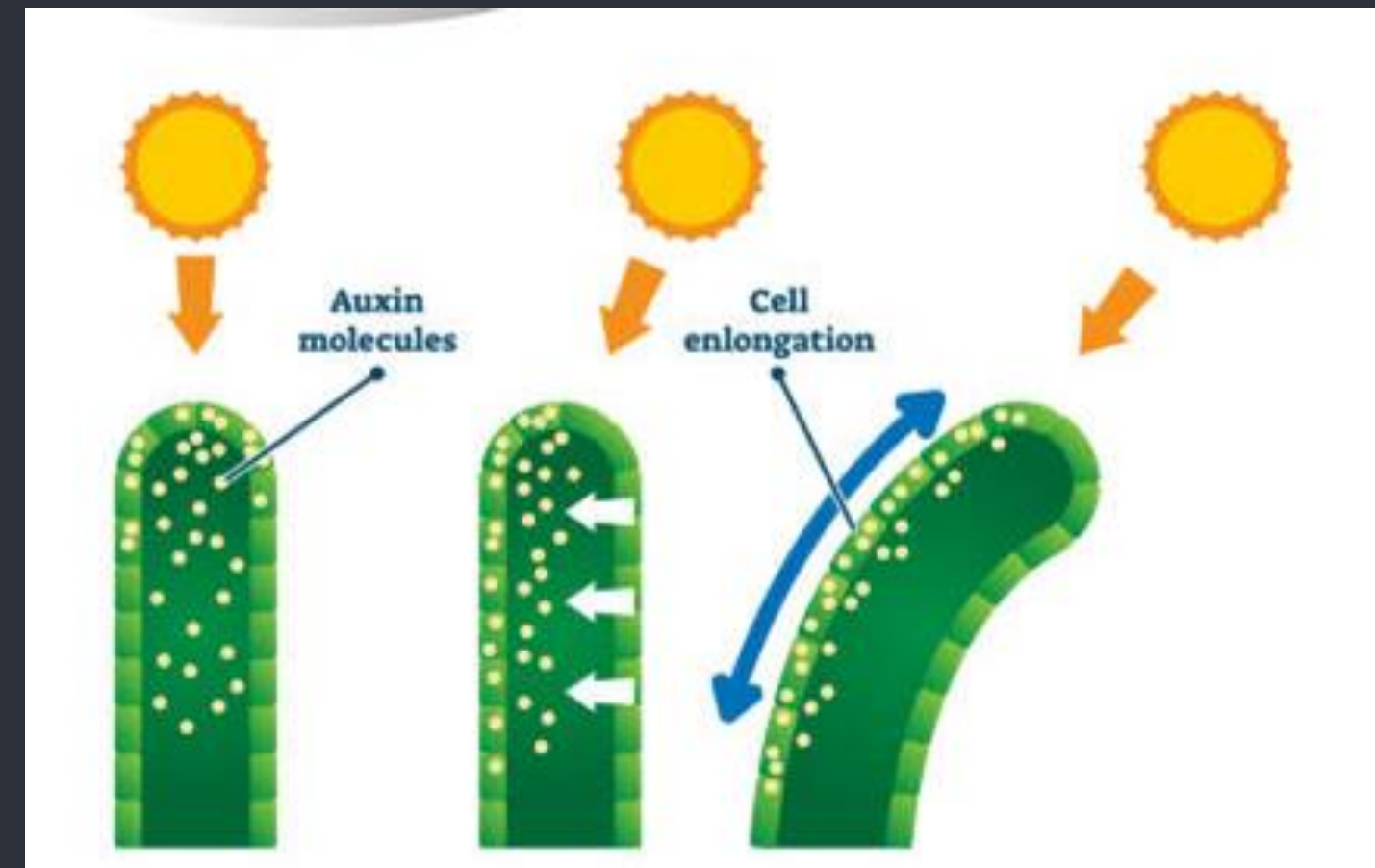




RESPONSE OF PLANTS TO LIGHT:

PHOTOTROPISM

- When the light falls only on the right side of the stem, then the auxin hormone collects in the left side of the stem, away from light. This is because auxin hormone prefers to stay in shade.

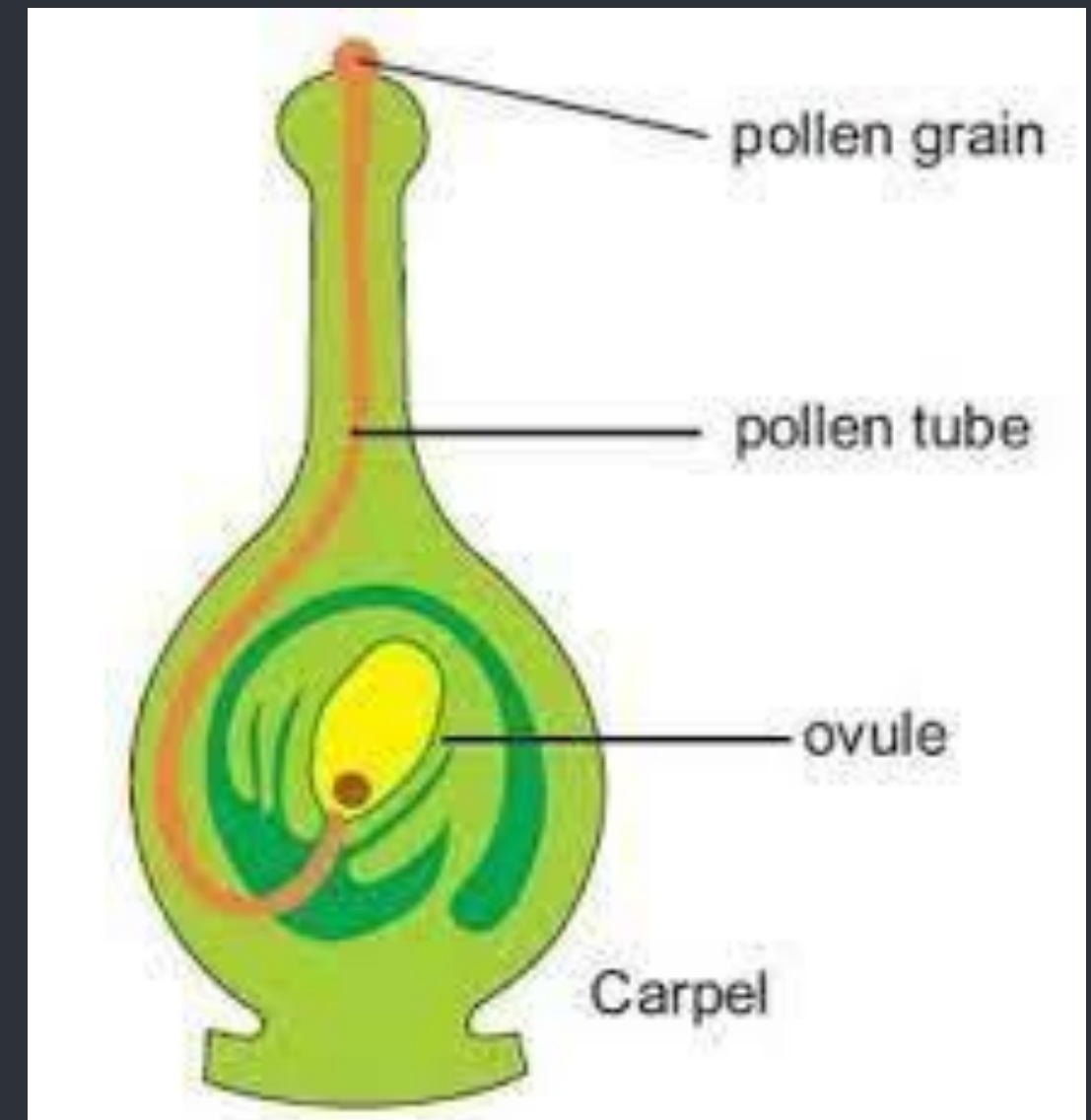




RESPONSE OF PLANTS TO LIGHT:

CHEMOTROPISM

- The growth (or movement) of a plant part due to chemical stimulus is known as chemotropism.
- The growth (or movement) of a pollen tube towards the ovule induced by a sugary substance as stimulus, is an example of chemotropism.
- The ripe stigma in the carpel of flower secretes a chemical substance (which is a sugary substance) into the style towards the ovary.





RESPONSE OF PLANTS TO LIGHT:

CHEMOTROPISM

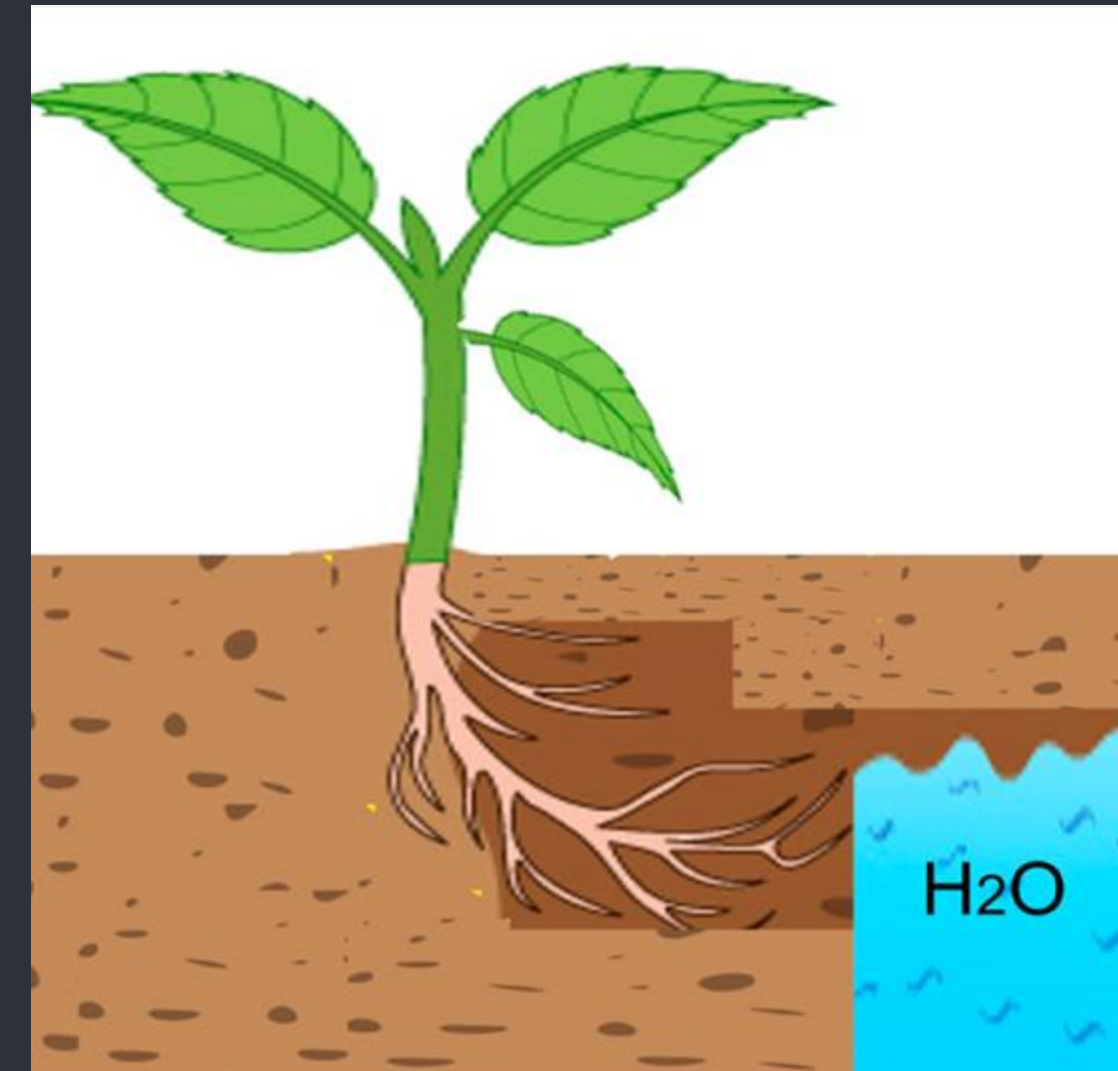
- This sugary substance acts as a stimulus for the pollen grains which fall on the stigma of the carpel.
- The pollen grain responds to this stimulus by growing a pollen tube in the downward direction into the style of the carpel and reaches the ovule in the ovary of the flower for carrying out fertilization.
- This growth of the pollen tube in response to a chemical substance secreted by the stigma of a flower is an example of Chemotropism.



RESPONSE OF PLANTS TO LIGHT:

HYDROTROPISM

- The roots of plants always go towards water, even if it means going against the pull of gravity.





THIGMOTROPISM

- Tendrils are the thin, thread-like growth on the stems or leaves of climbing plants.
- Thus, there are two types of tendrils : stem tendrils and leaf tendrils.
- Tendrils are sensitive to the touch (or contact) of other objects.
- Tendrils are positive thigmotrophic which means that they grow towards things they happen to touch.

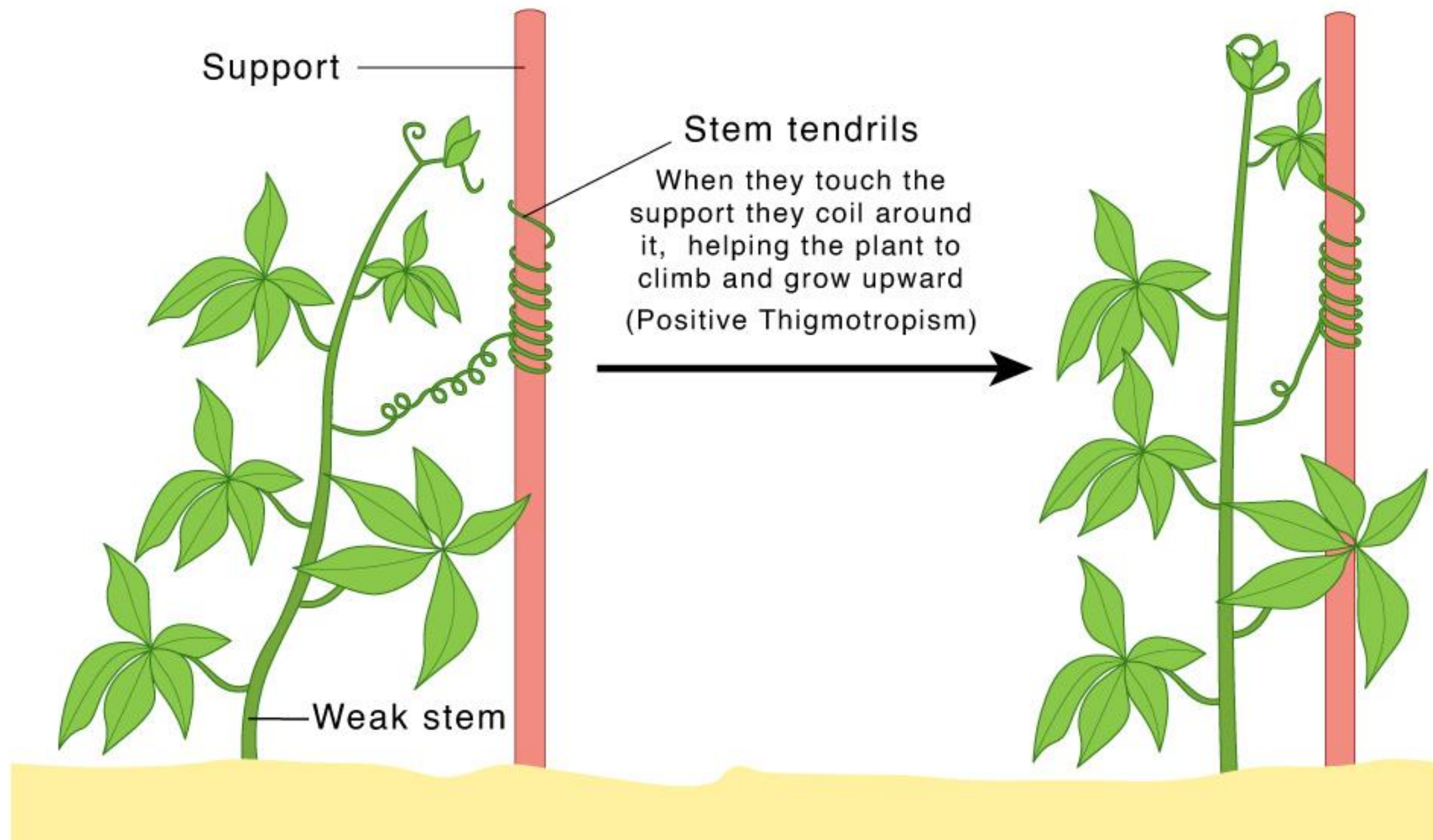




THIGMOTROPISM

Thigmotropism

ScienceFacts.net



NASTIES (OR NASTIC MOVEMENTS)



- The movement of a plant part in response to an external stimulus in which the direction of response is not determined by the direction of stimulus is called nastic movement.
- The folding up of the leaves of a sensitive plant (*Mimosa pudica*) on touching is an example of nastic movement. Here the stimulus is touch.

NASTIES (OR NASTIC MOVEMENTS)



- The opening up of the petals of dandelion flowers in morning in bright light and closing in the evening when the light fades is an example of nastic movement. In this case the stimulus is light.



NASTIES (OR NASTIC MOVEMENTS)



- The closing of the petals of moonflower in the morning in bright light and opening at dark when the light fades is also an example of nastic movement. In this case also the stimulus is light.





Tropic movements

Unidirectional response to the stimulus

Growth dependent movements

More or less permanent and irreversible

Found in all plants

Slow action

Nastic movements

Non-directional response to the stimulus

Growth independent movements

Temporary and reversible

Found only in a few specialized plants

Immediate action



THIGMONASTY

- The non-directional movement of a plant part in response to the touch of an object is called **thigmonasty**.
- The nastic movement in plants caused by touch (or thigmonasty) is provided by the sensitive plant (**Mimosa pudica**) which is also known as touch-me-not plant. It is called **chhui-mui** in Hindi.





THIGMONASTY

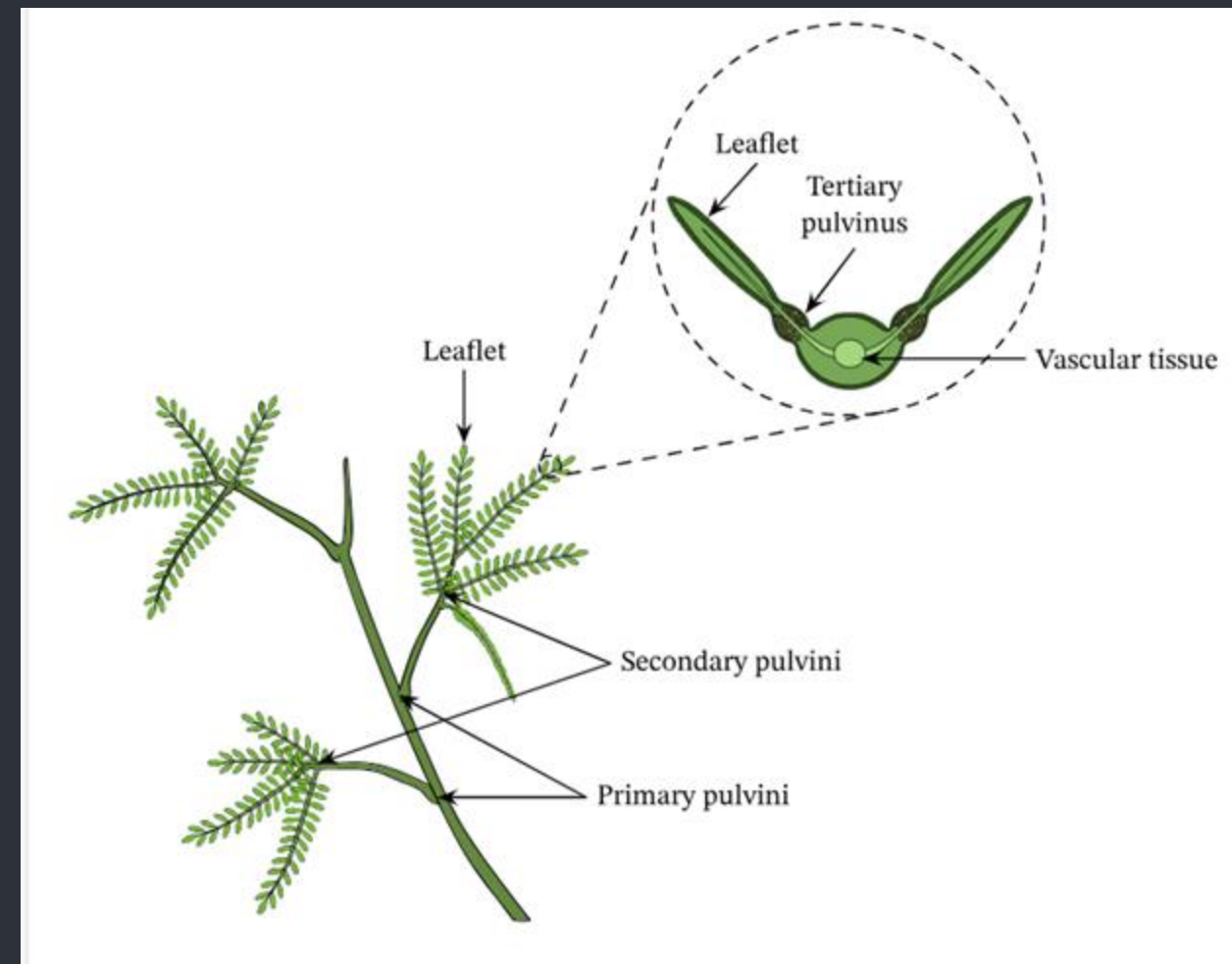
- If we touch the leaves (or rather leaflets) of the sensitive plant with our fingers, then its leaves fold up and droop almost immediately.





THIGMONASTY

- The folding up of the leaves of a sensitive plant on touching is due to the sudden loss of water from pad-like swellings called 'pulvini' present at the base of all leaves of the sensitive plant which make the pulvini lose their firmness causing the leaves to drop and fall.





PHOTONASTY

- The non-directional movement of a plant part (usually petals of flowers) in response to light is called photonasty.
- The opening and closing of flowers in response to light (or photonasty) are growth movements.
- Petals open when their inner surfaces grow more than their outer surfaces. On the other hand, petals close when their outer surfaces grow more than their inner surfaces.





FUNCTIONS OF PLANT HORMONES (OR PHYTOHORMONES)

- Germination of seeds (or Breaking the dormancy of seeds).



FUNCTIONS OF PLANT HORMONES (OR PHYTOHORMONES)

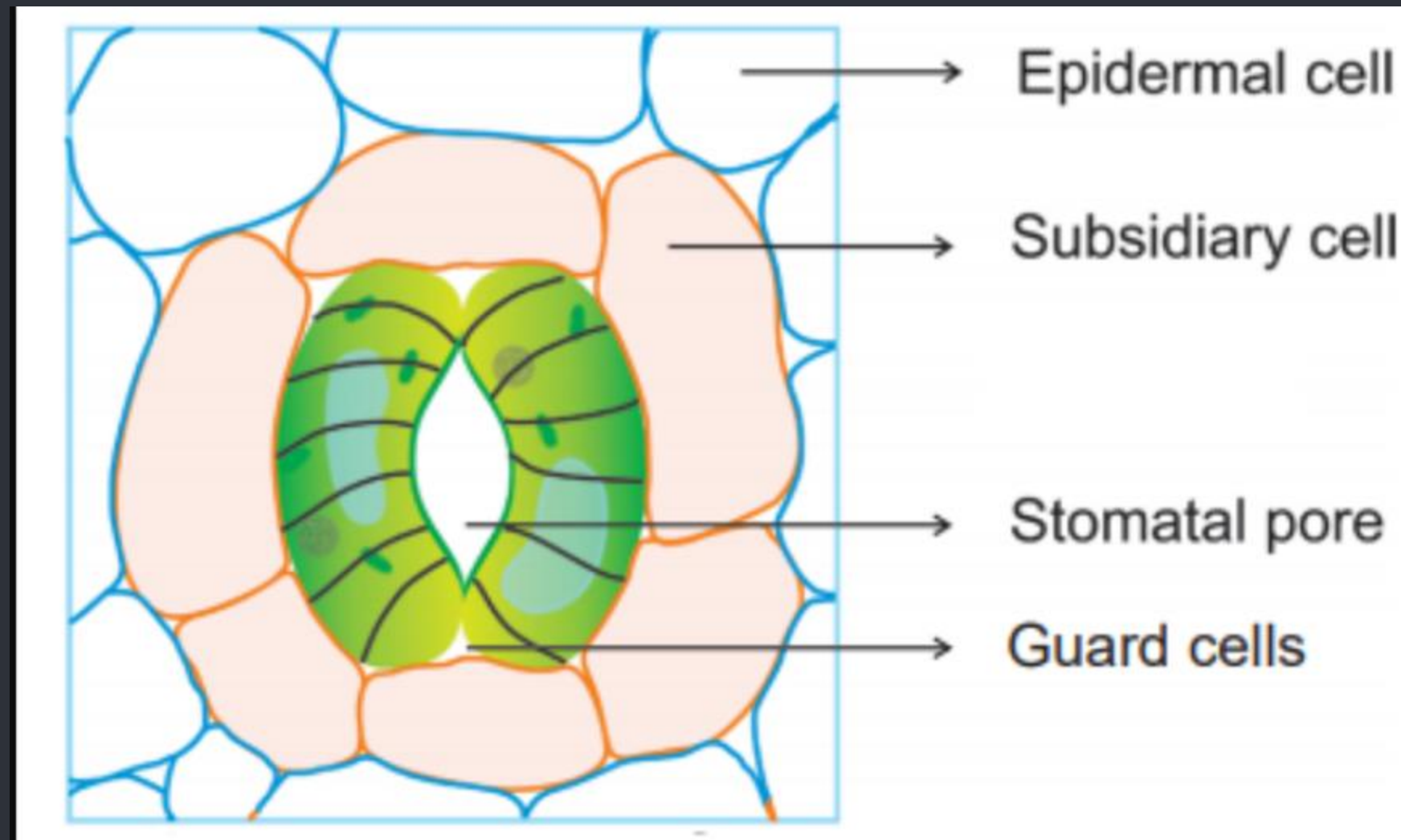
- Growth of root, stem and leaves





FUNCTIONS OF PLANT HORMONES (OR PHYTOHORMONES)

- Movement of stomata (or stomatal movement) in leaves





FUNCTIONS OF PLANT HORMONES (OR PHYTOHORMONES)

- Flowering of plants





FUNCTIONS OF PLANT HORMONES (OR PHYTOHORMONES)

- Ripening of fruits





THANK YOU.